

Single Column Model (SCM) Diagnostic Runs

GSPT Simulation Pipeline

July 25, 2026

1 Overview

This document compiles every current SCM case report TeX file under results/.

2 Case 1: cases99 80x2m 9h report

3 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/cases99/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`.

3.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/cases99`
- Number of saved time samples: 1081
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 108 / 0
- RHS evaluations: 27002
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reltol} = 1.000 \times 10^{-6}$

3.2 Forcing and Boundary Conditions

Table 1: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	$10.000000 \text{ m s}^{-1}$
Meridional geostrophic wind	V_g	$0.000000 \text{ m s}^{-1}$
Surface roughness (momentum)	z_{0m}	0.020000 m
Surface roughness (heat)	z_{0h}	0.005000 m
Coriolis parameter	f	$1.000\text{e-}04 \text{ s}^{-1}$
Reference potential temperature	θ_a	280.000000 K
Deep soil temperature	T_{deep}	276.000000 K
Downward longwave forcing	$R_{\downarrow}$	$300.000000 \text{ W m}^{-2}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{\text{min,surf}}$	$0.001000 \text{ m}^2 \text{ s}^{-1}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 s^{-1}

3.3 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 7.678076%
- Diffusivity bounds: $K_m \in [13.399989, 94.108839] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [-4.024 \times 10^{-6}, 1551.592844]$

3.4 Generated Artifacts

- Time-series CSV: `results/cases99/time_series.csv`
- Diagnostic summary JSON: `results/cases99/summary.json`
- Payload JLD2: `results/cases99/payload.jld2`

3.5 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times
5. Surface energy budget components (R_n , H , G , storage)
6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Eddy diffusivity response (K_m , K_h) versus stability/Richardson number
8. Fold proximity diagnostics versus time

3.6 Included Plots

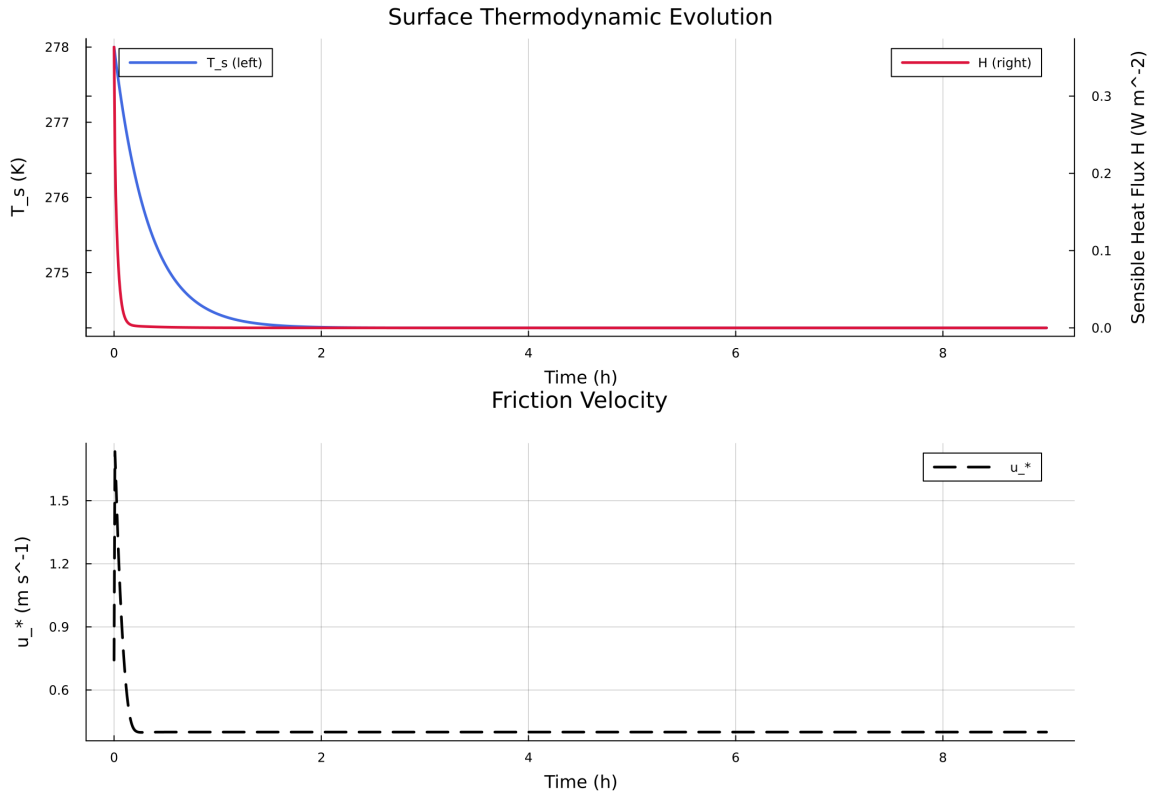


Figure 1: Time series of T_s , sensible heat flux, and friction velocity

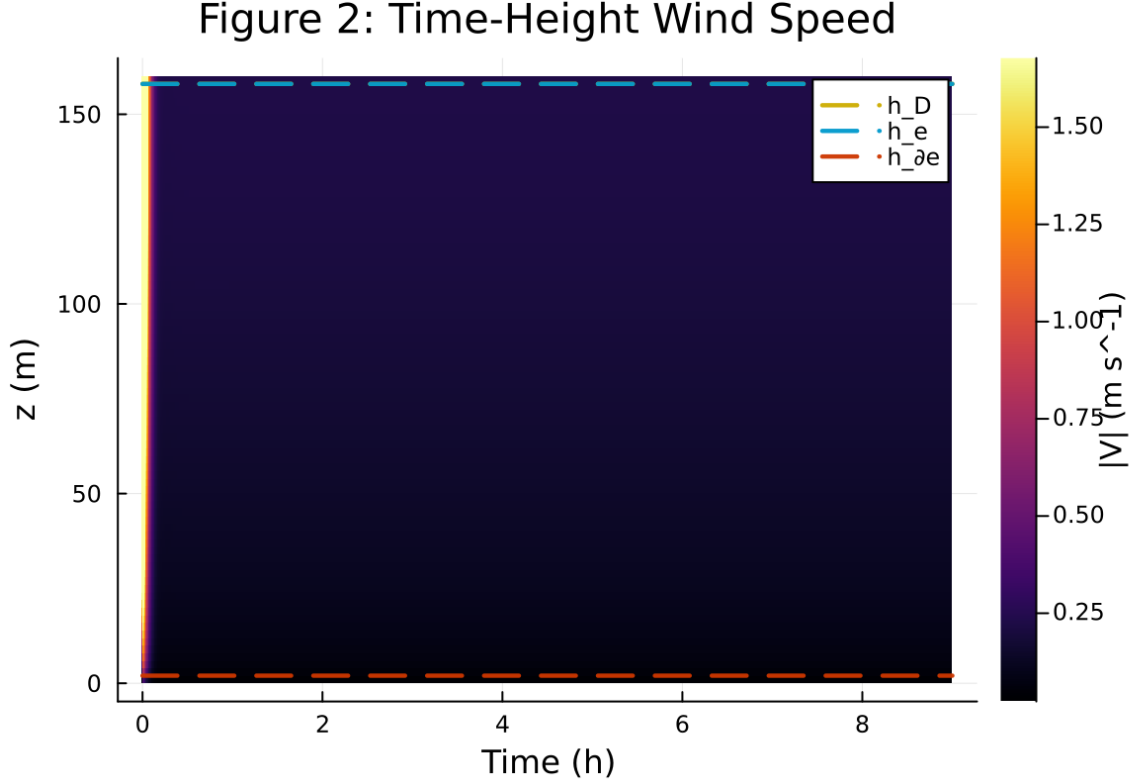


Figure 2: Time-height contour of wind speed with LLJ evolution

4 Case 2: floss 80x2m 9h report

5 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/floss/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`.

5.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/floss`
- Number of saved time samples: 1081
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 108 / 0
- RHS evaluations: 27002
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

Figure 3: Time-Height Potential Temperature

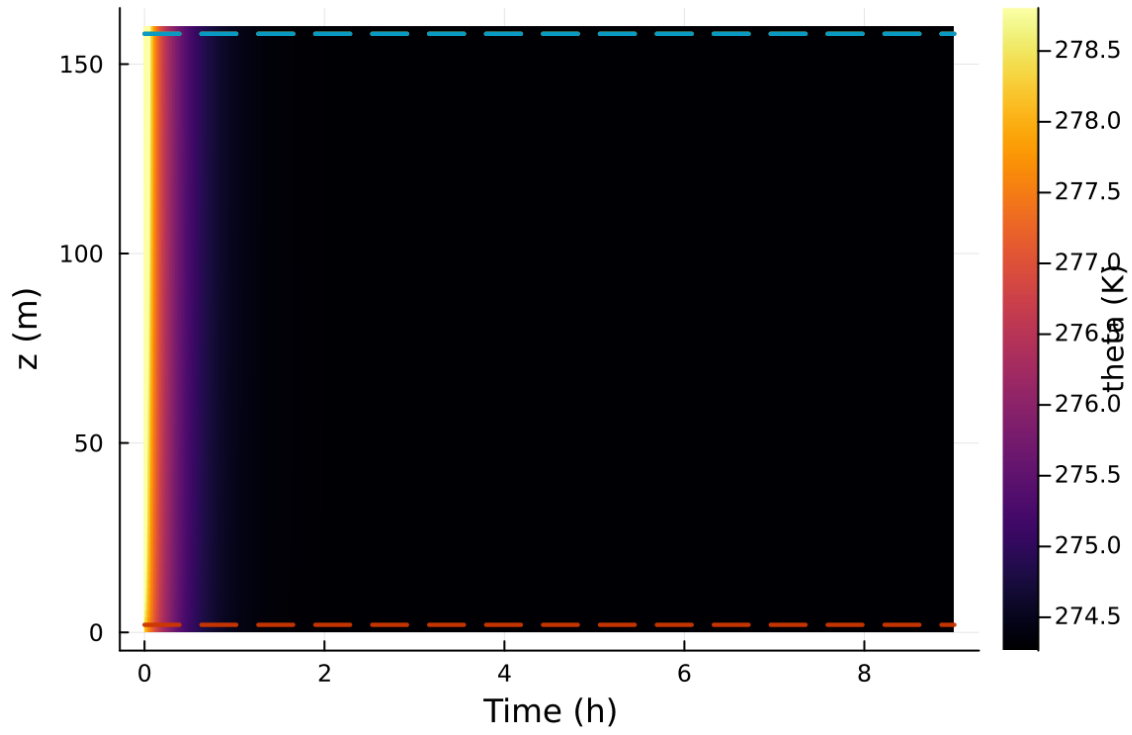


Figure 3: Time-height contour of potential temperature and inversion growth

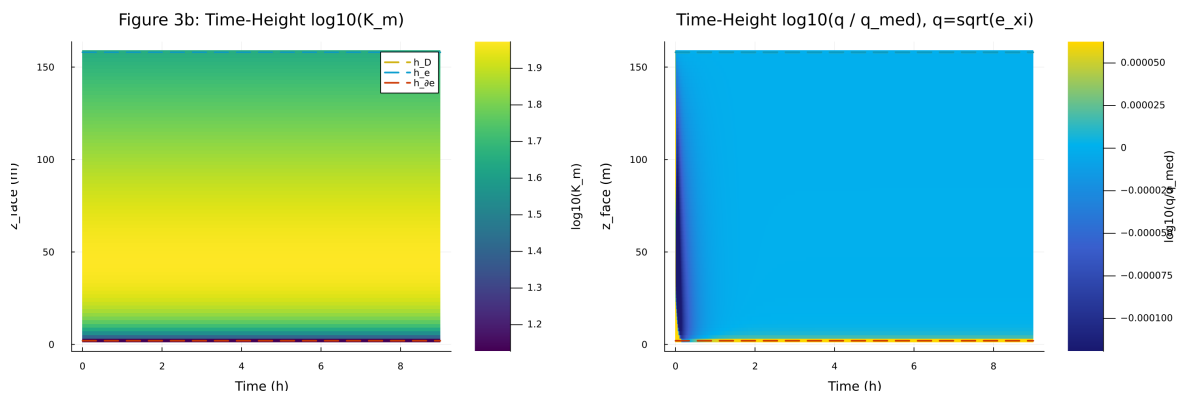


Figure 4: Vertical profiles of U, theta, and Km at representative times

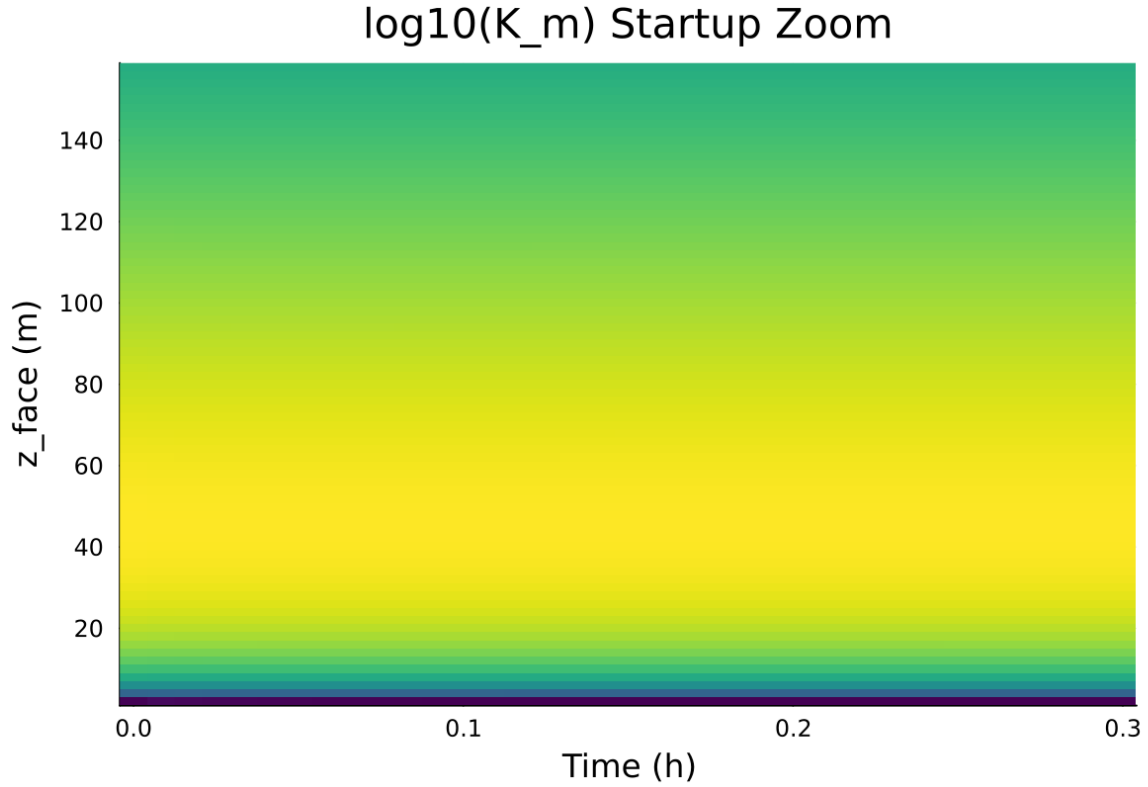


Figure 5: Surface energy budget components (R_n , H , G , storage)

Table 2: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	$10.000000 \text{ m s}^{-1}$
Meridional geostrophic wind	V_g	$0.000000 \text{ m s}^{-1}$
Surface roughness (momentum)	z_{0m}	0.020000 m
Surface roughness (heat)	z_{0h}	0.005000 m
Coriolis parameter	f	$1.000\text{e-}04 \text{ s}^{-1}$
Reference potential temperature	θ_a	280.000000 K
Deep soil temperature	T_{deep}	276.000000 K
Downward longwave forcing	$R_{\downarrow}$	$300.000000 \text{ W m}^{-2}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{min,surf}$	$0.001000 \text{ m}^2 \text{ s}^{-1}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 s^{-1}

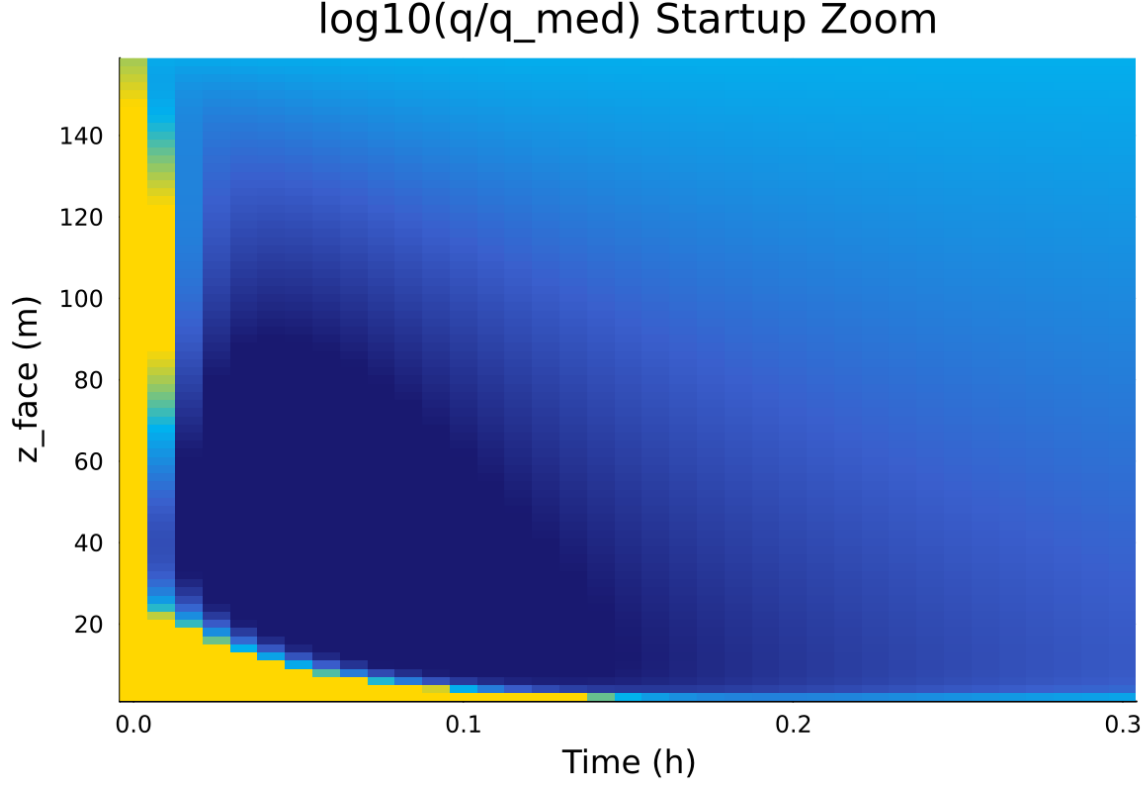


Figure 6: Phase portrait on regularized manifold (Delta vs e_{xi})

5.2 Forcing and Boundary Conditions

5.3 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 7.678076%
- Diffusivity bounds: $K_m \in [13.399989, 94.108839] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [-4.024 \times 10^{-6}, 1551.592844]$

5.4 Generated Artifacts

- Time-series CSV: `results/floss/time_series.csv`
- Diagnostic summary JSON: `results/floss/summary.json`
- Payload JLD2: `results/floss/payload.jld2`

5.5 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times

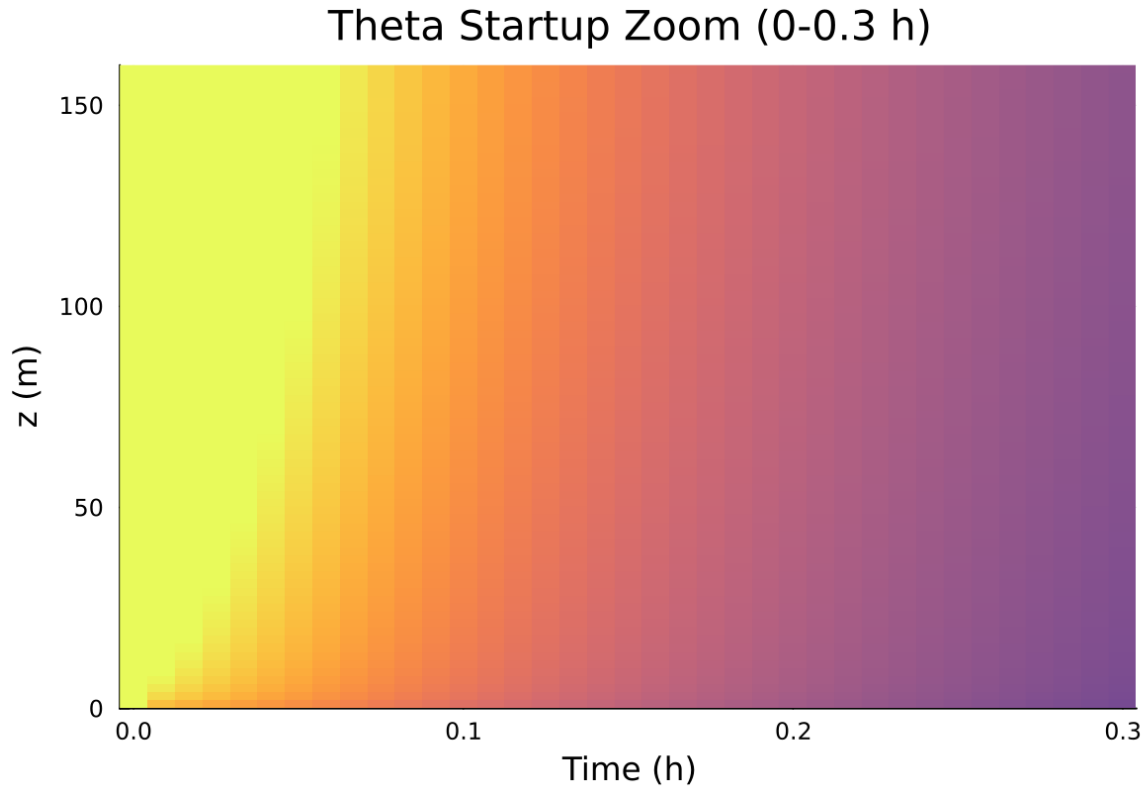


Figure 7: Eddy diffusivity response (K_m , K_h) versus stability/Richardson number

5. Surface energy budget components (R_n , H , G , storage)
6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Eddy diffusivity response (K_m , K_h) versus stability/Richardson number
8. Fold proximity diagnostics versus time

5.6 Included Plots

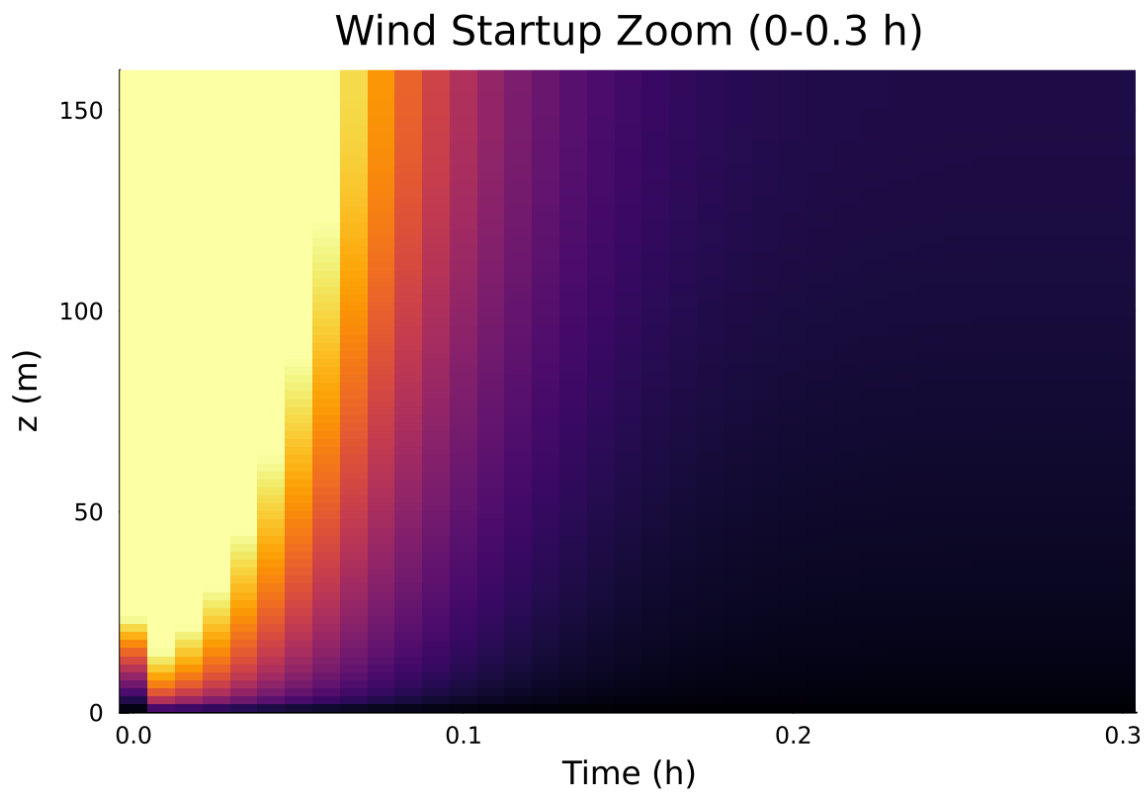


Figure 8: Fold proximity diagnostics versus time

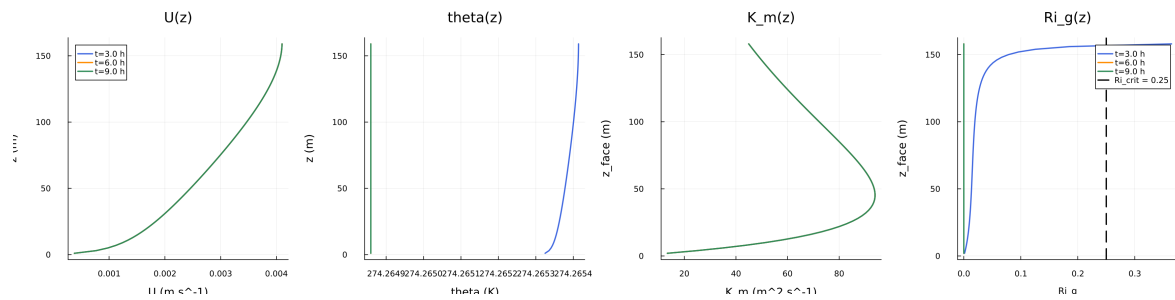


Figure 9: fig04 profiles u theta km.png

Figure 5: Surface Energy Budget

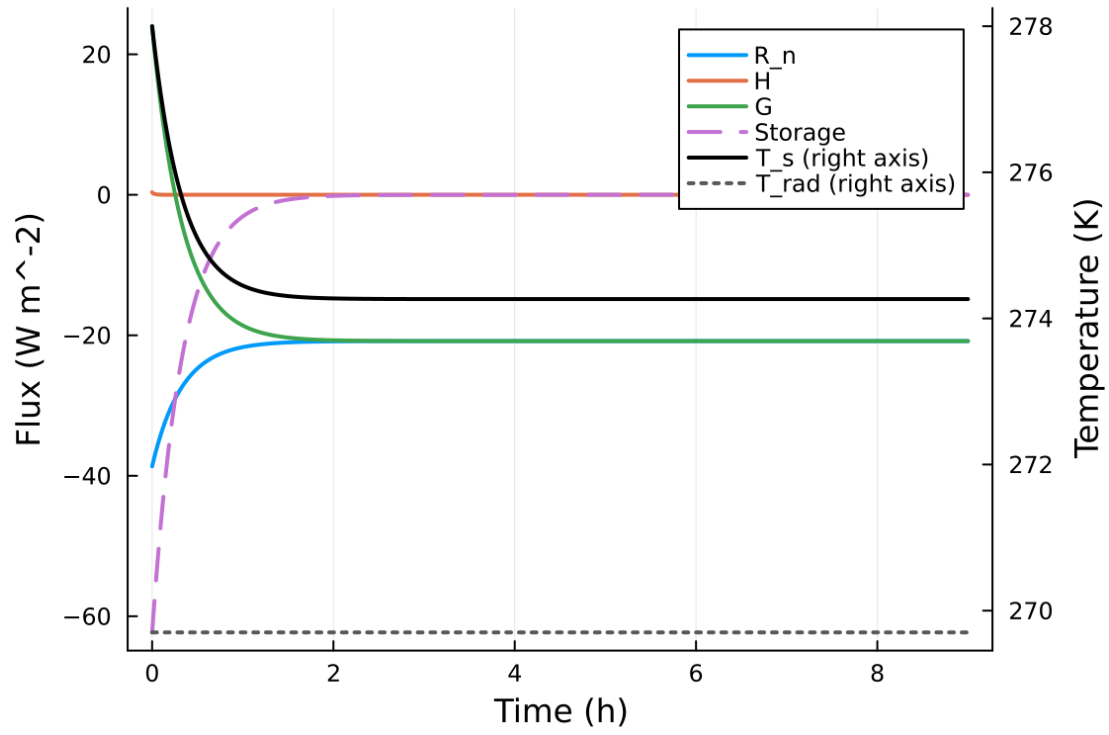


Figure 10: fig05 surface energy budget.png

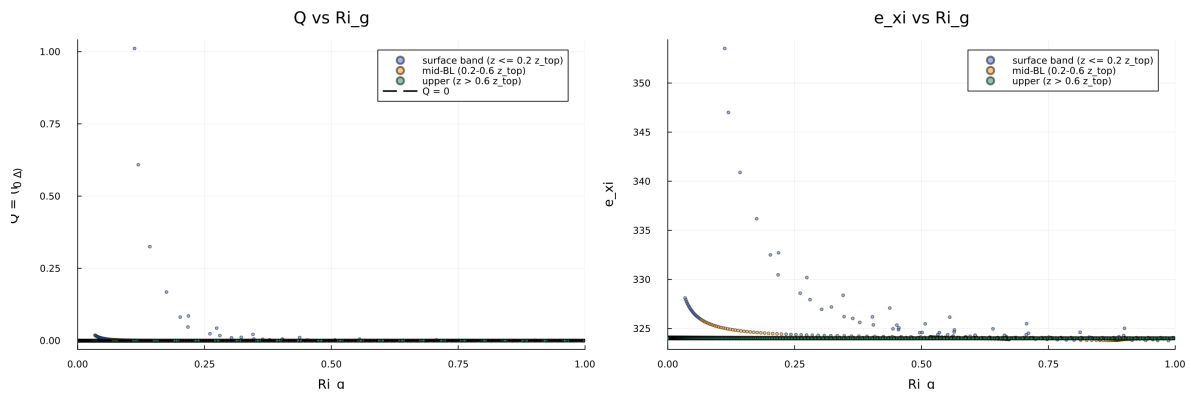


Figure 11: fig06 phase delta exi.png

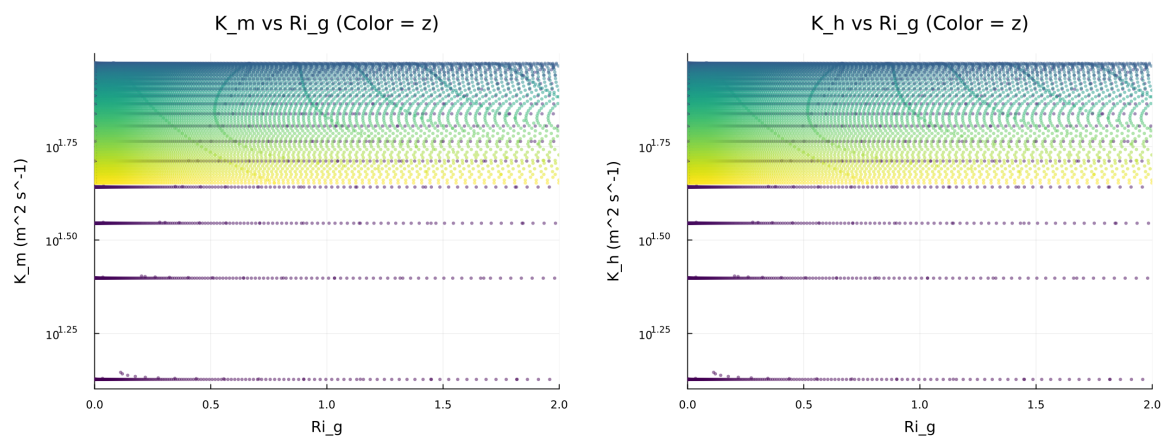


Figure 12: fig07 diffusivity vs ri.png

Figure 8: Quadratic Fold-Distance Diagnostic

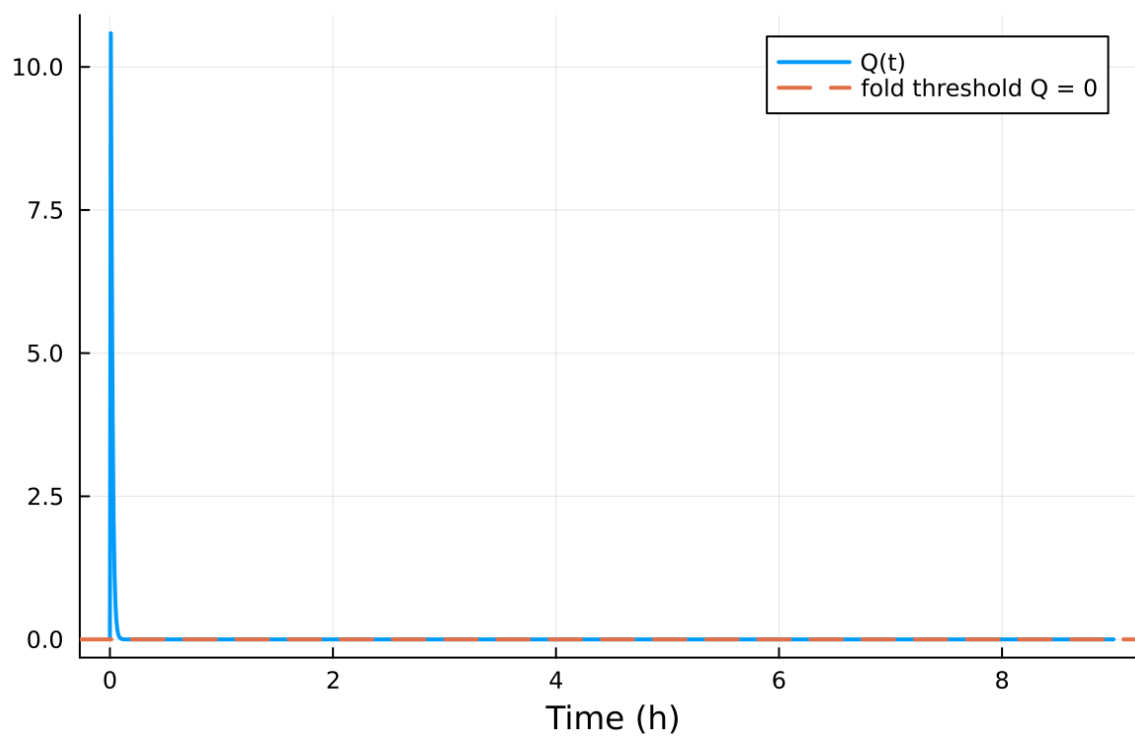


Figure 13: fig08 fold proximity.png

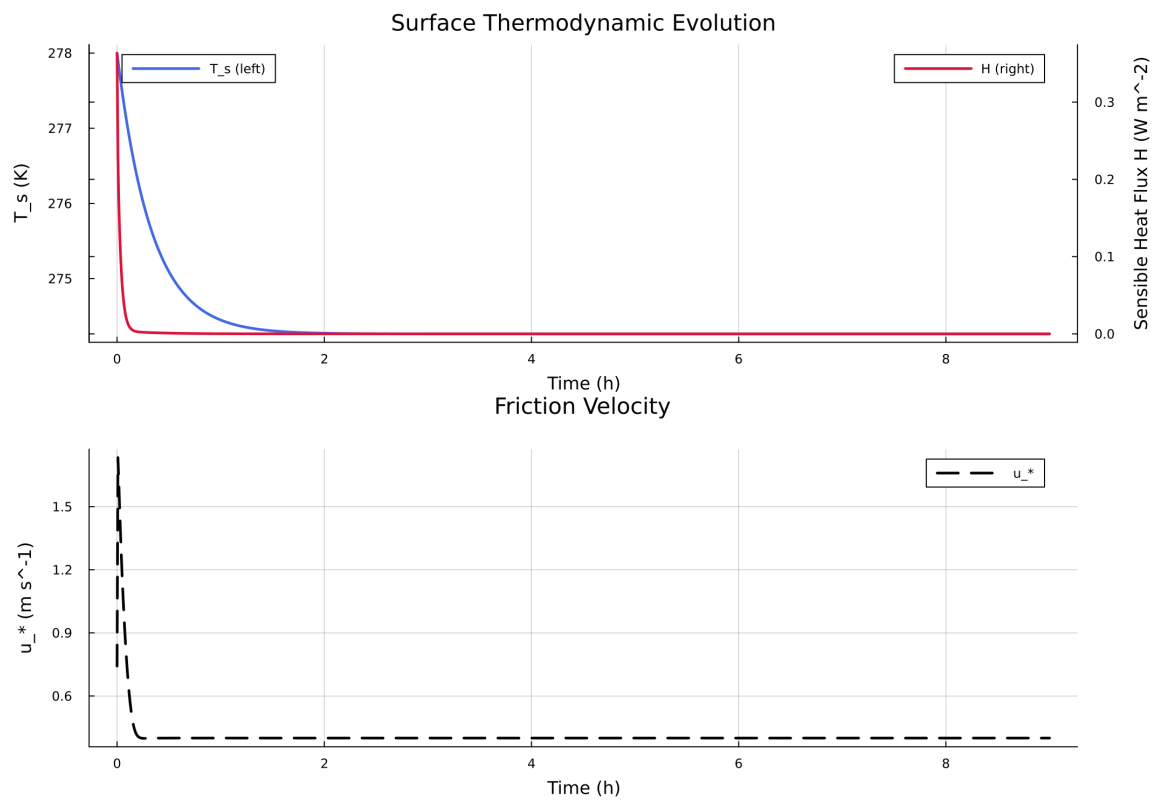


Figure 14: Time series of T_s , sensible heat flux, and friction velocity

Figure 2: Time-Height Wind Speed

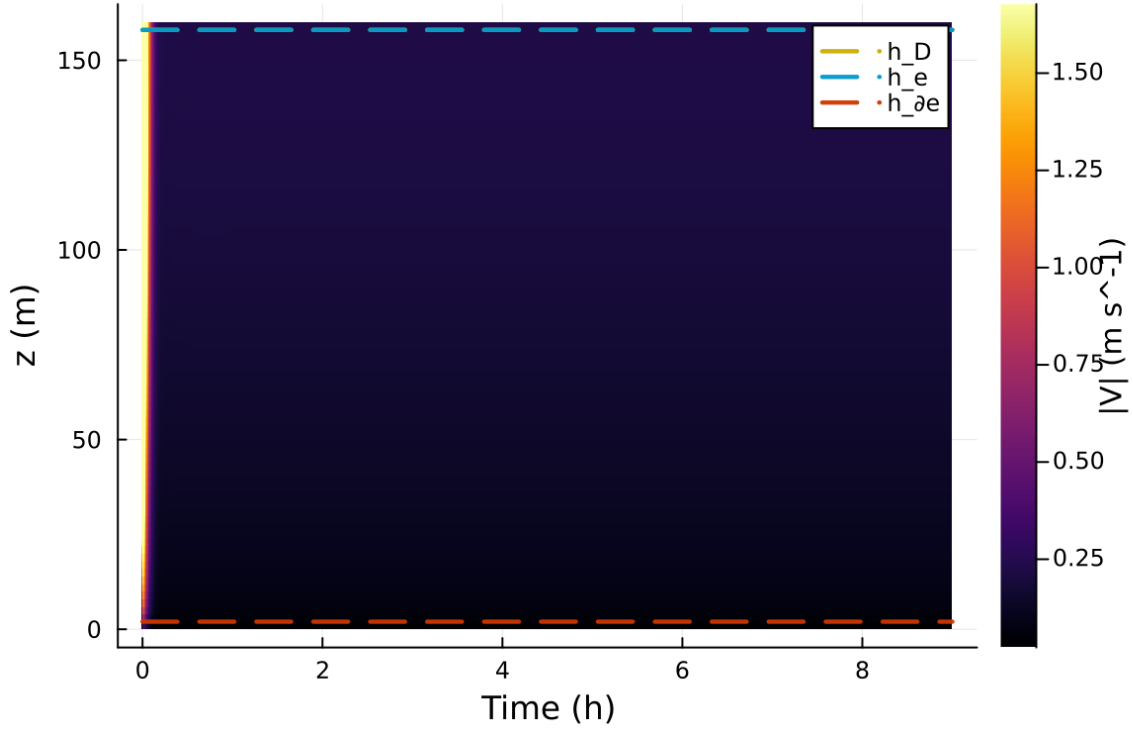


Figure 15: Time-height contour of wind speed with LLJ evolution

6 Case 3: sheba 160x2m 12h report

7 SCM Case Summary: sheba

This section was generated automatically from the SCM diagnostics artifact `results/sheba/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`.

7.1 Core Run Metadata

- Case identifier: `sheba`
- Output directory: `results/sheba`
- Number of saved time samples: 4321
- Number of profile snapshots: 25
- Solver accepted/rejected steps: 100 / 0
- RHS evaluations: 49002
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

Figure 3: Time-Height Potential Temperature

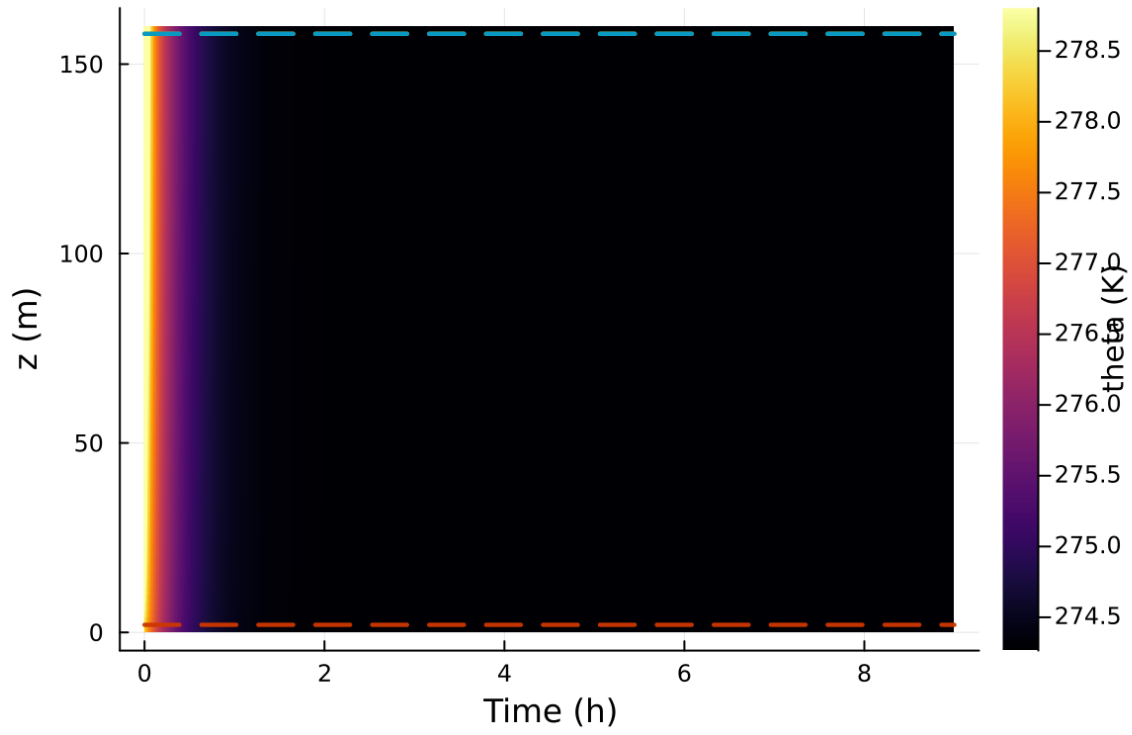


Figure 16: Time-height contour of potential temperature and inversion growth

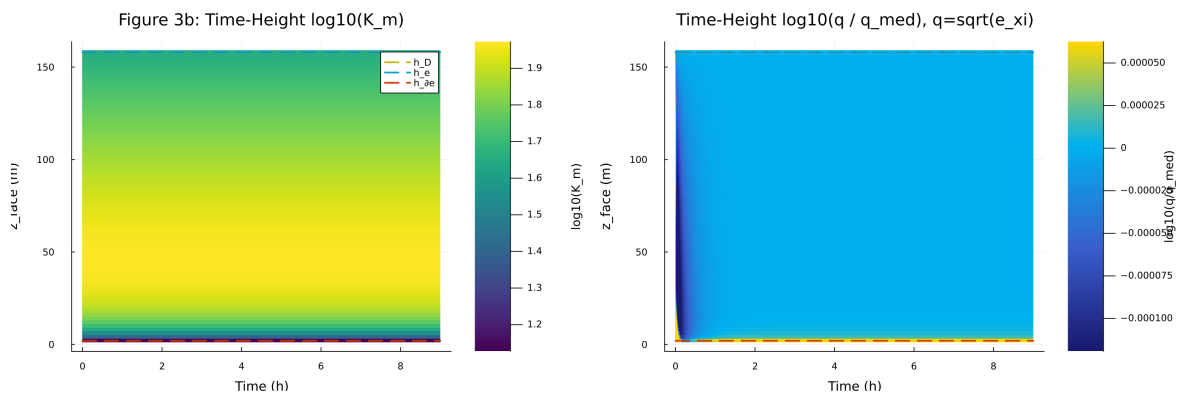


Figure 17: Vertical profiles of U , θ , and K_m at representative times

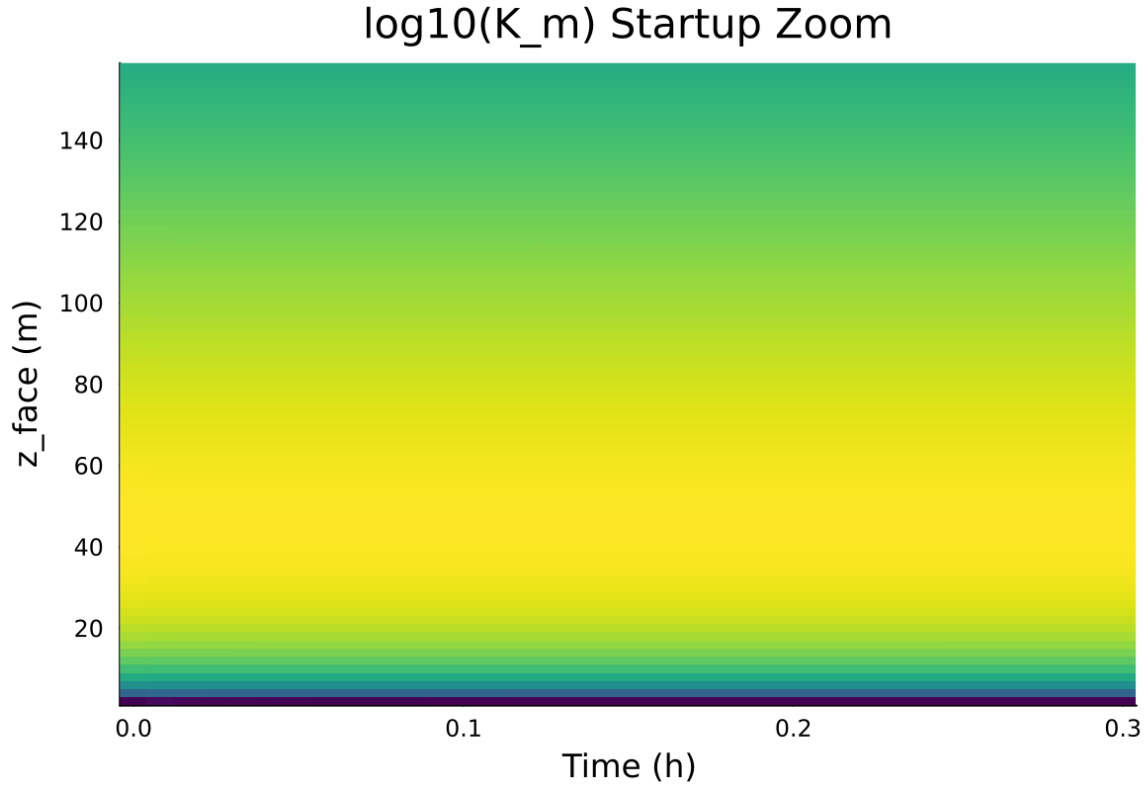


Figure 18: Surface energy budget components (R_n , H , G , storage)

Table 3: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	$7.000000 \text{ m s}^{-1}$
Meridional geostrophic wind	V_g	$0.000000 \text{ m s}^{-1}$
Surface roughness (momentum)	z_{0m}	$1.000\text{e-}04 \text{ m}$
Surface roughness (heat)	z_{0h}	$1.000\text{e-}05 \text{ m}$
Coriolis parameter	f	$1.000\text{e-}04 \text{ s}^{-1}$
Reference potential temperature	θ_a	257.000000 K
Deep soil temperature	T_{deep}	255.500000 K
Downward longwave forcing	$R_{\downarrow}$	$190.000000 \text{ W m}^{-2}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{\min, \text{surf}}$	$0.001000 \text{ m}^2 \text{ s}^{-1}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	257.000000 K
Top relaxation coefficient	λ_{top}	0.000000 s^{-1}

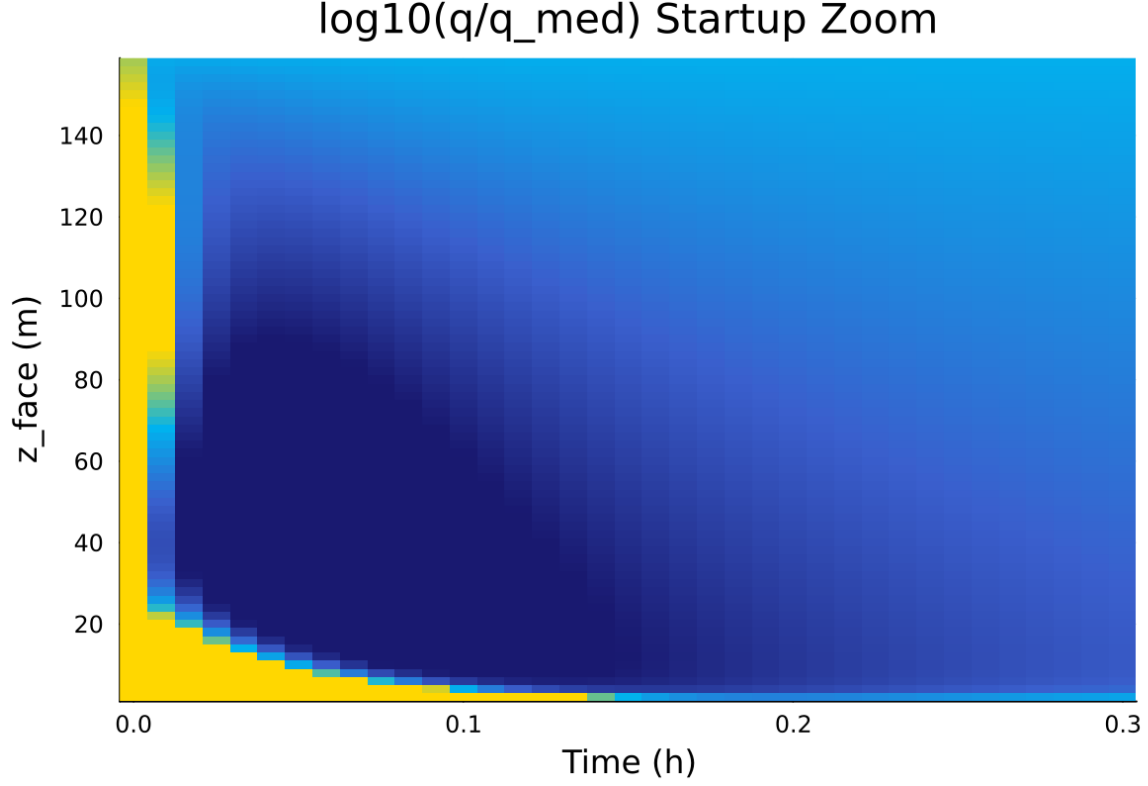


Figure 19: Phase portrait on regularized manifold (Delta vs e_xi)

7.2 Forcing and Boundary Conditions

7.3 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [13.028339, 141.433744] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [-7.243 \times 10^{-6}, 3569.000203]$

7.4 Generated Artifacts

- Time-series CSV: `results/sheba/time_series.csv`
- Diagnostic summary JSON: `results/sheba/summary.json`
- Payload JLD2: `results/sheba/payload.jld2`

7.5 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times

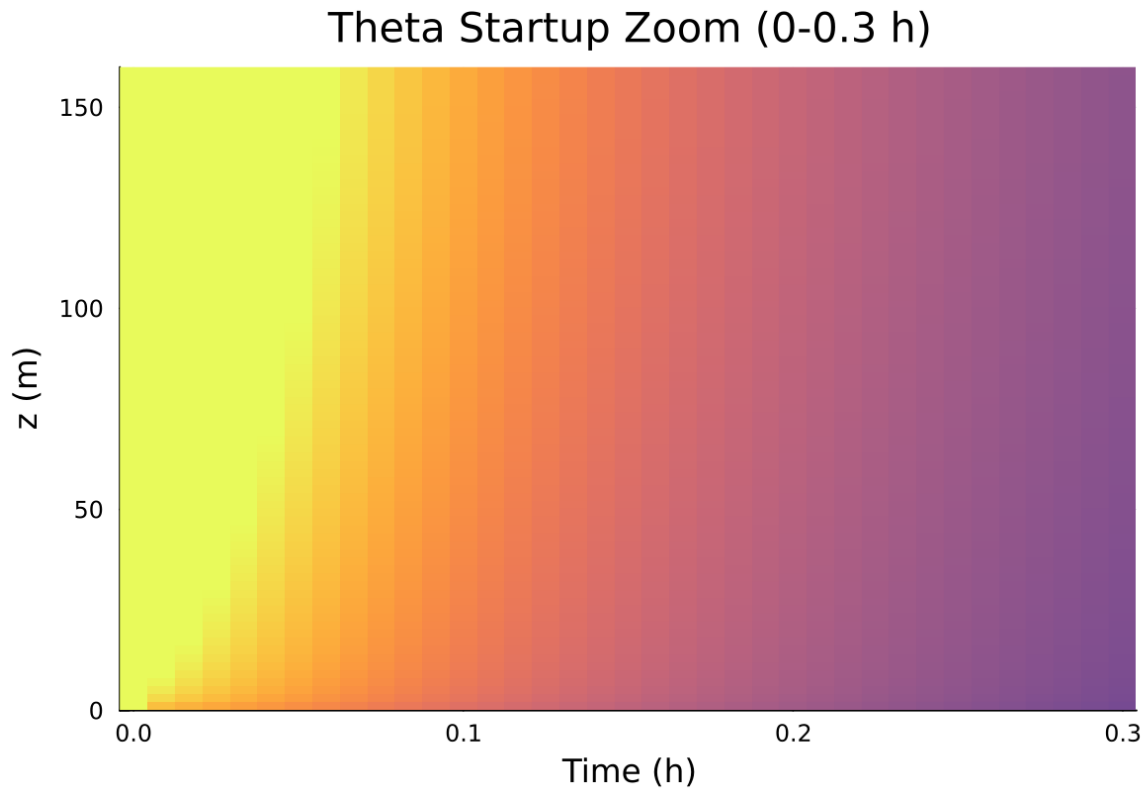


Figure 20: Eddy diffusivity response (K_m , K_h) versus stability/Richardson number

5. Surface energy budget components (R_n , H , G , storage)
6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Eddy diffusivity response (K_m , K_h) versus stability/Richardson number
8. Fold proximity diagnostics versus time

7.6 Included Plots

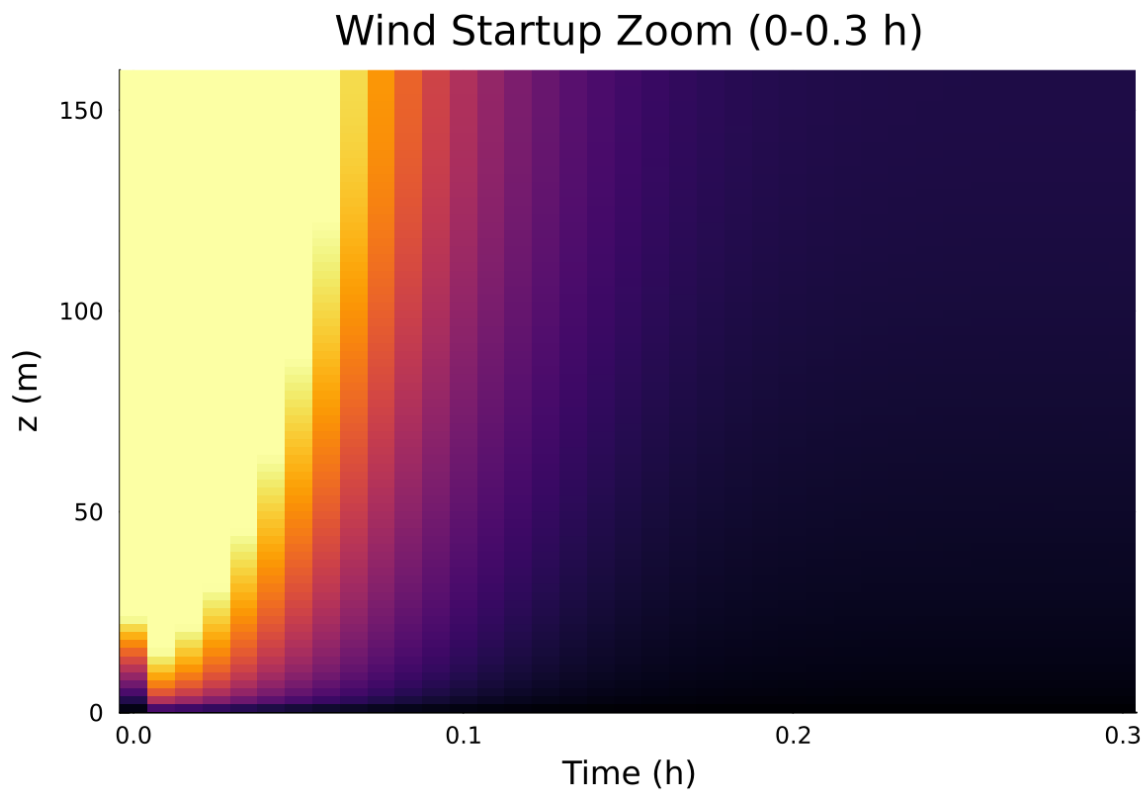


Figure 21: Fold proximity diagnostics versus time

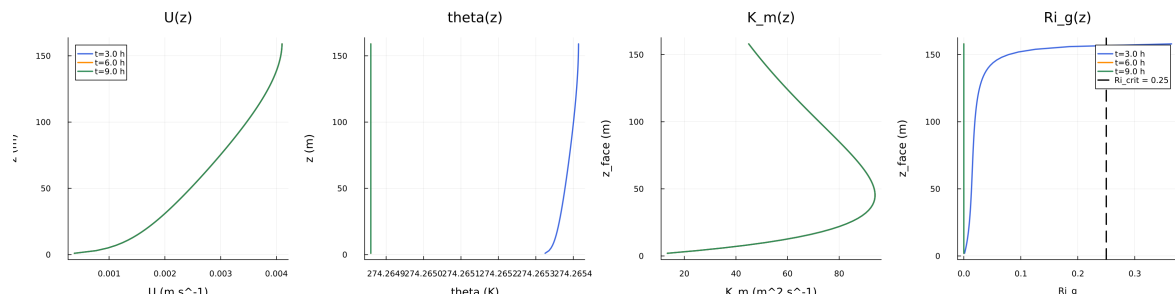


Figure 22: fig04 profiles u theta km.png

Figure 5: Surface Energy Budget

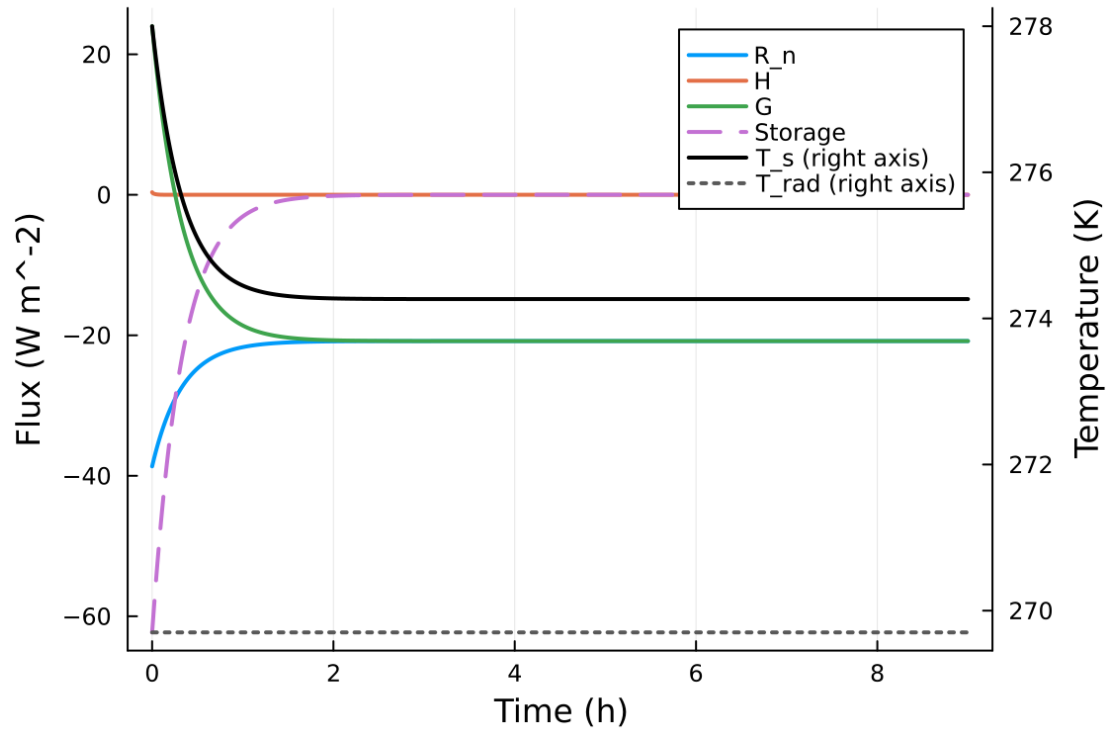


Figure 23: fig05 surface energy budget.png

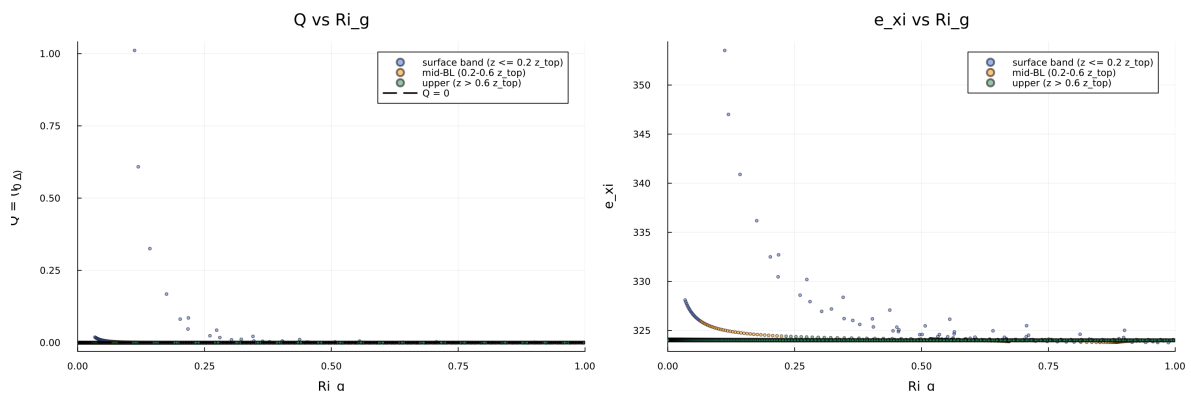


Figure 24: fig06 phase delta exi.png

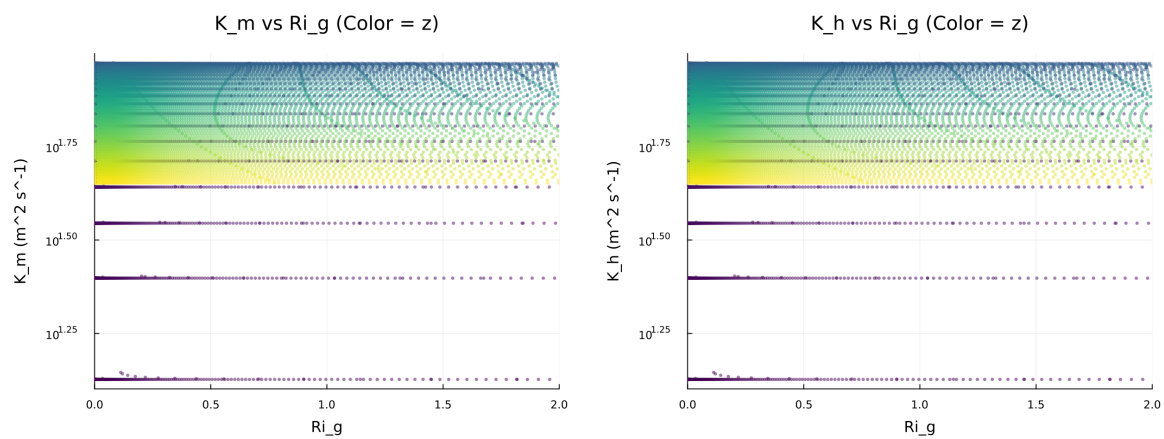


Figure 25: fig07 diffusivity vs ri.png

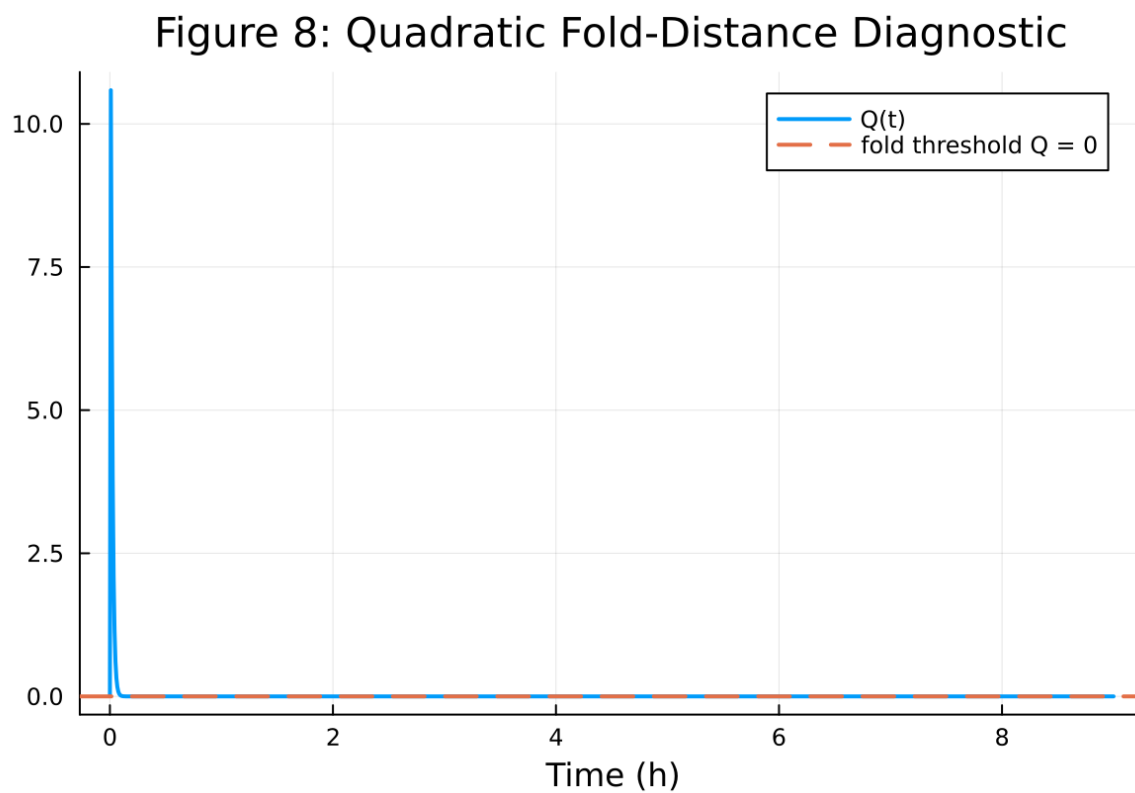


Figure 26: fig08 fold proximity.png

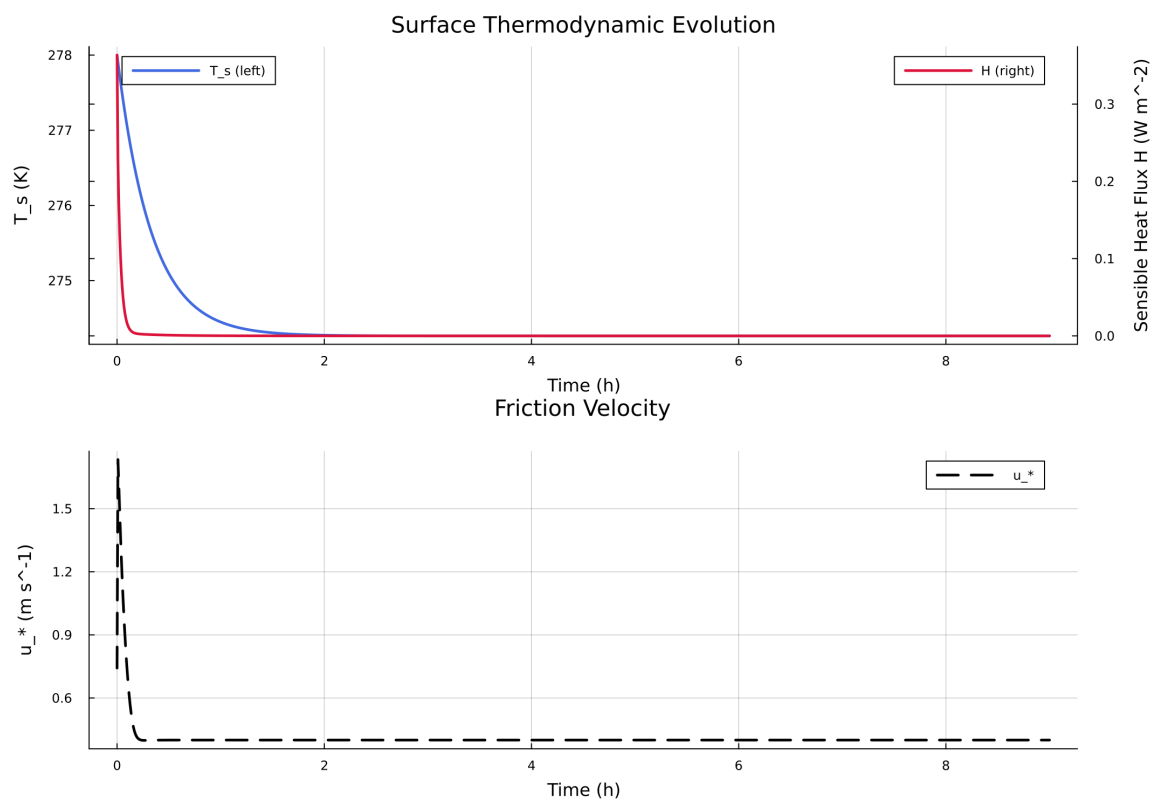


Figure 27: Time series of T_s , sensible heat flux, and friction velocity

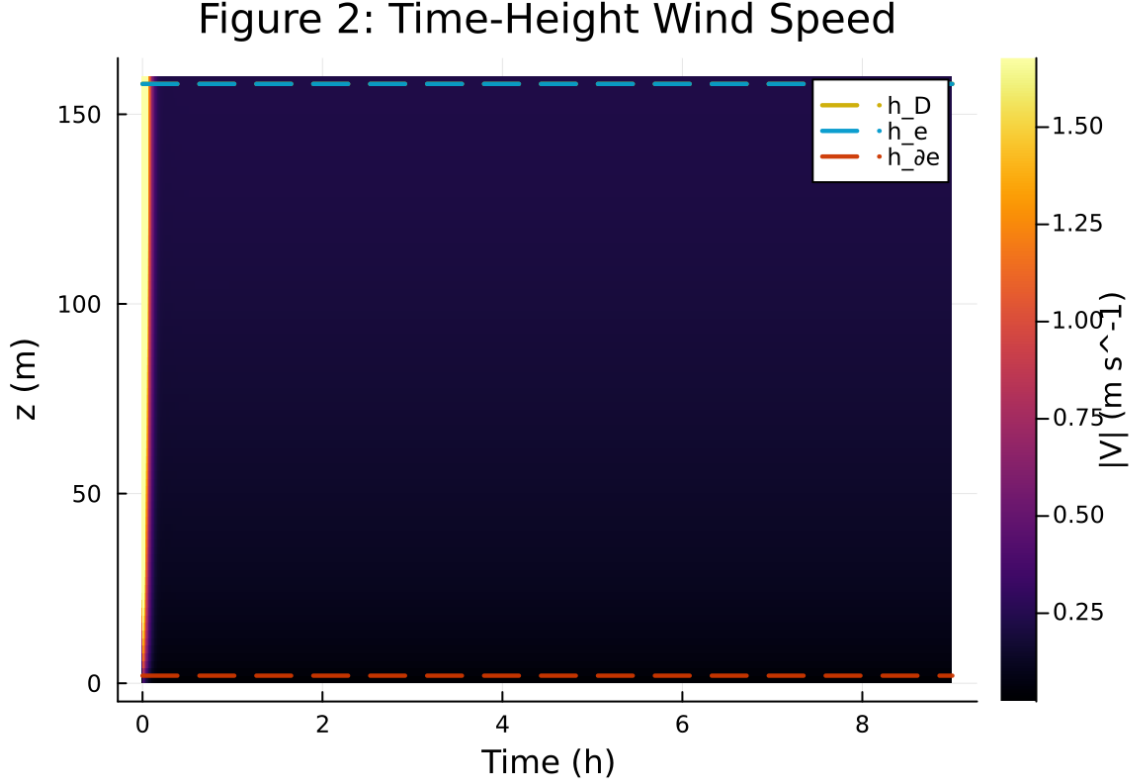


Figure 28: Time-height contour of wind speed with LLJ evolution

8 Case 4: gabls1 200x2m 12h report

9 SCM Case Summary: gabls1

This section was generated automatically from the SCM diagnostics artifact `results/gabls1/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`.

9.1 Core Run Metadata

- Case identifier: `gabls1`
- Output directory: `results/gabls1`
- Number of saved time samples: 1441
- Number of profile snapshots: 25
- Solver accepted/rejected steps: 99 / 0
- RHS evaluations: 60392
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

Figure 3: Time-Height Potential Temperature

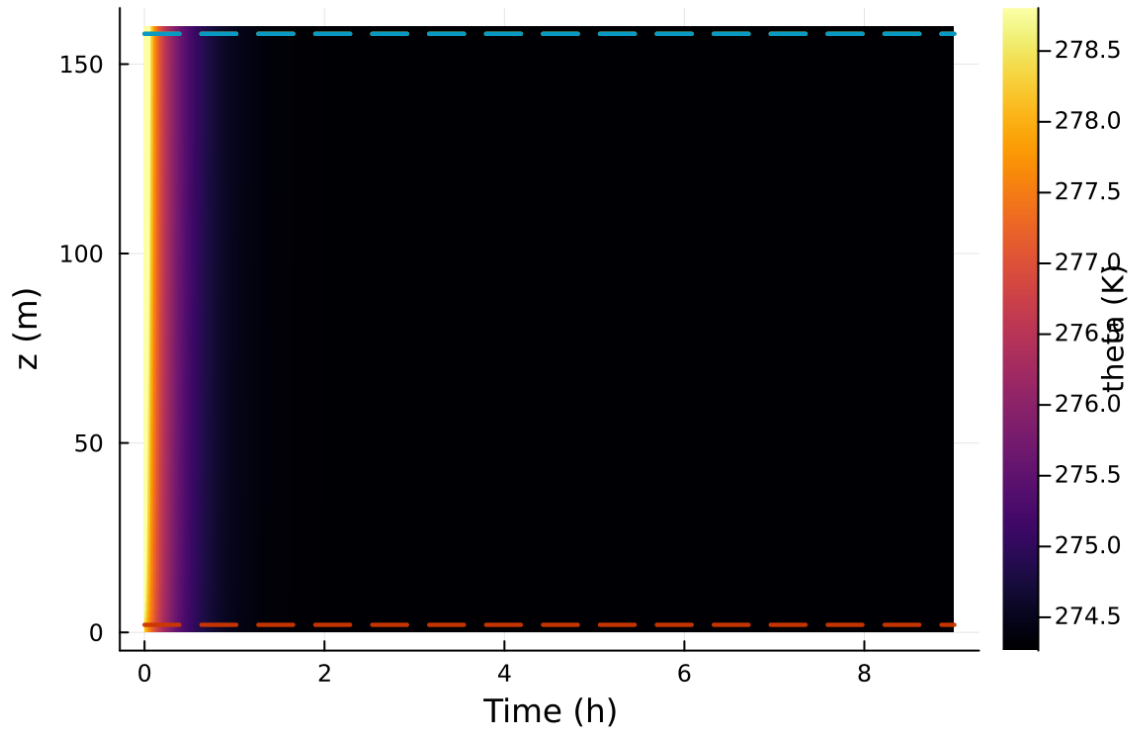


Figure 29: Time-height contour of potential temperature and inversion growth

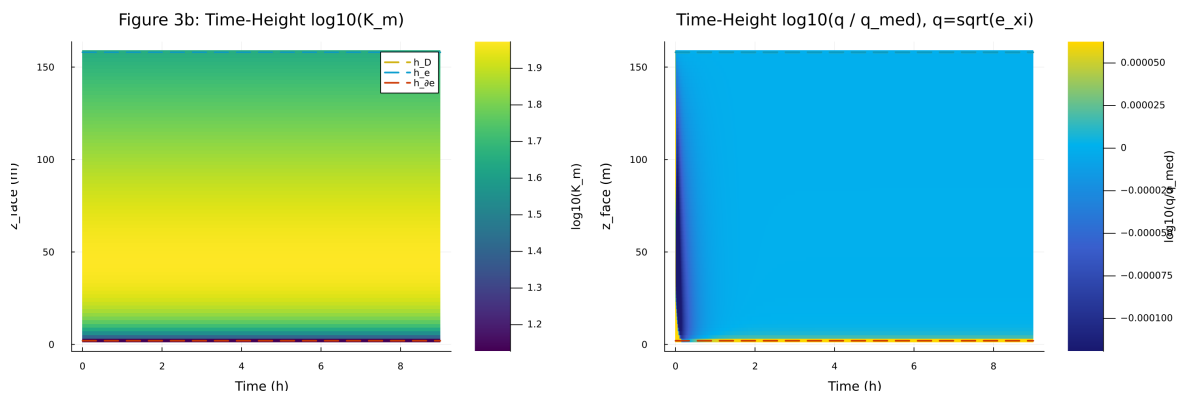


Figure 30: Vertical profiles of U , θ , and K_m at representative times

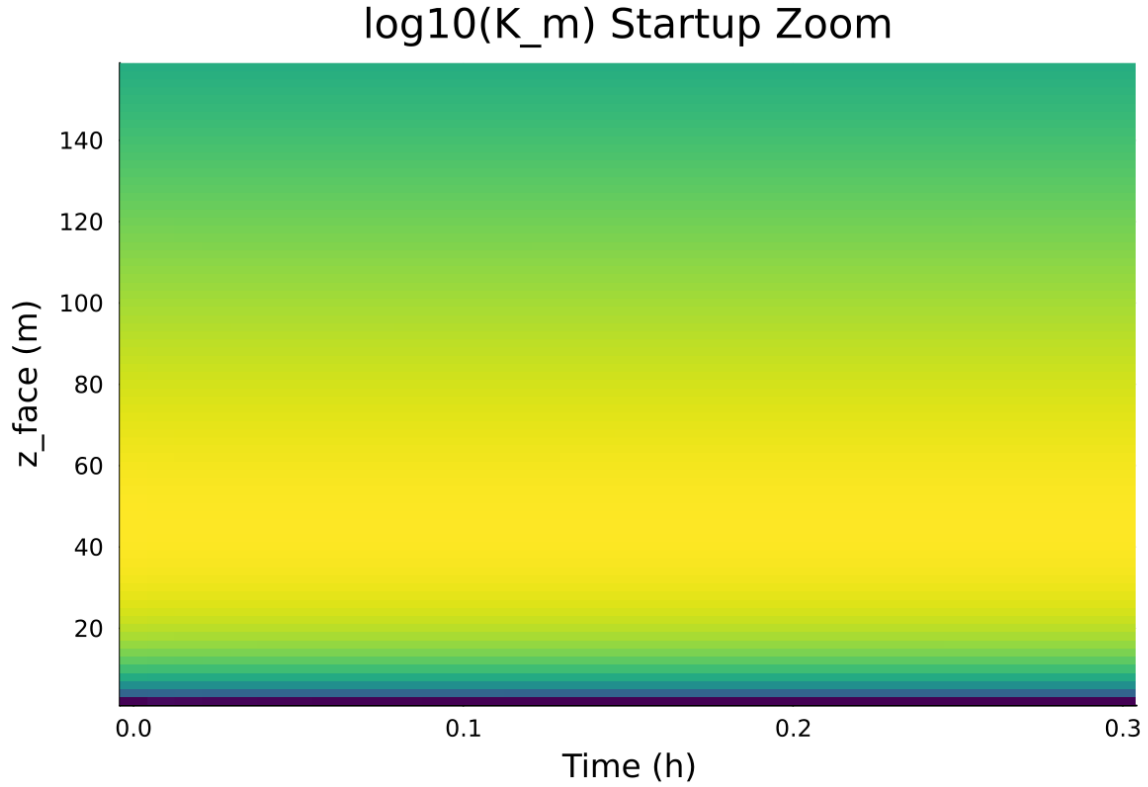


Figure 31: Surface energy budget components (R_n , H , G , storage)

Table 4: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	$8.000000 \text{ m s}^{-1}$
Meridional geostrophic wind	V_g	$0.000000 \text{ m s}^{-1}$
Surface roughness (momentum)	z_{0m}	0.100000 m
Surface roughness (heat)	z_{0h}	0.010000 m
Coriolis parameter	f	$1.000\text{e-}04 \text{ s}^{-1}$
Reference potential temperature	θ_a	265.000000 K
Deep soil temperature	T_{deep}	262.000000 K
Downward longwave forcing	$R_{\downarrow}$	$245.000000 \text{ W m}^{-2}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{min,surf}$	$0.001000 \text{ m}^2 \text{ s}^{-1}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 s^{-1}

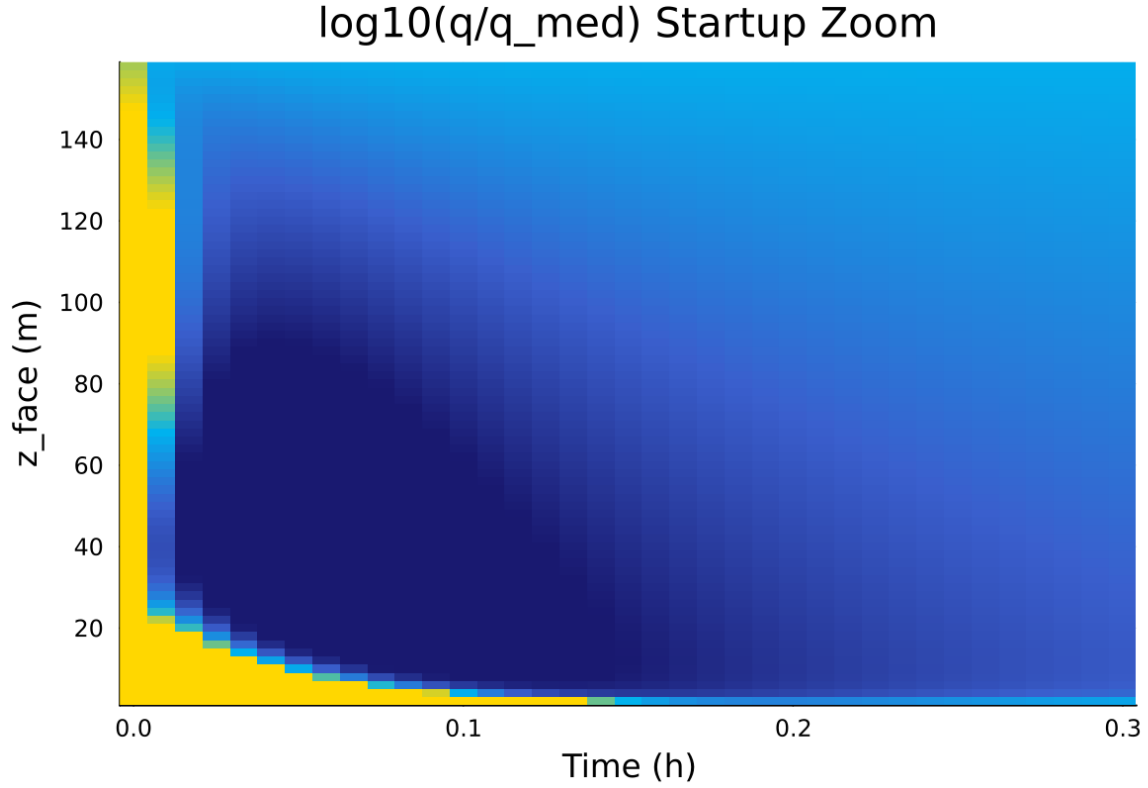


Figure 32: Phase portrait on regularized manifold (Delta vs e_{xi})

9.2 Forcing and Boundary Conditions

9.3 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [3.844877, 78.288180] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [6.468 \times 10^{-9}, 843.105654]$

9.4 Generated Artifacts

- Time-series CSV: `results/gabls1/time_series.csv`
- Diagnostic summary JSON: `results/gabls1/summary.json`
- Payload JLD2: `results/gabls1/payload.jld2`

9.5 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times

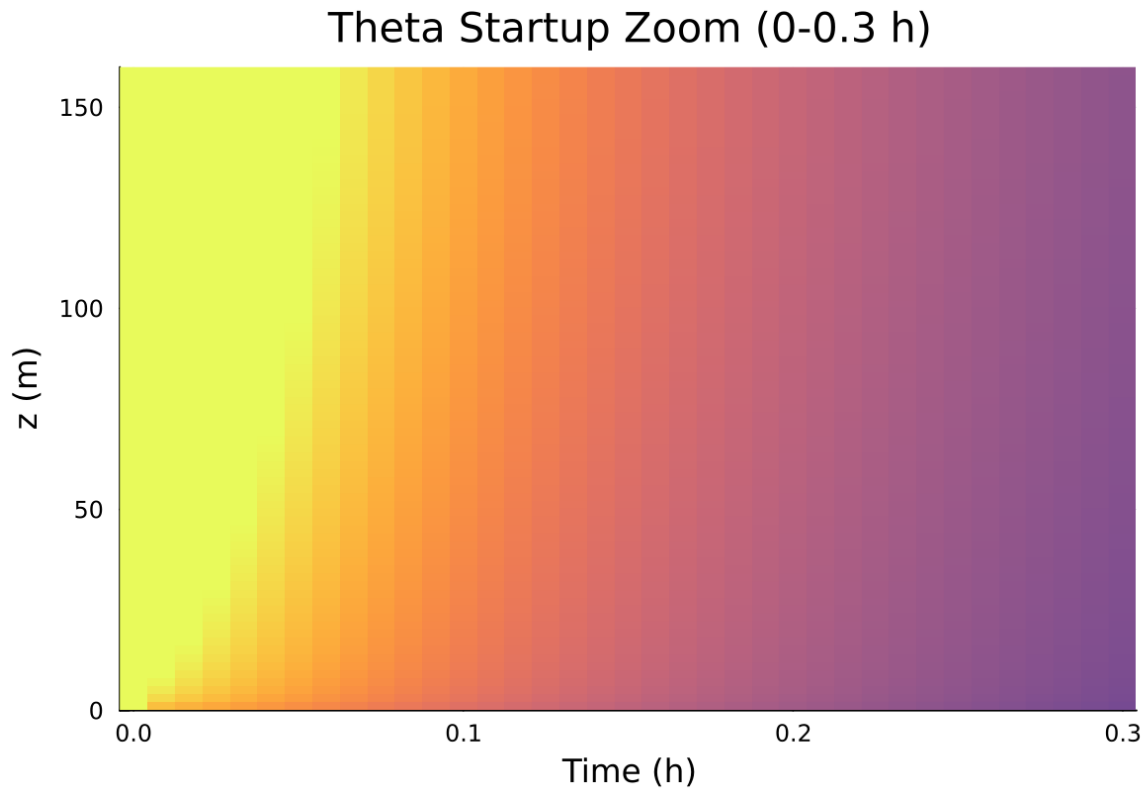


Figure 33: Eddy diffusivity response (K_m , K_h) versus stability/Richardson number

5. Surface energy budget components (R_n , H , G , storage)
6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Eddy diffusivity response (K_m , K_h) versus stability/Richardson number
8. Fold proximity diagnostics versus time

9.6 Included Plots

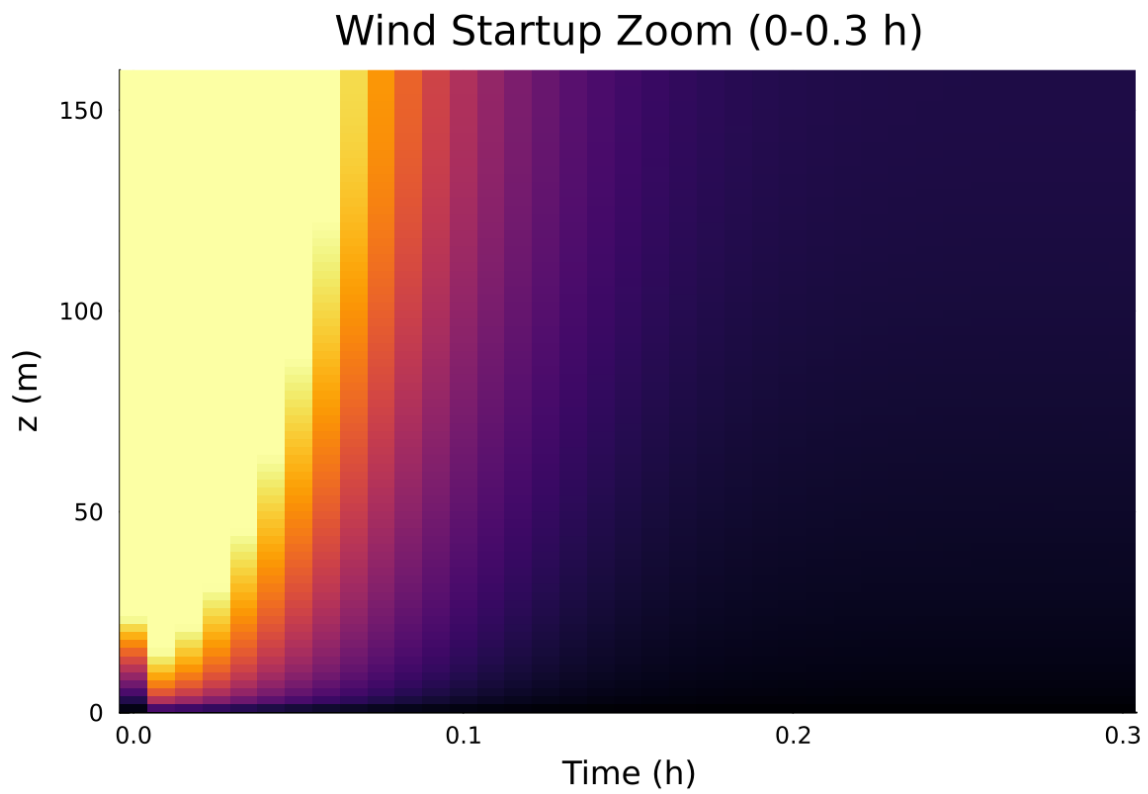


Figure 34: Fold proximity diagnostics versus time

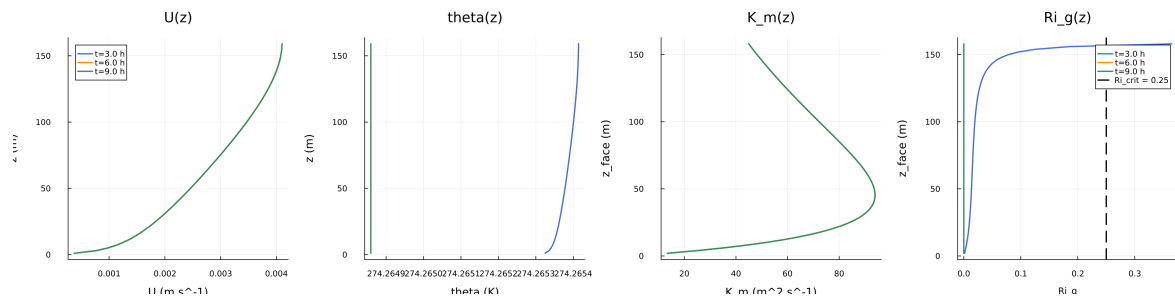


Figure 35: fig04 profiles u theta km.png

Figure 5: Surface Energy Budget

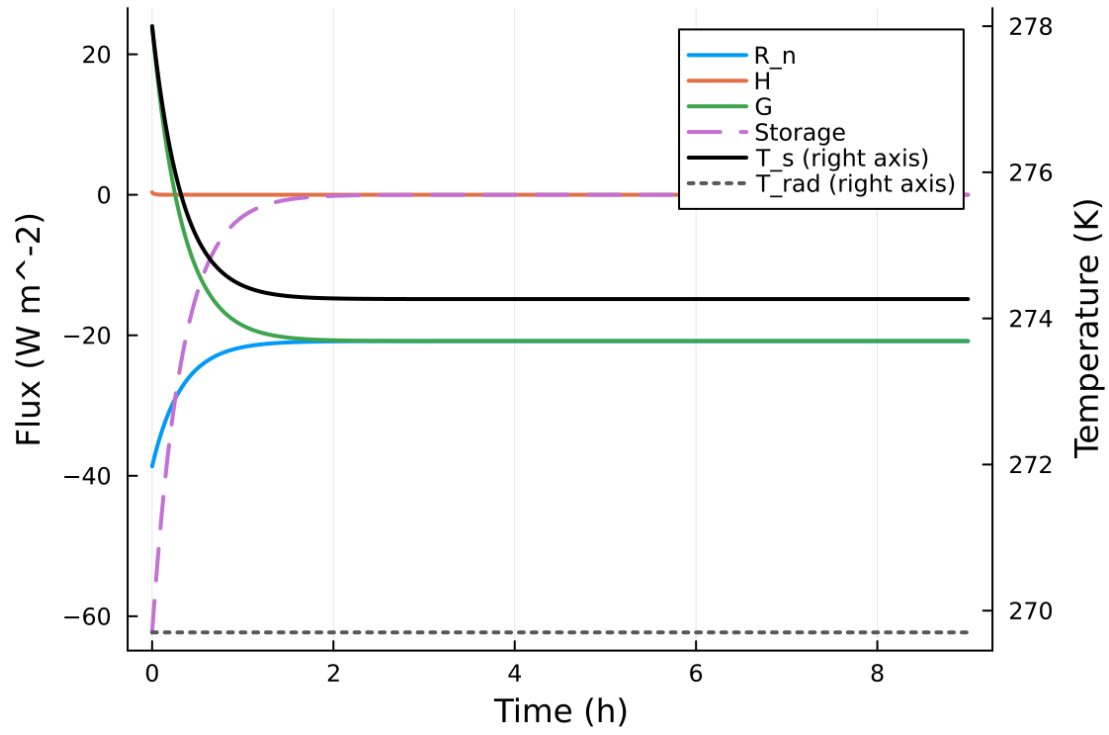


Figure 36: fig05 surface energy budget.png

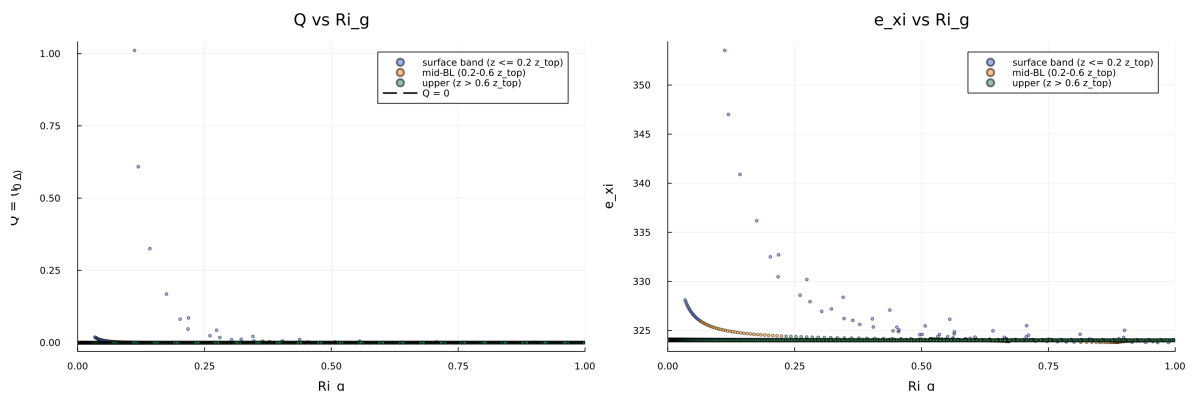


Figure 37: fig06 phase delta exi.png

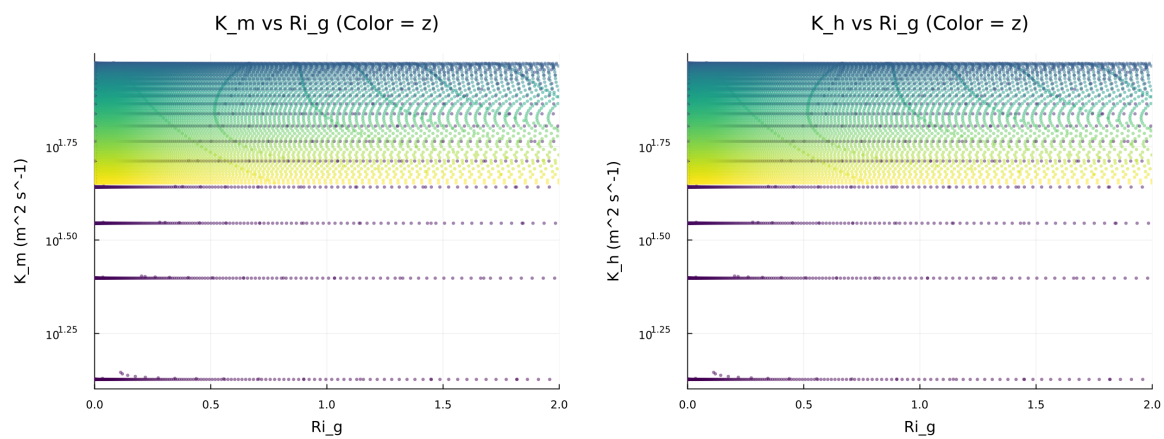


Figure 38: fig07 diffusivity vs ri.png

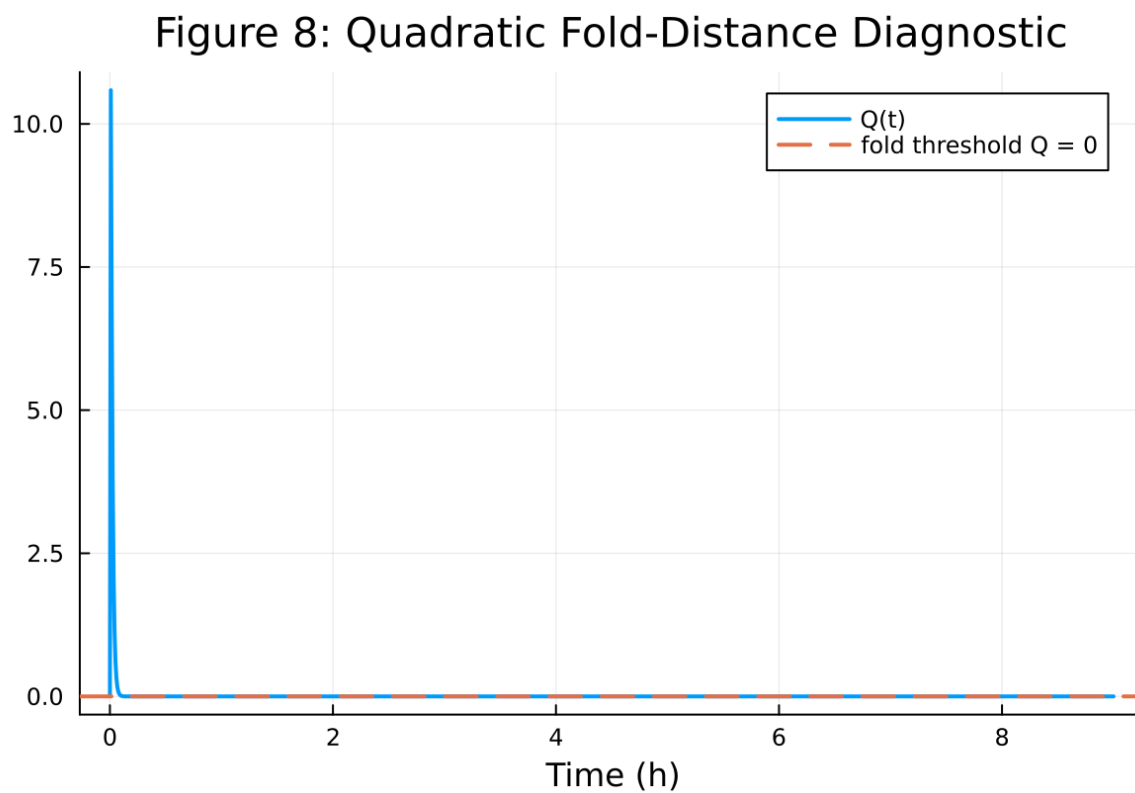


Figure 39: fig08 fold proximity.png

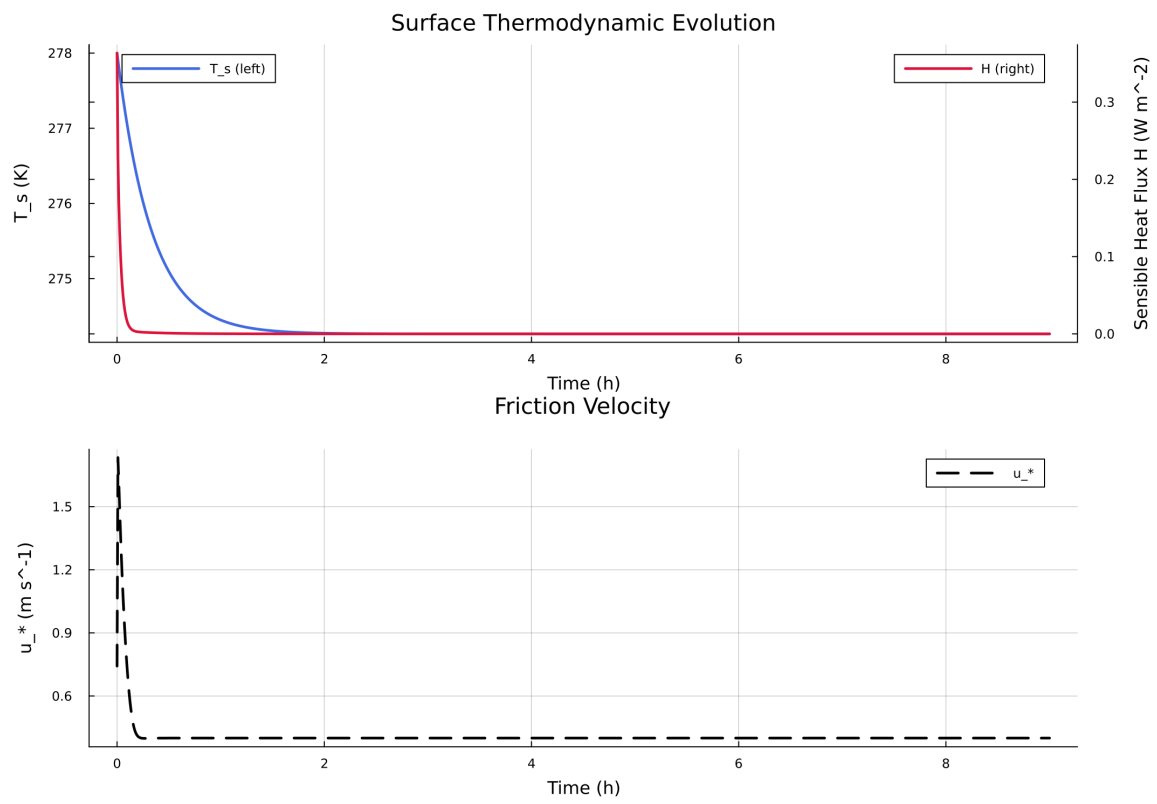


Figure 40: Time series of T_s , sensible heat flux, and friction velocity

Figure 2: Time-Height Wind Speed

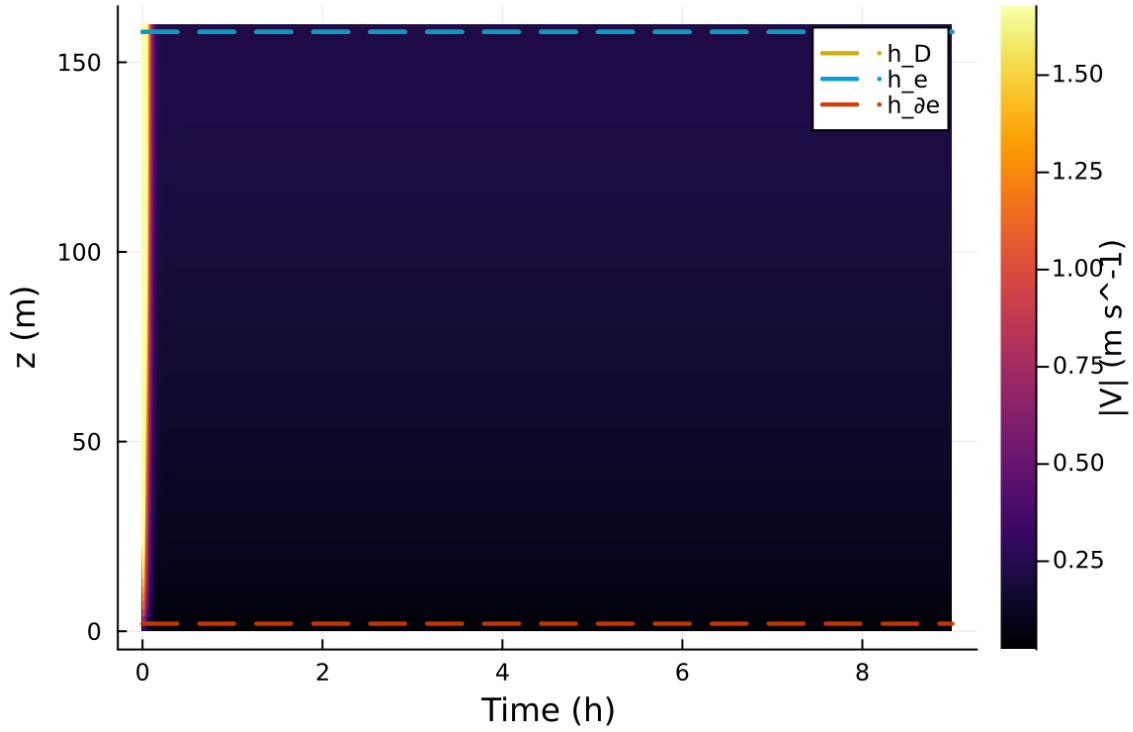


Figure 41: Time-height contour of wind speed with LLJ evolution

10 Case 5: gabls1 80x2m 9h report

11 SCM Case Summary: gabls1

This section was generated automatically from the SCM diagnostics artifact `results/gabls1/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`.

11.1 Core Run Metadata

- Case identifier: `gabls1`
- Output directory: `results/gabls1`
- Number of saved time samples: 1081
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 102 / 0
- RHS evaluations: 25502
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

Figure 3: Time-Height Potential Temperature

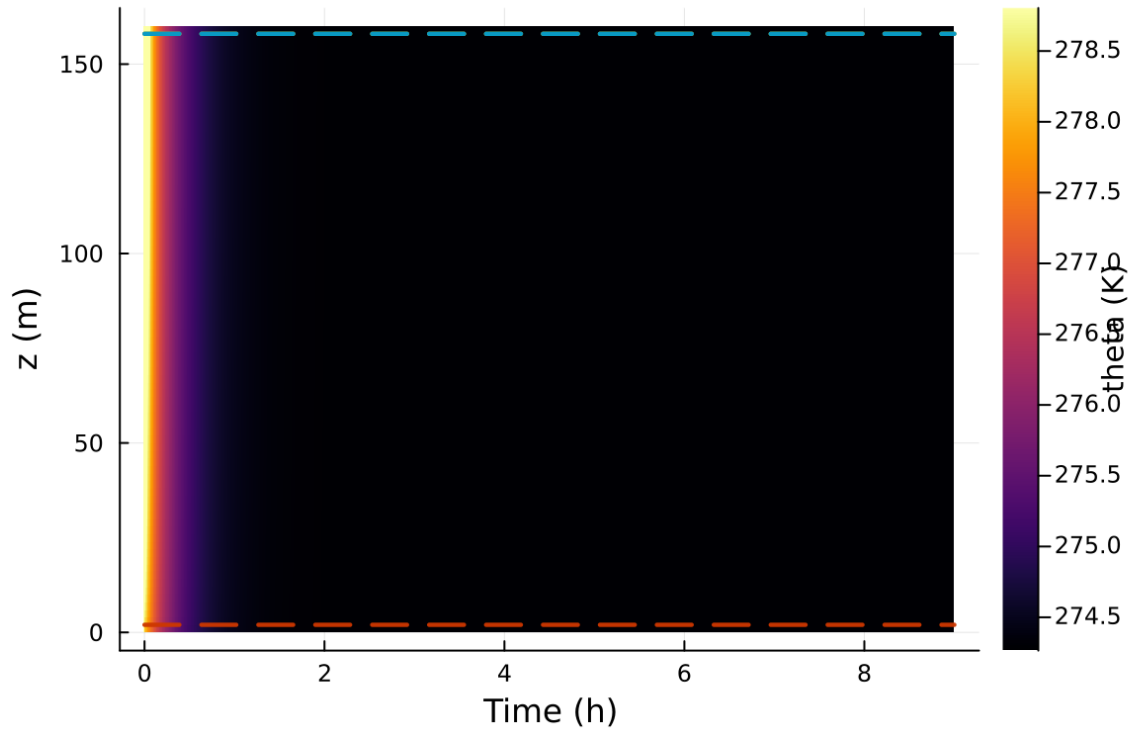


Figure 42: Time-height contour of potential temperature and inversion growth

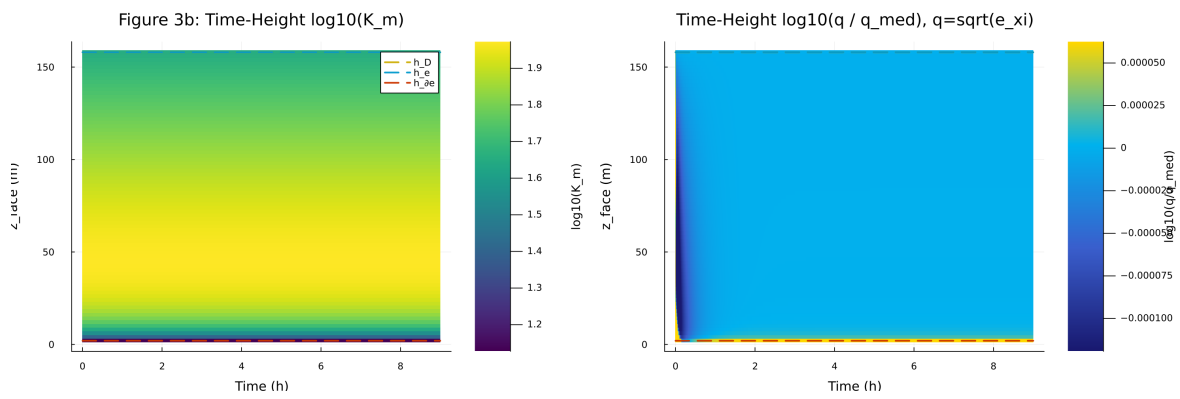


Figure 43: Vertical profiles of U, theta, and Km at representative times

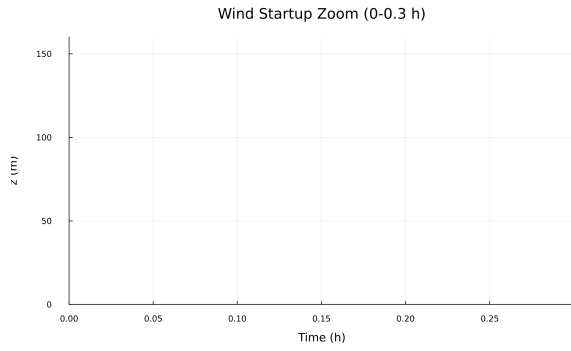


Figure 44: Surface energy budget components (R_n , H , G , storage)

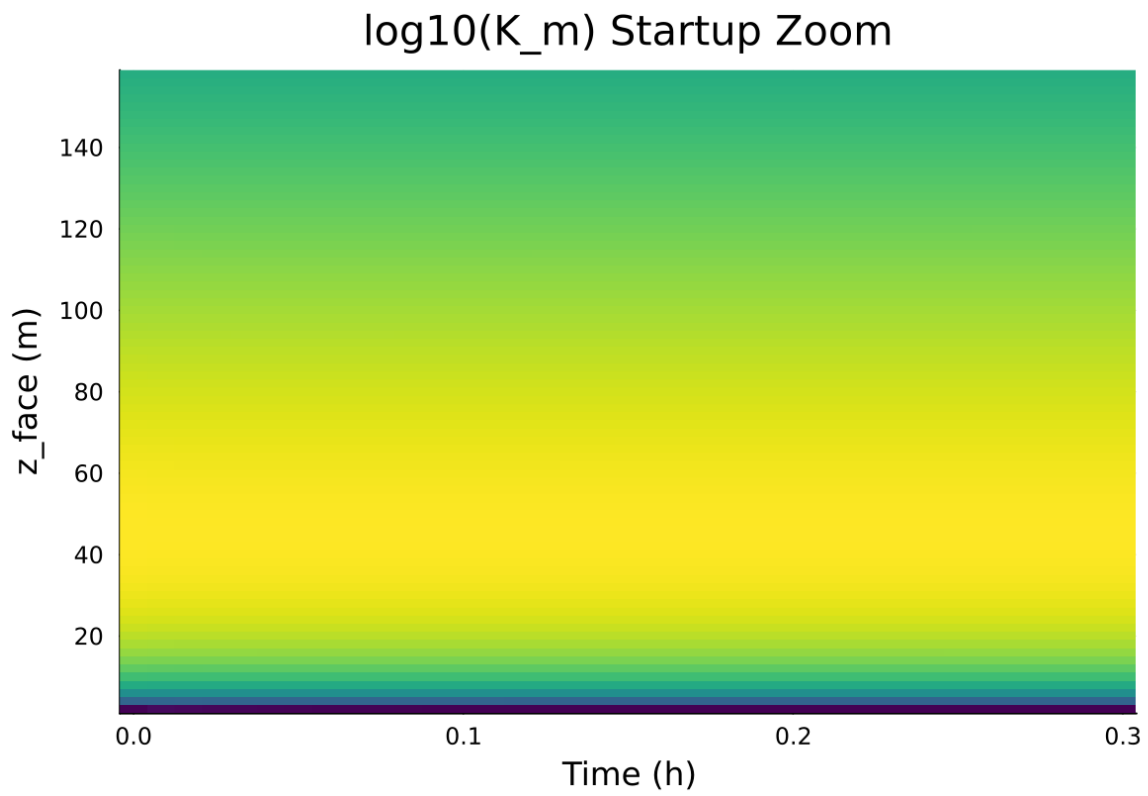


Figure 45: Phase portrait on regularized manifold (Δ vs e_{xi})

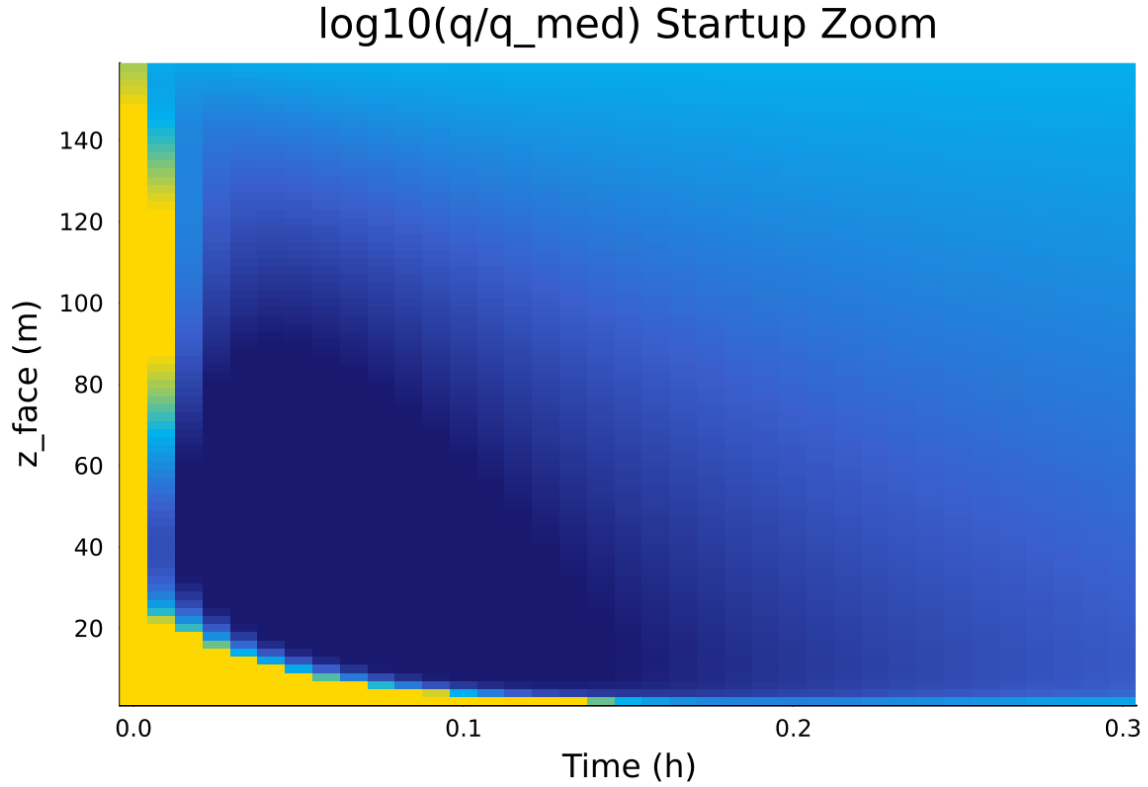


Figure 46: Eddy diffusivity response (K_m , K_h) versus stability/Richardson number

Table 5: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	$8.000000 \text{ m s}^{-1}$
Meridional geostrophic wind	V_g	$0.000000 \text{ m s}^{-1}$
Surface roughness (momentum)	z_{0m}	0.100000 m
Surface roughness (heat)	z_{0h}	0.010000 m
Coriolis parameter	f	$1.000\text{e-}04 \text{ s}^{-1}$
Reference potential temperature	θ_a	265.000000 K
Deep soil temperature	T_{deep}	262.000000 K
Downward longwave forcing	$R_{\downarrow}$	$245.000000 \text{ W m}^{-2}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{min,surf}$	$0.001000 \text{ m}^2 \text{ s}^{-1}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 s^{-1}

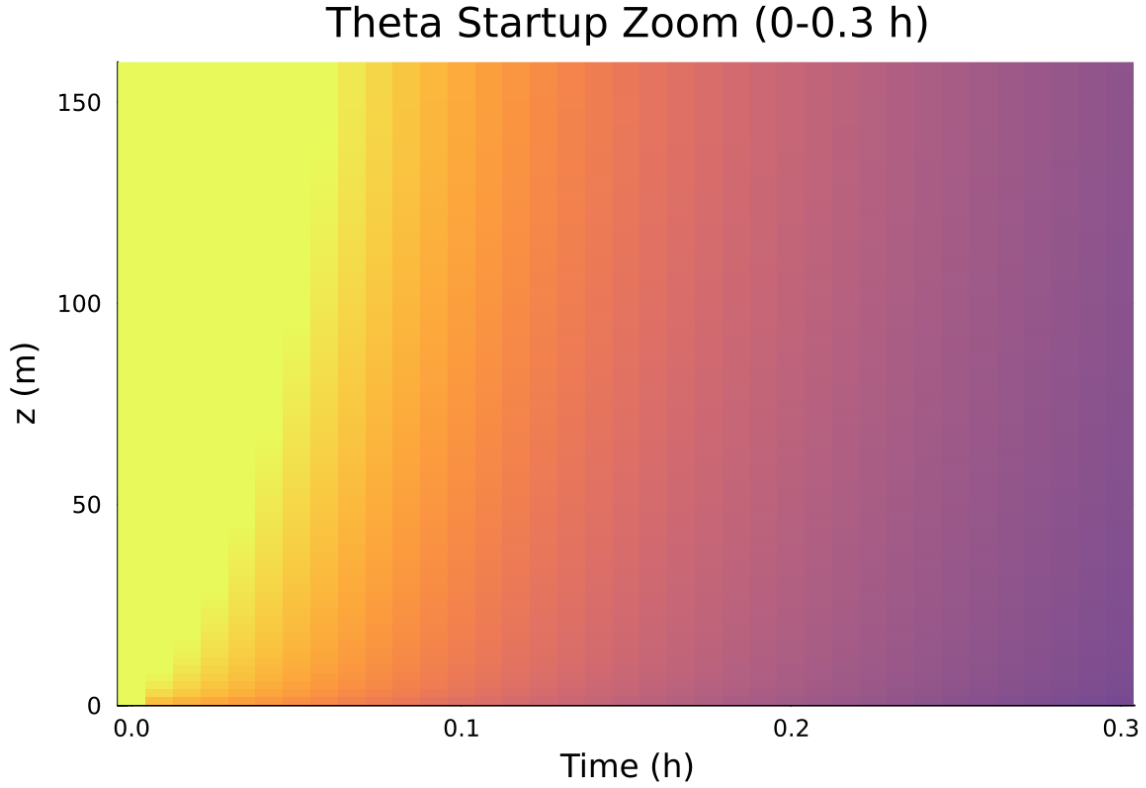


Figure 47: Fold proximity diagnostics versus time

11.2 Forcing and Boundary Conditions

11.3 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 94.819611%
- Diffusivity bounds: $K_m \in [11.165526, 78.288180] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [-1.154 \times 10^{-6}, 1887.093112]$

11.4 Generated Artifacts

- Time-series CSV: `results/gabls1/time_series.csv`
- Diagnostic summary JSON: `results/gabls1/summary.json`
- Payload JLD2: `results/gabls1/payload.jld2`

11.5 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times

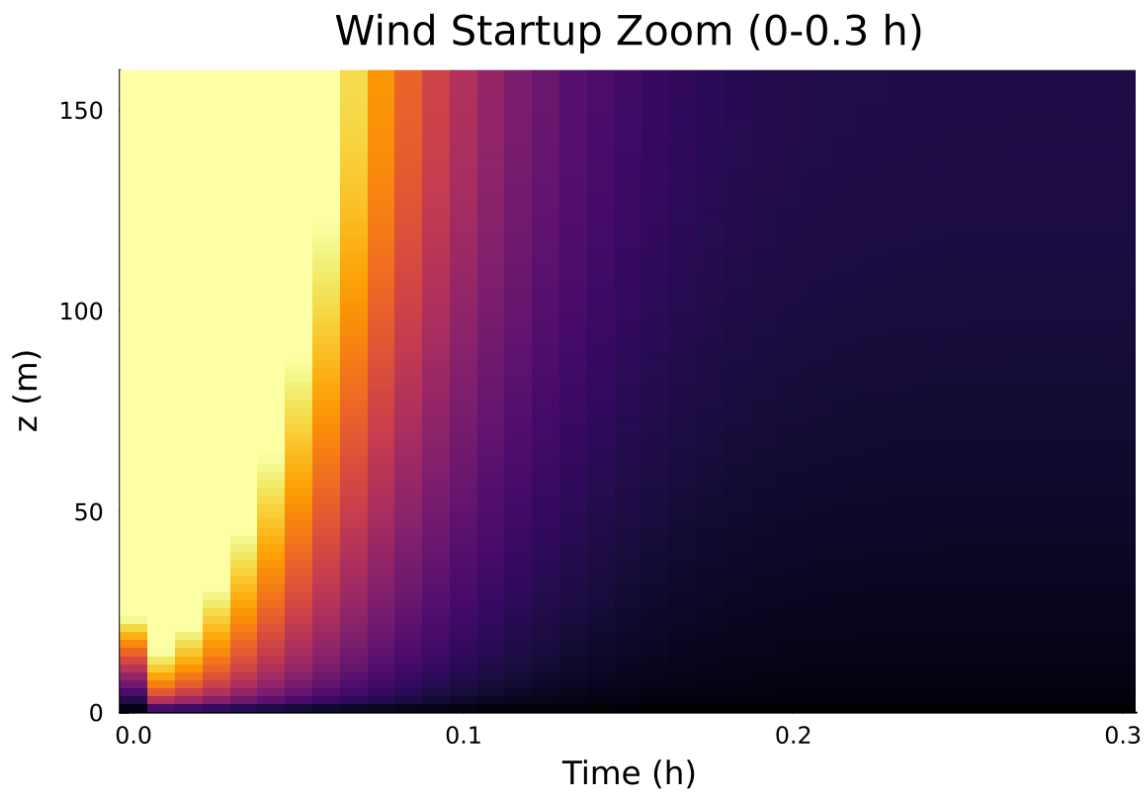


Figure 48: fig03c wind startup zoom.png

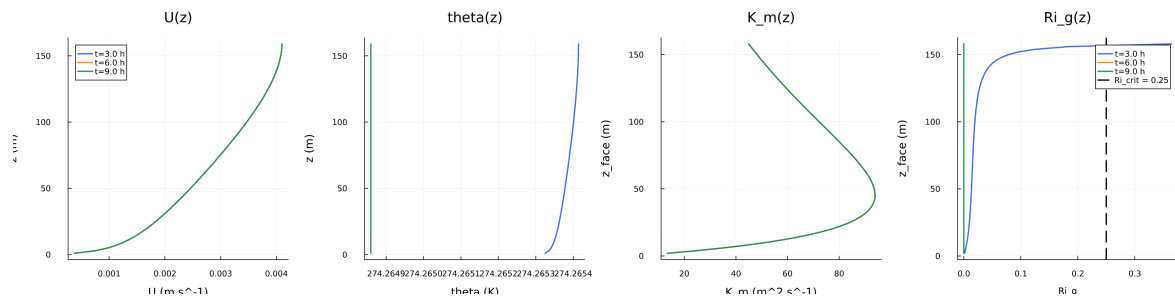


Figure 49: fig04 profiles u theta km.png

Figure 5: Surface Energy Budget

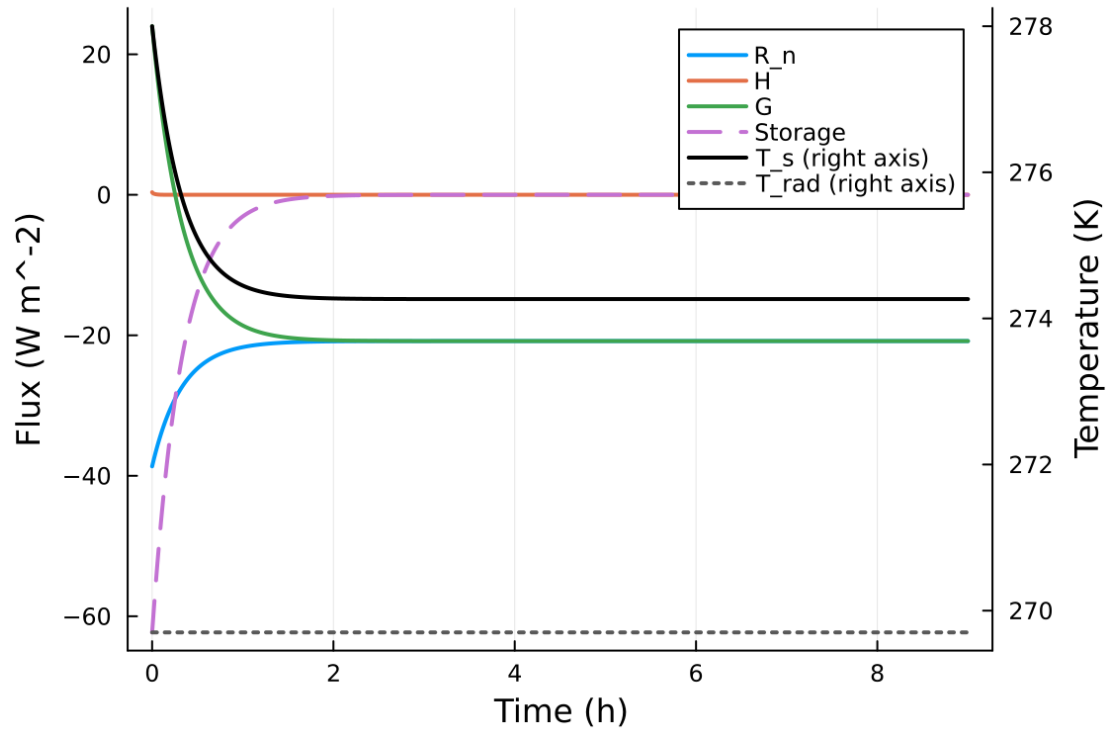


Figure 50: fig05 surface energy budget.png

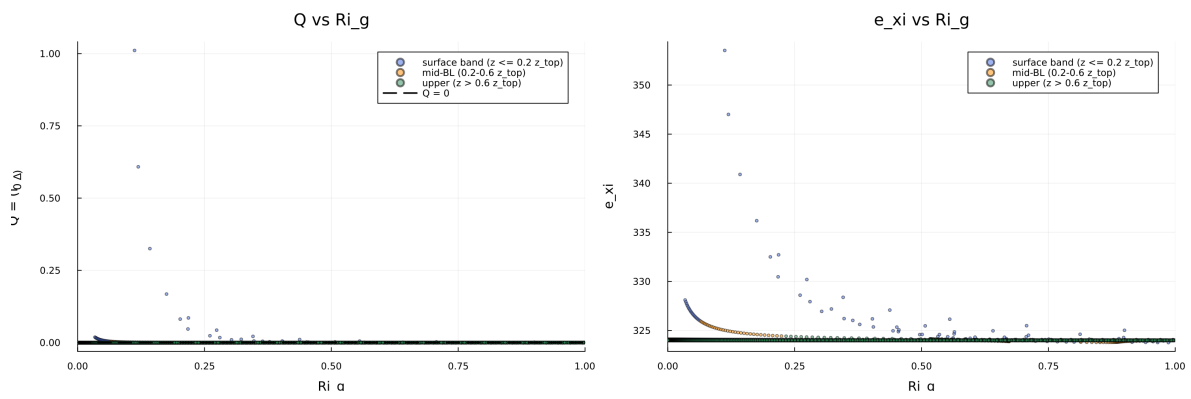


Figure 51: fig06 phase delta exi.png

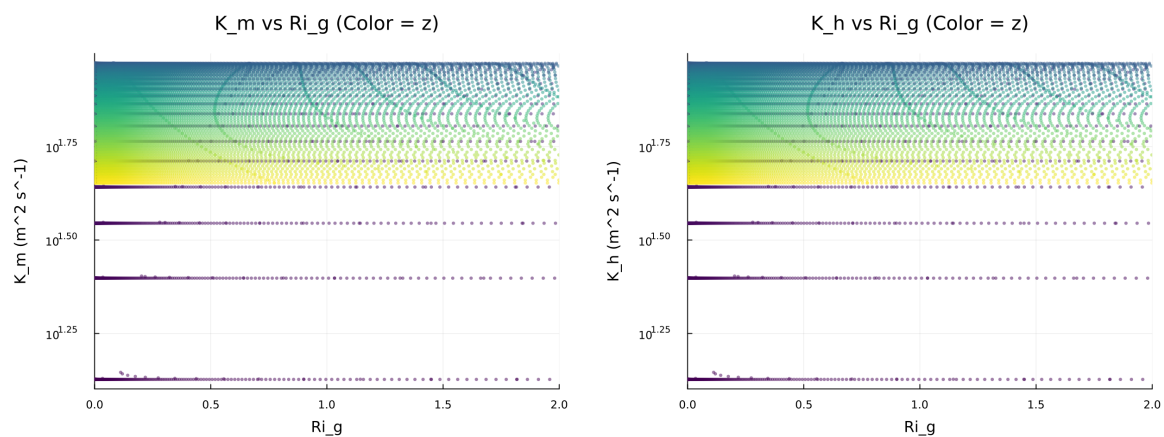


Figure 52: fig07 diffusivity vs ri.png

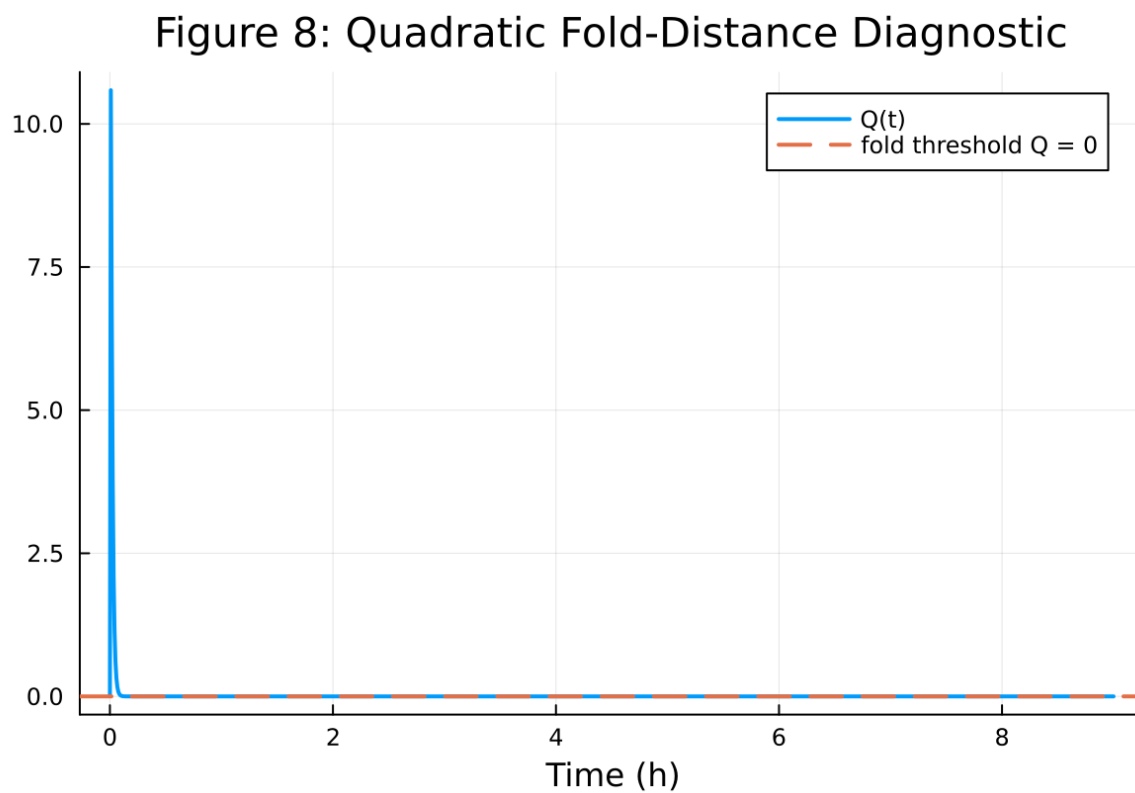


Figure 53: fig08 fold proximity.png

5. Surface energy budget components (R_n , H , G , storage)
6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Eddy diffusivity response (K_m , K_h) versus stability/Richardson number
8. Fold proximity diagnostics versus time

11.6 Included Plots

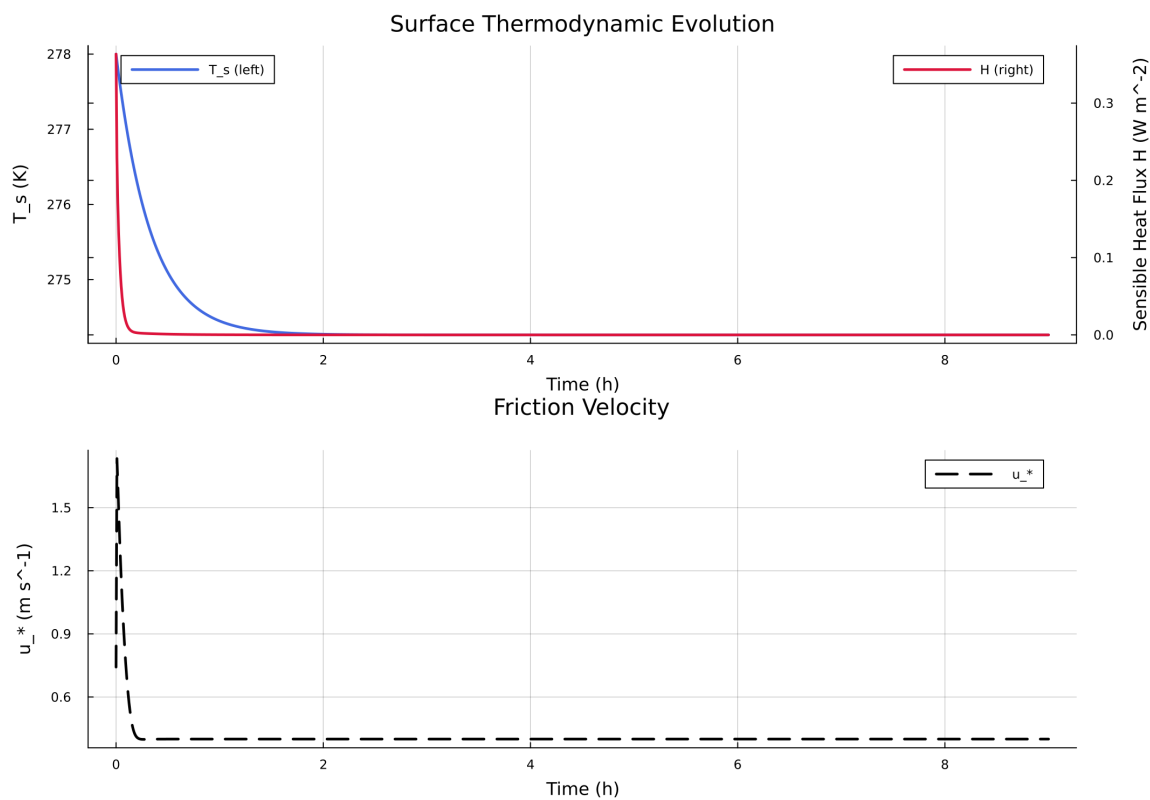


Figure 54: Time series of T_s , sensible heat flux, and friction velocity

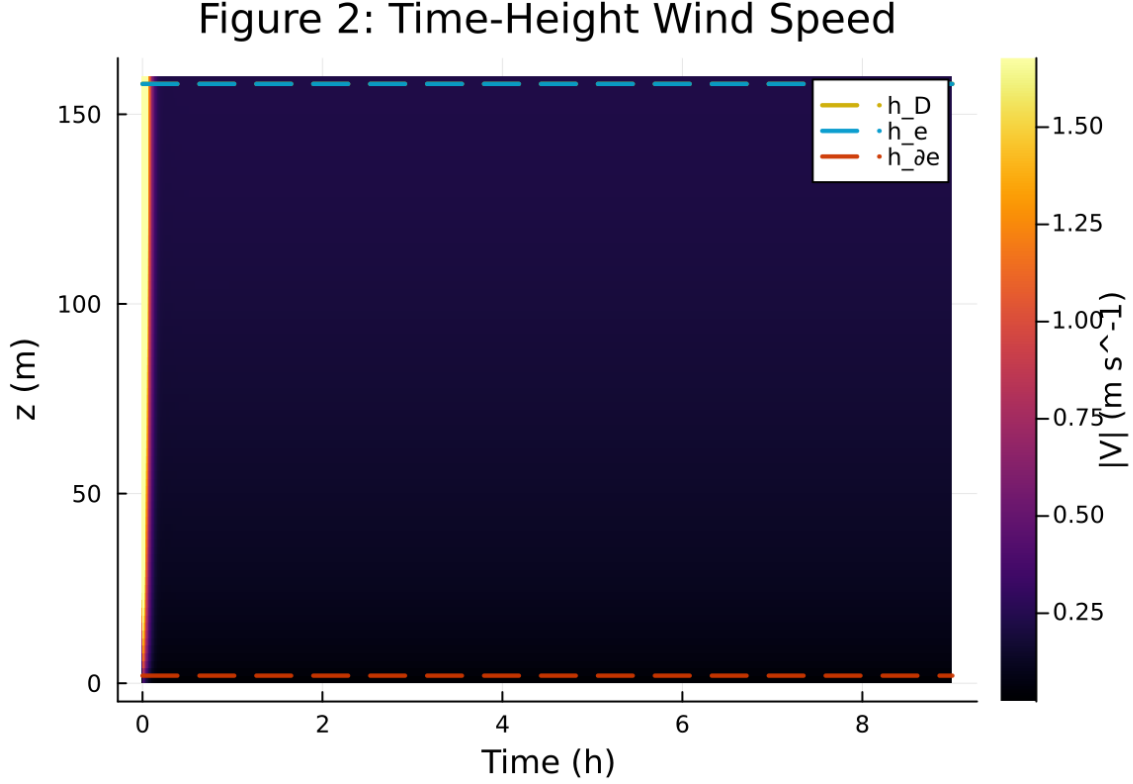


Figure 55: Time-height contour of wind speed with LLJ evolution

12 Case 6: idealized_sbl 80x2m 9h report

13 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/idealized_sbl/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`.

13.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/idealized_sbl`
- Number of saved time samples: 1081
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 108 / 0
- RHS evaluations: 27002
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

Figure 3: Time-Height Potential Temperature

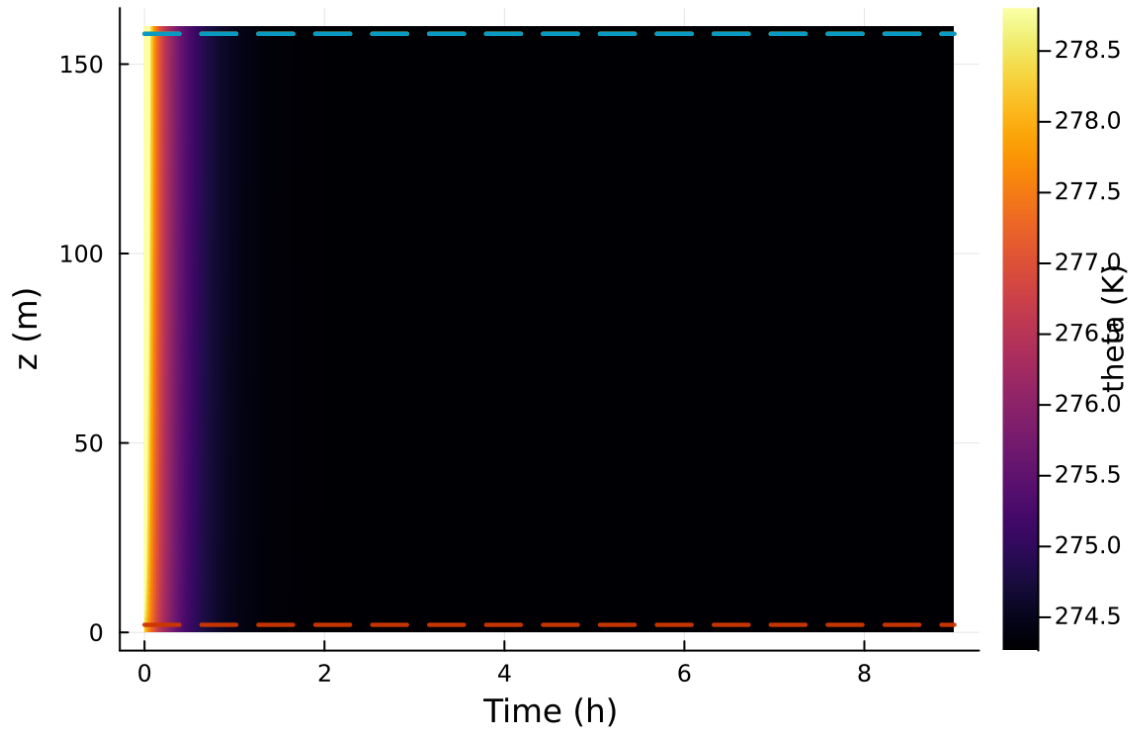


Figure 56: Time-height contour of potential temperature and inversion growth

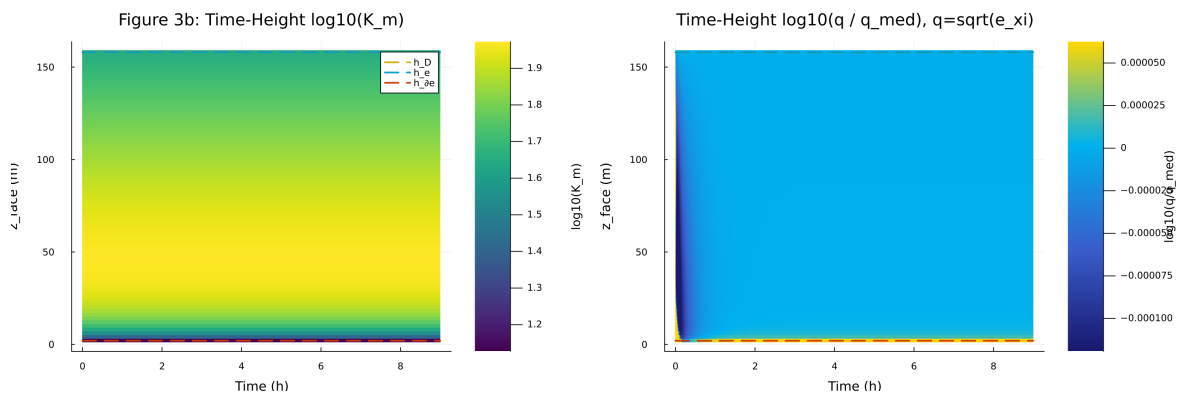


Figure 57: Vertical profiles of U , θ , and K_m at representative times

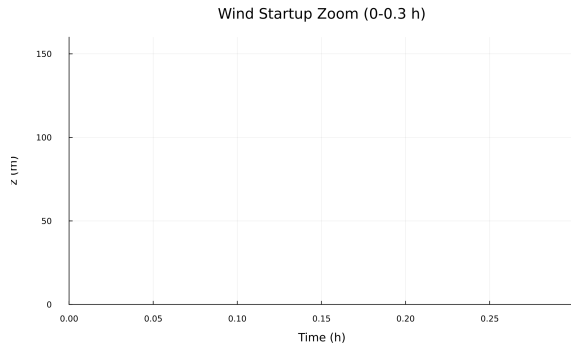


Figure 58: Surface energy budget components (R_n , H , G , storage)

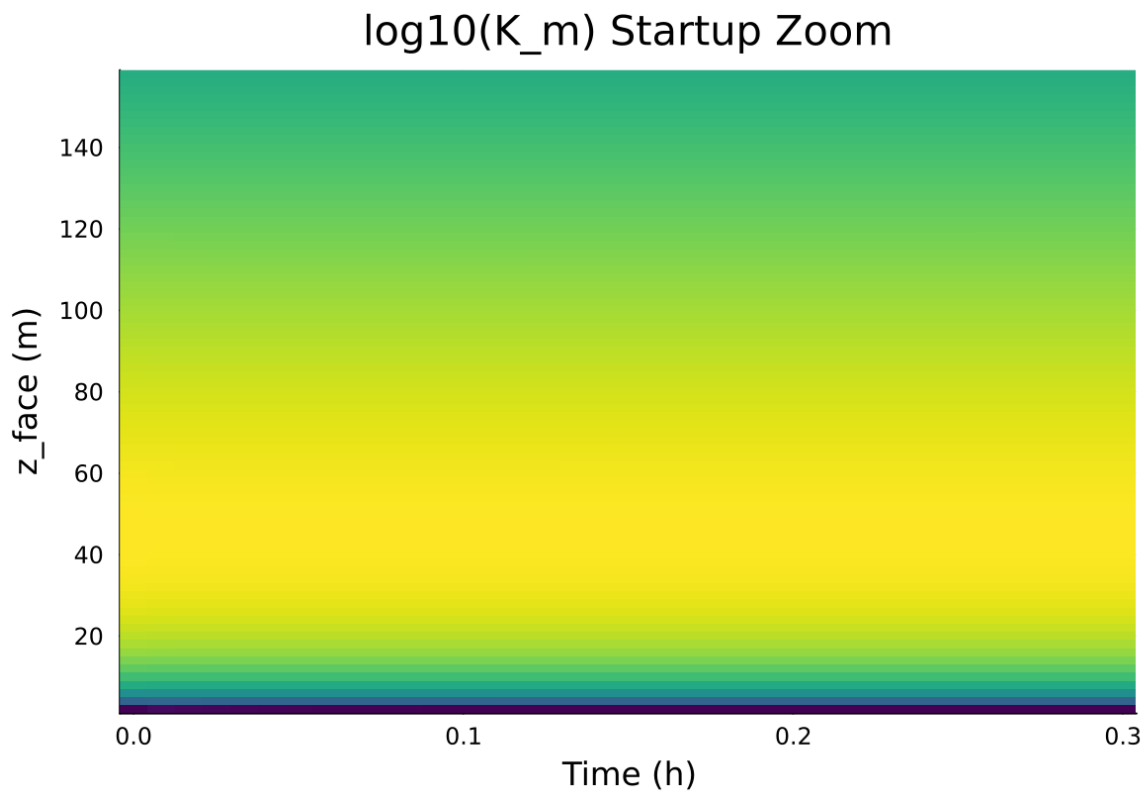


Figure 59: Phase portrait on regularized manifold (Δ vs e_{xi})

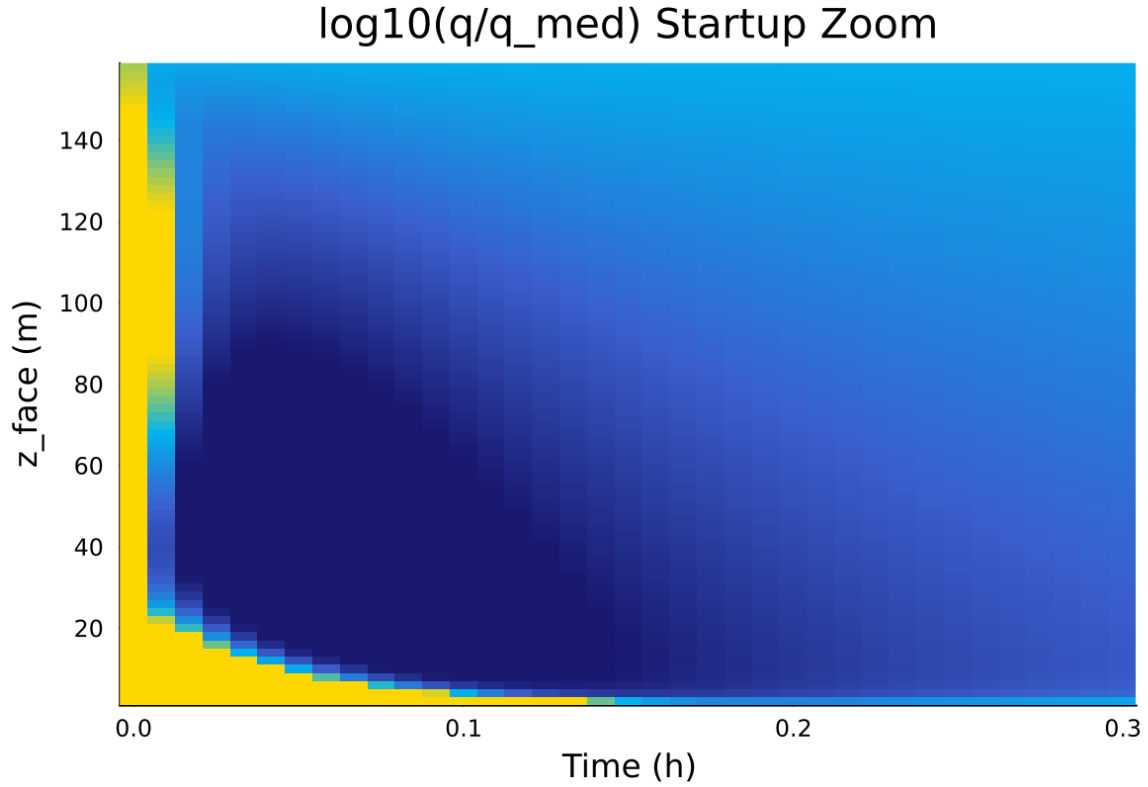


Figure 60: Eddy diffusivity response (K_m , K_h) versus stability/Richardson number

Table 6: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	$10.000000 \text{ m s}^{-1}$
Meridional geostrophic wind	V_g	$0.000000 \text{ m s}^{-1}$
Surface roughness (momentum)	z_{0m}	0.020000 m
Surface roughness (heat)	z_{0h}	0.005000 m
Coriolis parameter	f	$1.000\text{e-}04 \text{ s}^{-1}$
Reference potential temperature	θ_a	280.000000 K
Deep soil temperature	T_{deep}	276.000000 K
Downward longwave forcing	$R_{\downarrow}$	$300.000000 \text{ W m}^{-2}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{min,surf}$	$0.001000 \text{ m}^2 \text{ s}^{-1}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 s^{-1}

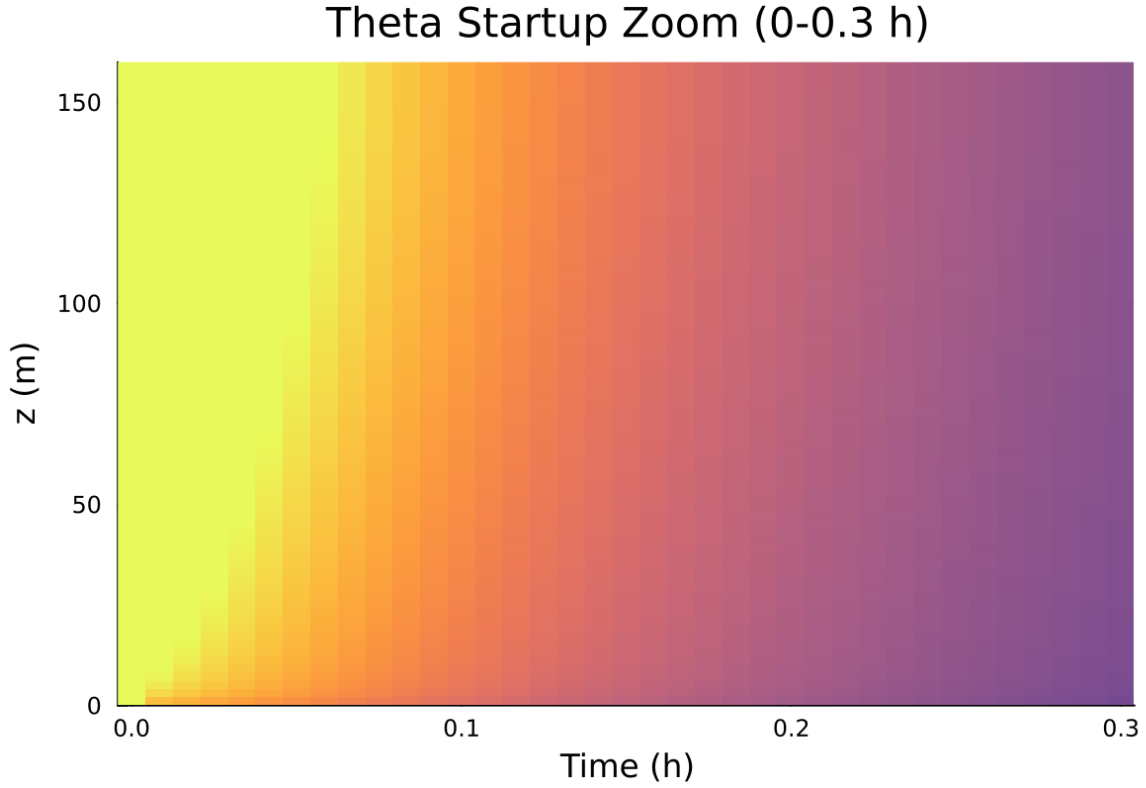


Figure 61: Fold proximity diagnostics versus time

13.2 Forcing and Boundary Conditions

13.3 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 7.678076%
- Diffusivity bounds: $K_m \in [13.399989, 94.108839] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [-4.024 \times 10^{-6}, 1551.592844]$

13.4 Generated Artifacts

- Time-series CSV: `results/idealized_sbl/time_series.csv`
- Diagnostic summary JSON: `results/idealized_sbl/summary.json`
- Payload JLD2: `results/idealized_sbl/payload.jld2`

13.5 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , theta, and K_m at representative times

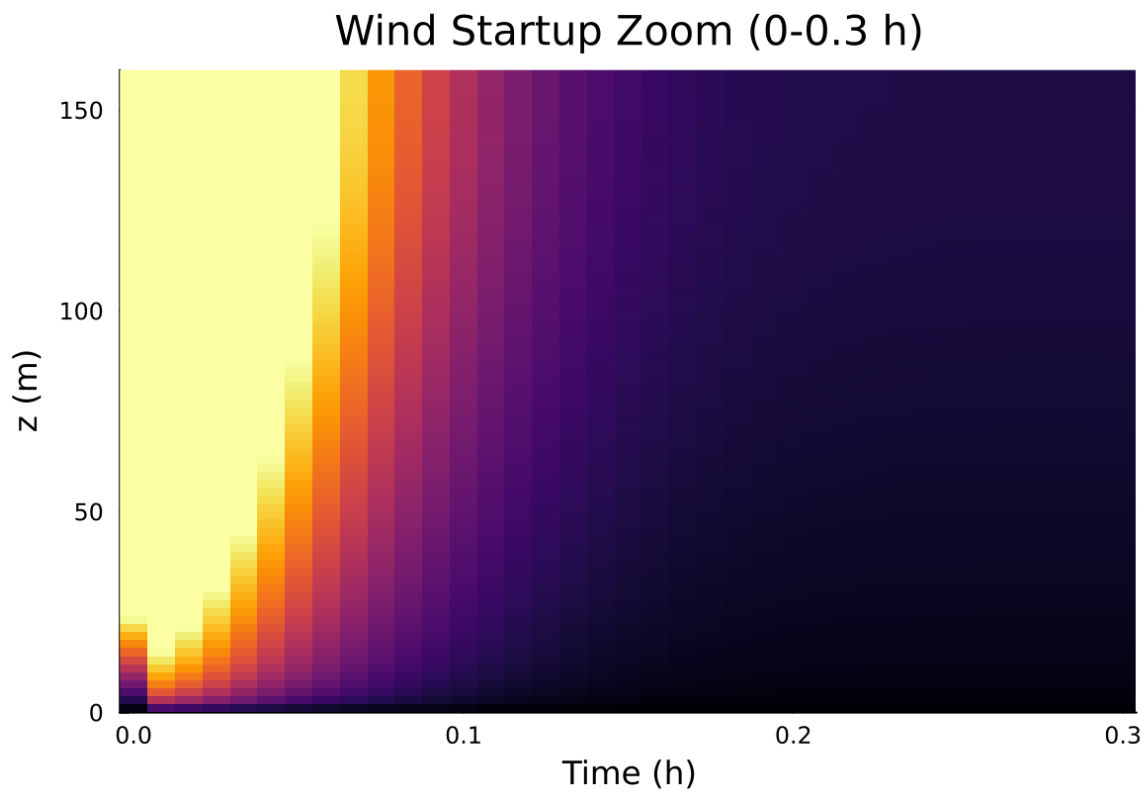


Figure 62: fig03c wind startup zoom.png

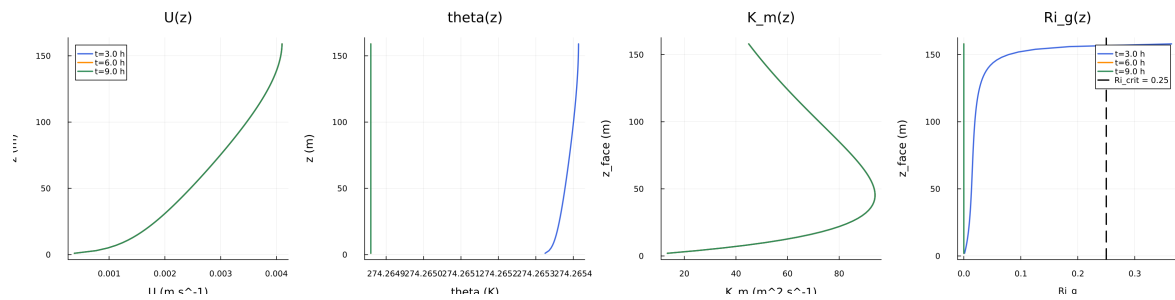


Figure 63: fig04 profiles u theta km.png

Figure 5: Surface Energy Budget

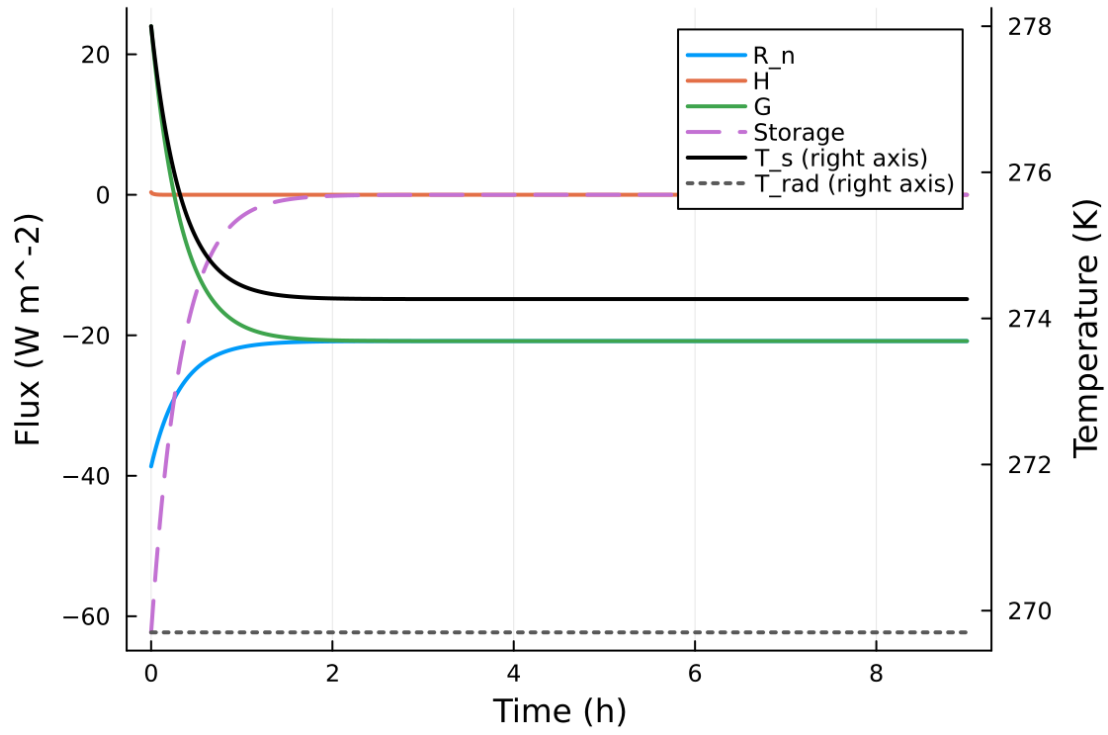


Figure 64: fig05 surface energy budget.png

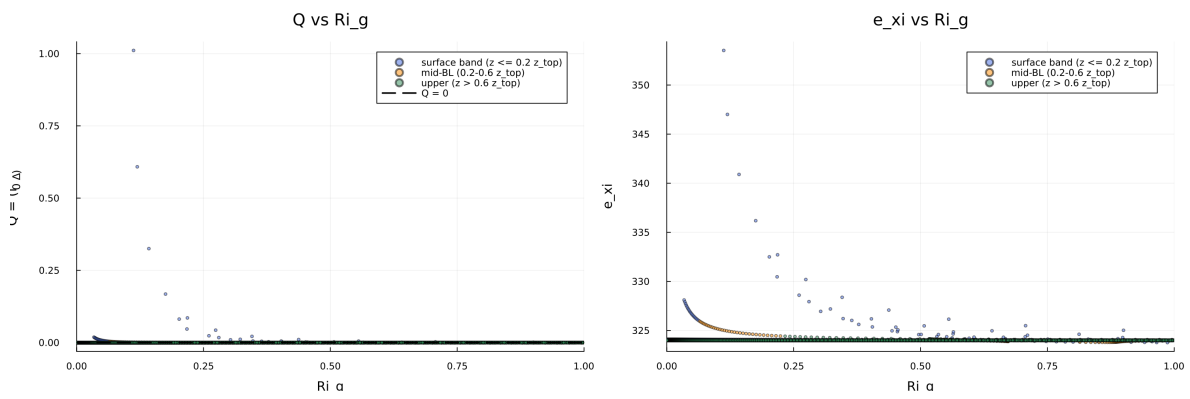


Figure 65: fig06 phase delta exi.png

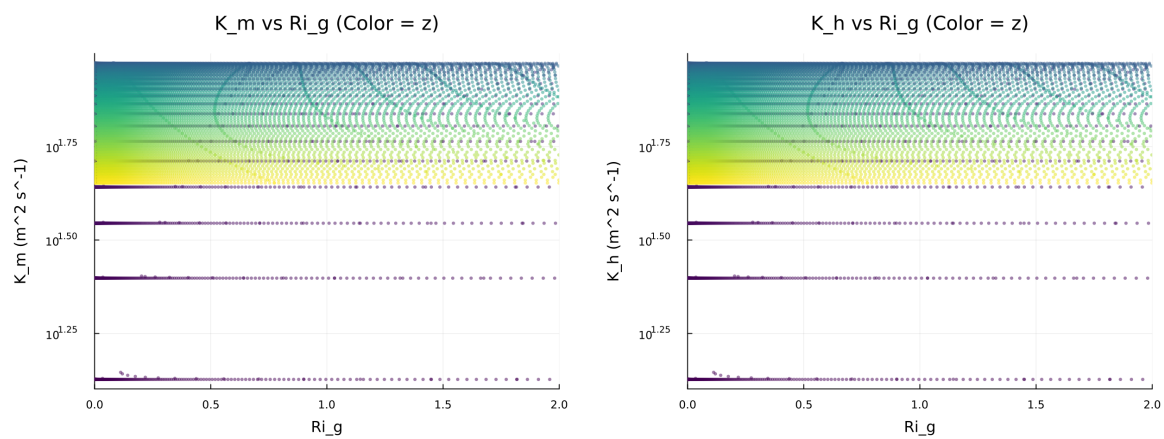


Figure 66: fig07 diffusivity vs ri.png

Figure 8: Quadratic Fold-Distance Diagnostic

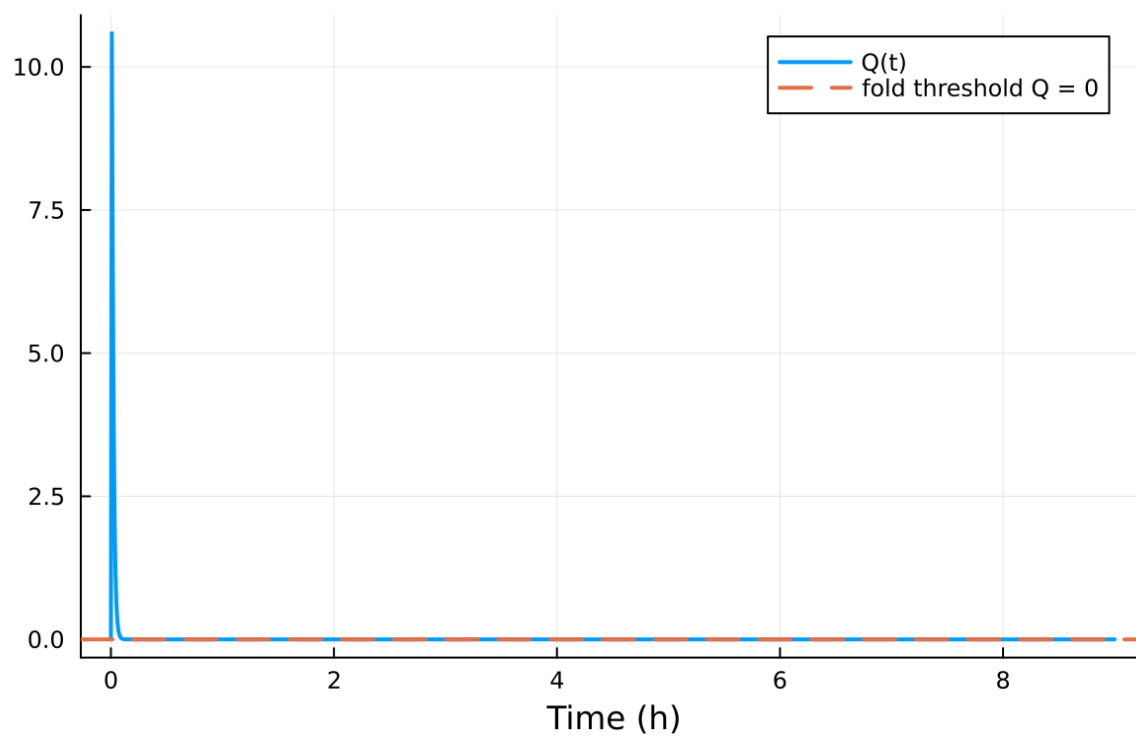


Figure 67: fig08 fold proximity.png

5. Surface energy budget components (R_n , H , G , storage)
6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Eddy diffusivity response (K_m , K_h) versus stability/Richardson number
8. Fold proximity diagnostics versus time

13.6 Included Plots

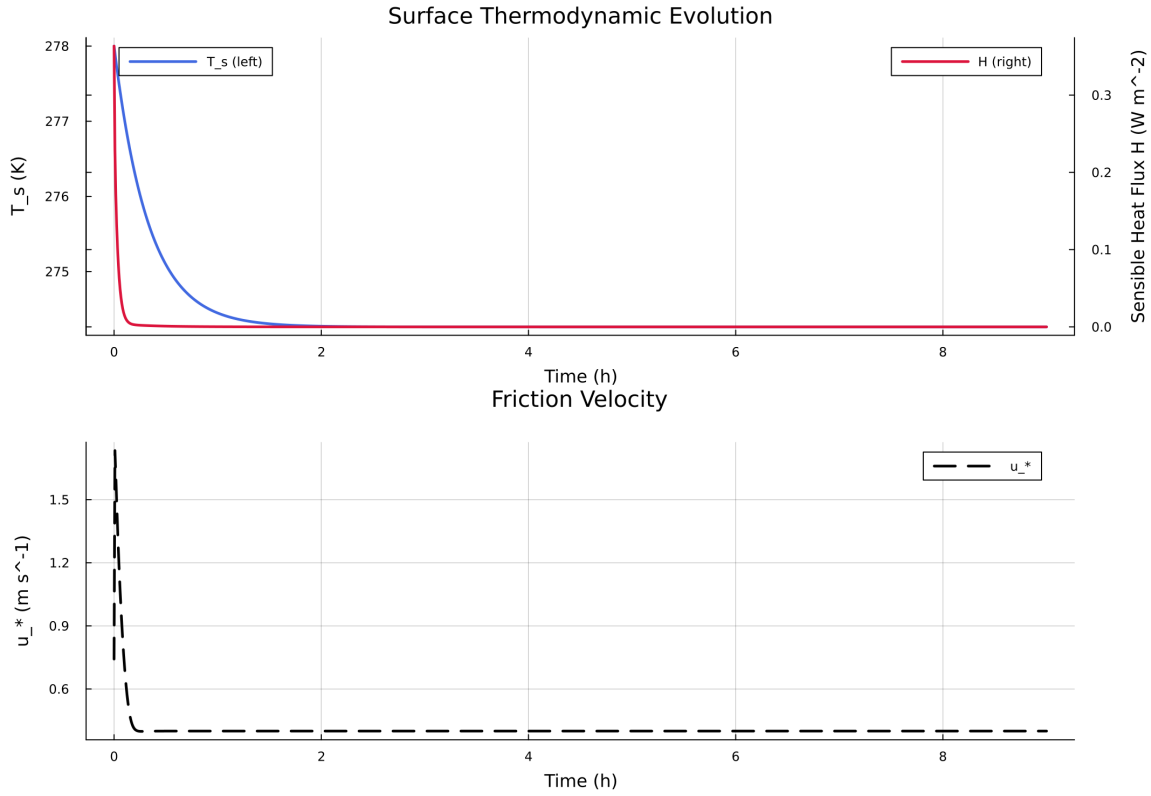


Figure 68: Time series of T_s , sensible heat flux, and friction velocity

Figure 2: Time-Height Wind Speed

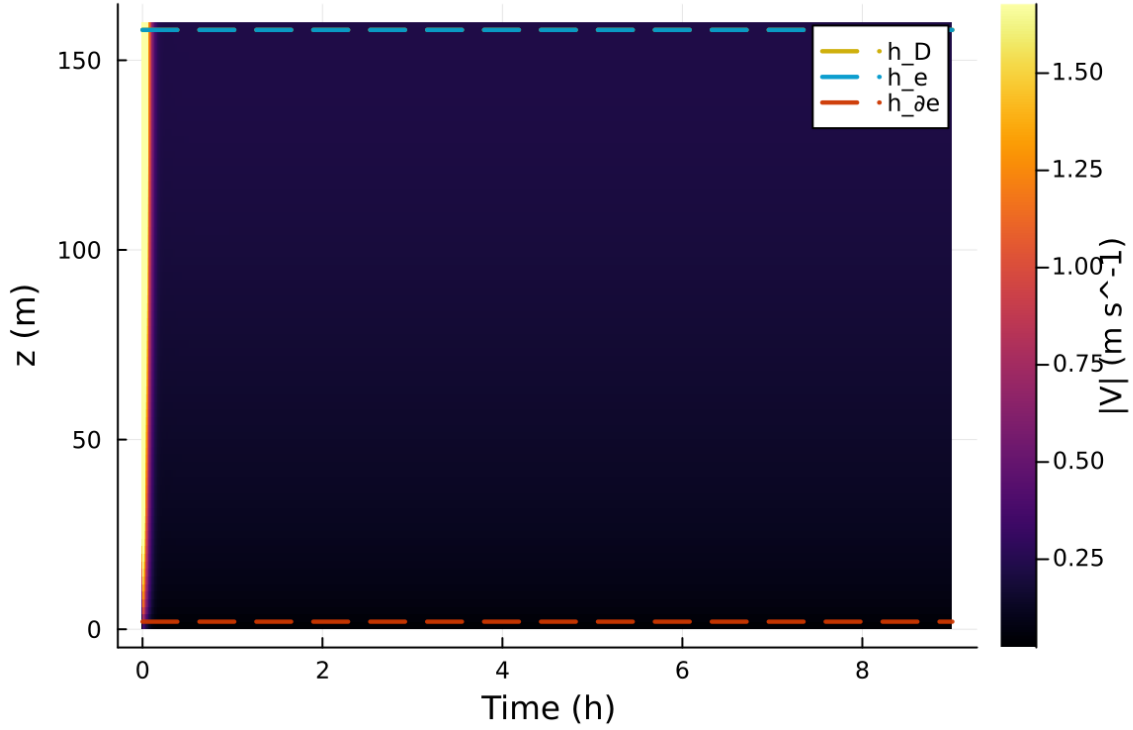


Figure 69: Time-height contour of wind speed with LLJ evolution

14 Case 7: scm verify 40x4m 0h report

15 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/scm_verify/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`.

15.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/scm_verify`
- Number of saved time samples: 13
- Number of profile snapshots: 1
- Solver accepted/rejected steps: 52 / 0
- RHS evaluations: 6762
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

Figure 3: Time-Height Potential Temperature

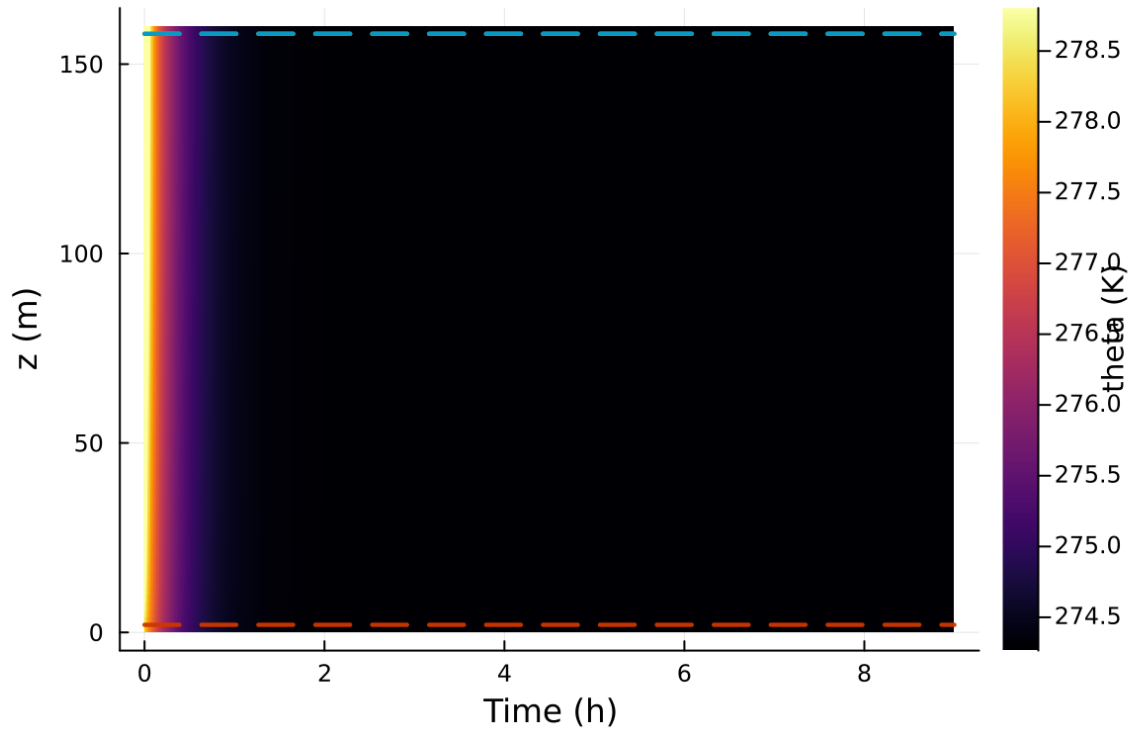


Figure 70: Time-height contour of potential temperature and inversion growth

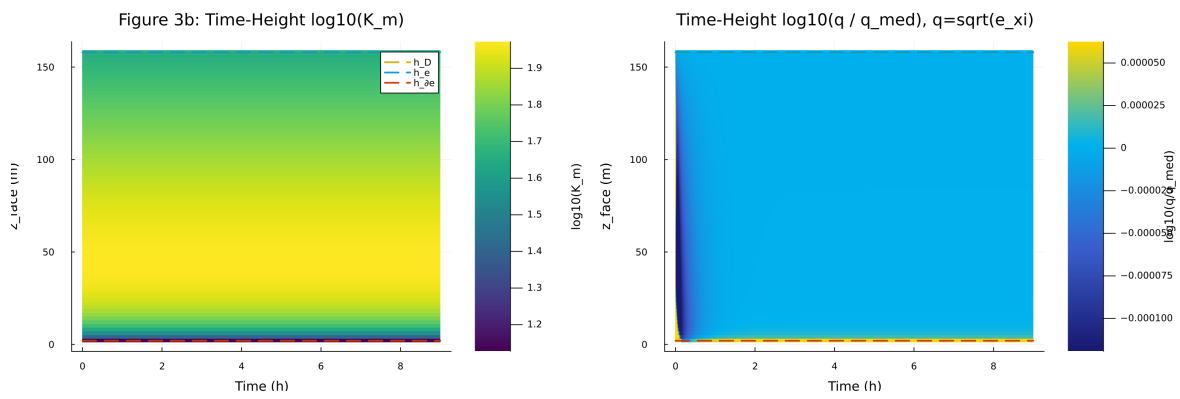


Figure 71: Vertical profiles of U, theta, and Km at representative times

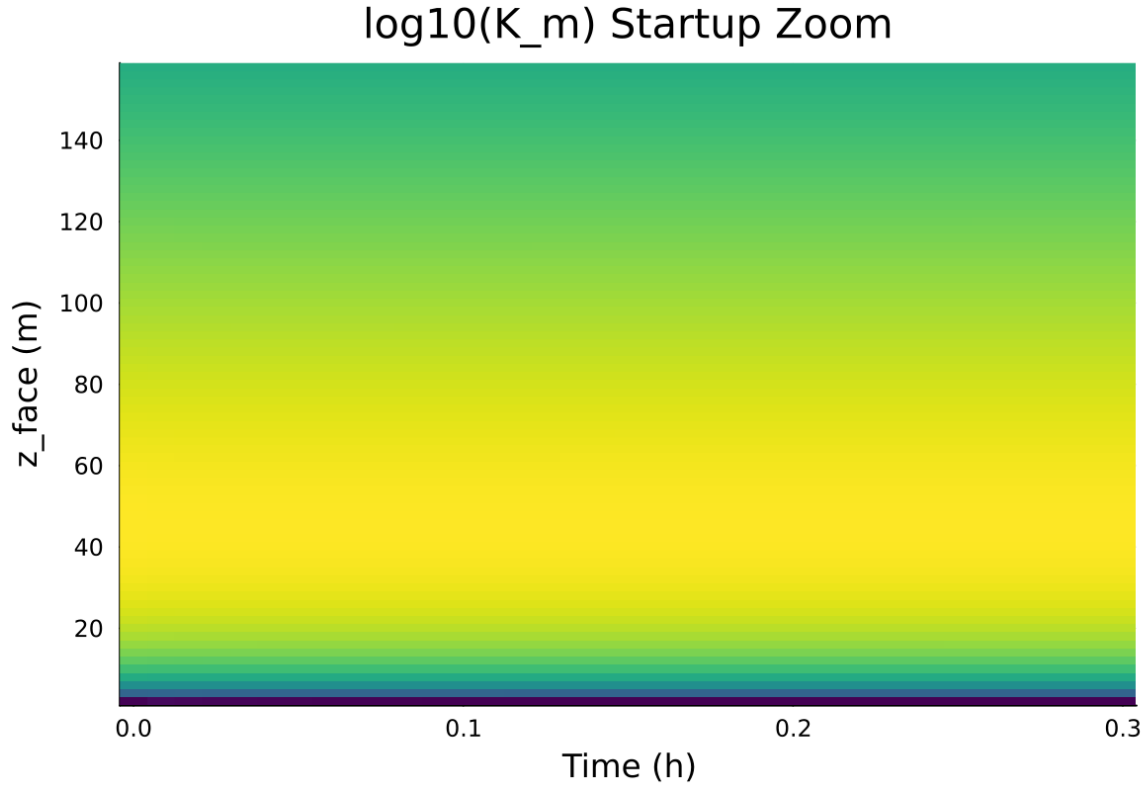


Figure 72: Surface energy budget components (R_n , H , G , storage)

Table 7: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	$10.000000 \text{ m s}^{-1}$
Meridional geostrophic wind	V_g	$0.000000 \text{ m s}^{-1}$
Surface roughness (momentum)	z_{0m}	0.020000 m
Surface roughness (heat)	z_{0h}	0.005000 m
Coriolis parameter	f	$1.000\text{e-}04 \text{ s}^{-1}$
Reference potential temperature	θ_a	280.000000 K
Deep soil temperature	T_{deep}	276.000000 K
Downward longwave forcing	$R_{\downarrow}$	$300.000000 \text{ W m}^{-2}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{min,surf}$	$0.001000 \text{ m}^2 \text{ s}^{-1}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 s^{-1}

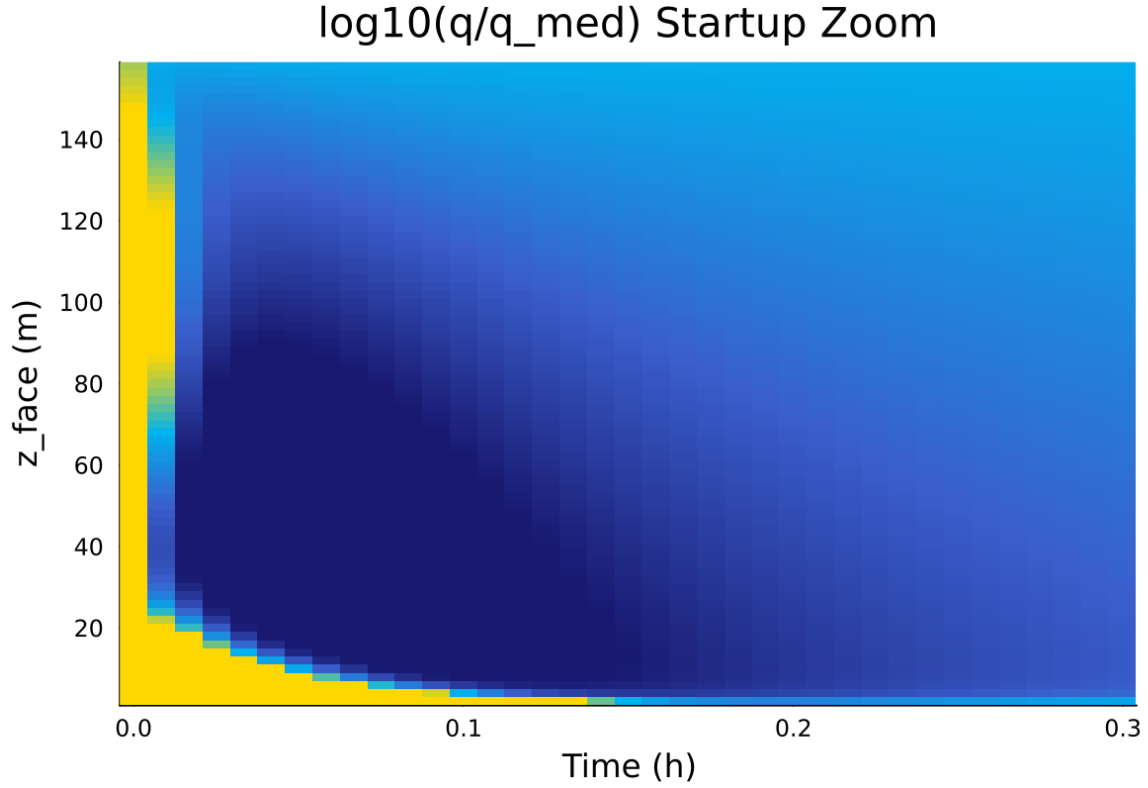


Figure 73: Phase portrait on regularized manifold (Delta vs e_{xi})

15.2 Forcing and Boundary Conditions

15.3 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [25.002377, 94.099418] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [0.035577, 103.656525]$

15.4 Generated Artifacts

- Time-series CSV: `results/scm_verify/time_series.csv`
- Diagnostic summary JSON: `results/scm_verify/summary.json`
- Payload JLD2: `results/scm_verify/payload.jld2`

15.5 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times

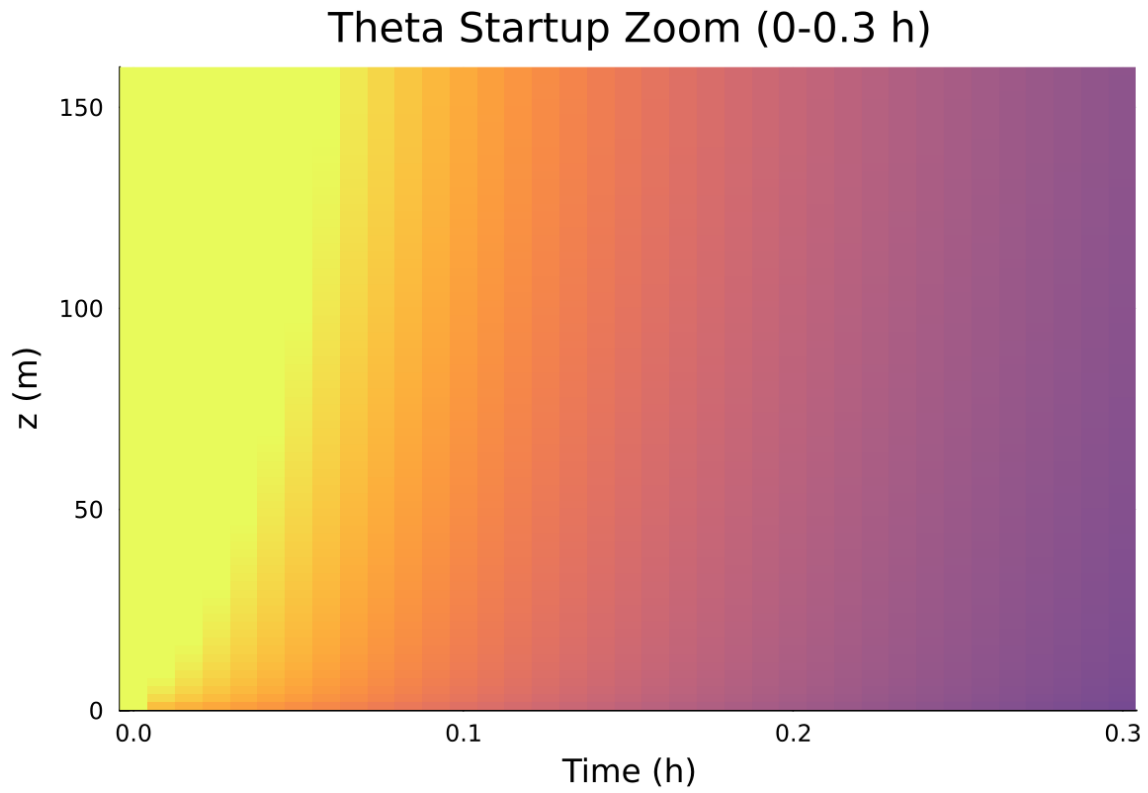


Figure 74: Eddy diffusivity response (K_m , K_h) versus stability/Richardson number

5. Surface energy budget components (R_n , H , G , storage)
6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Eddy diffusivity response (K_m , K_h) versus stability/Richardson number
8. Fold proximity diagnostics versus time

15.6 Included Plots

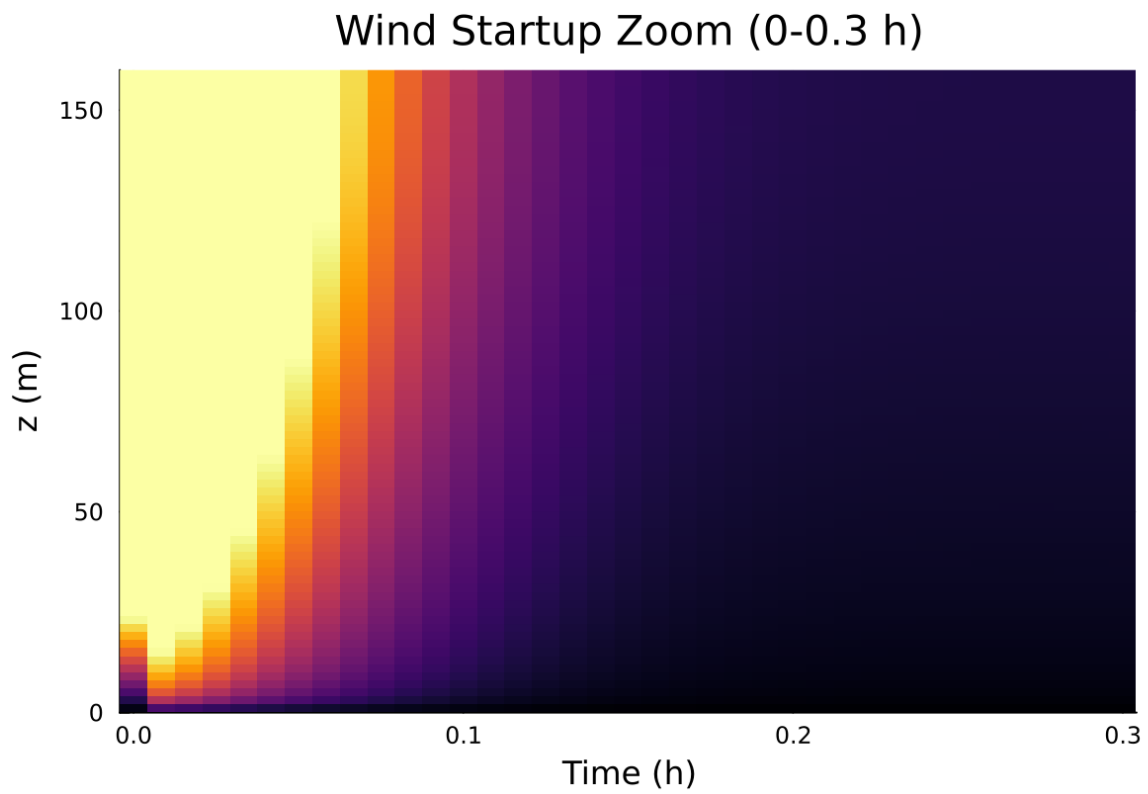


Figure 75: Fold proximity diagnostics versus time

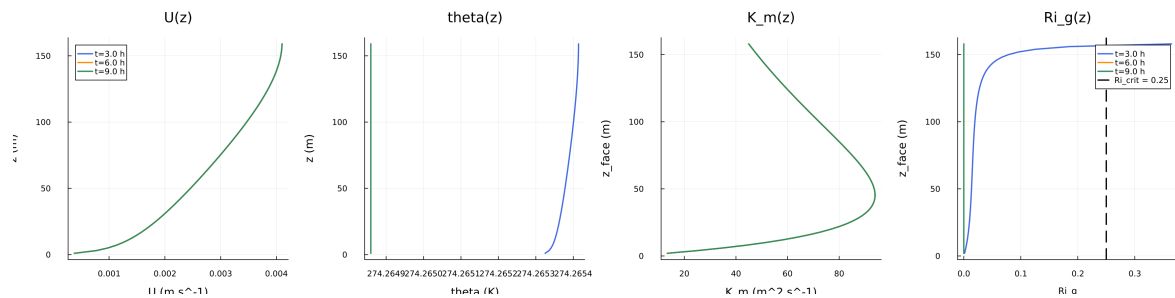


Figure 76: fig04 profiles u theta km.png

Figure 5: Surface Energy Budget

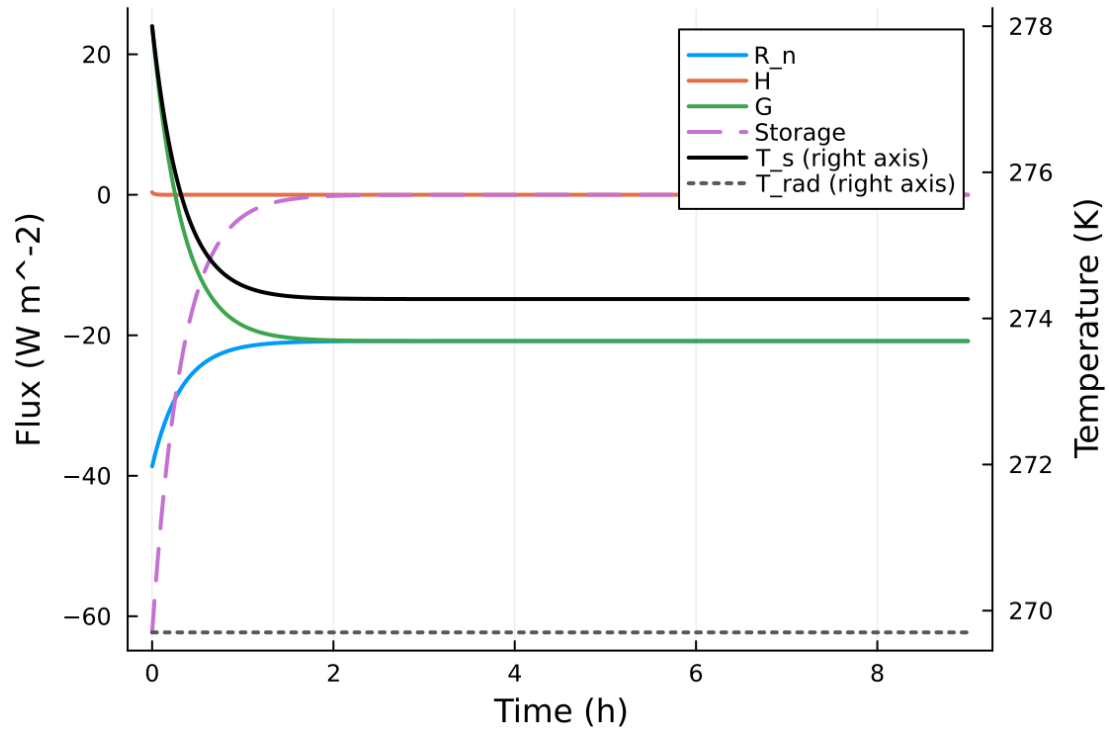


Figure 77: fig05 surface energy budget.png

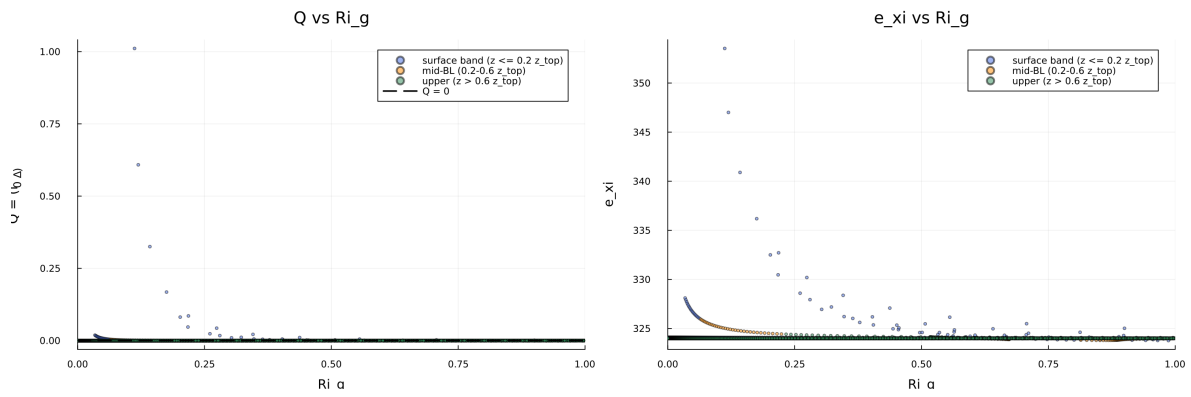


Figure 78: fig06 phase delta exi.png

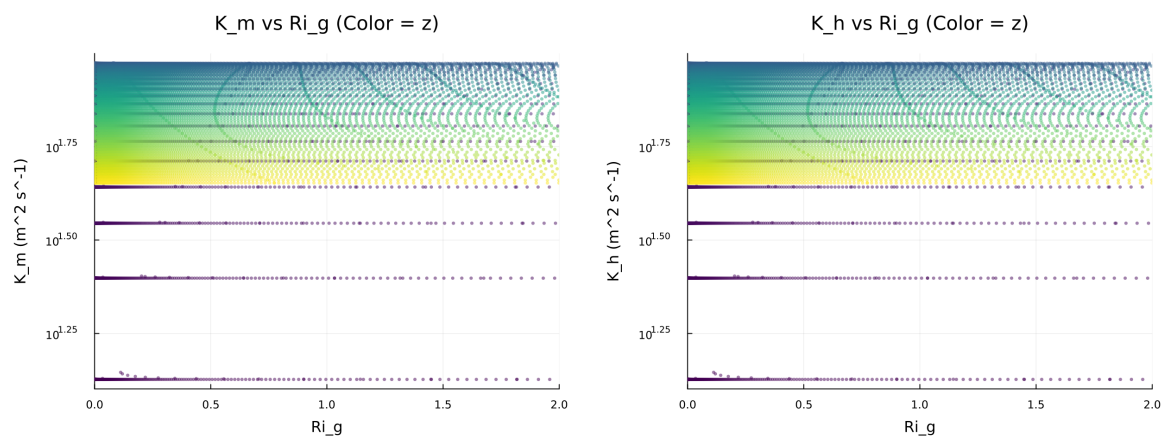


Figure 79: fig07 diffusivity vs ri.png

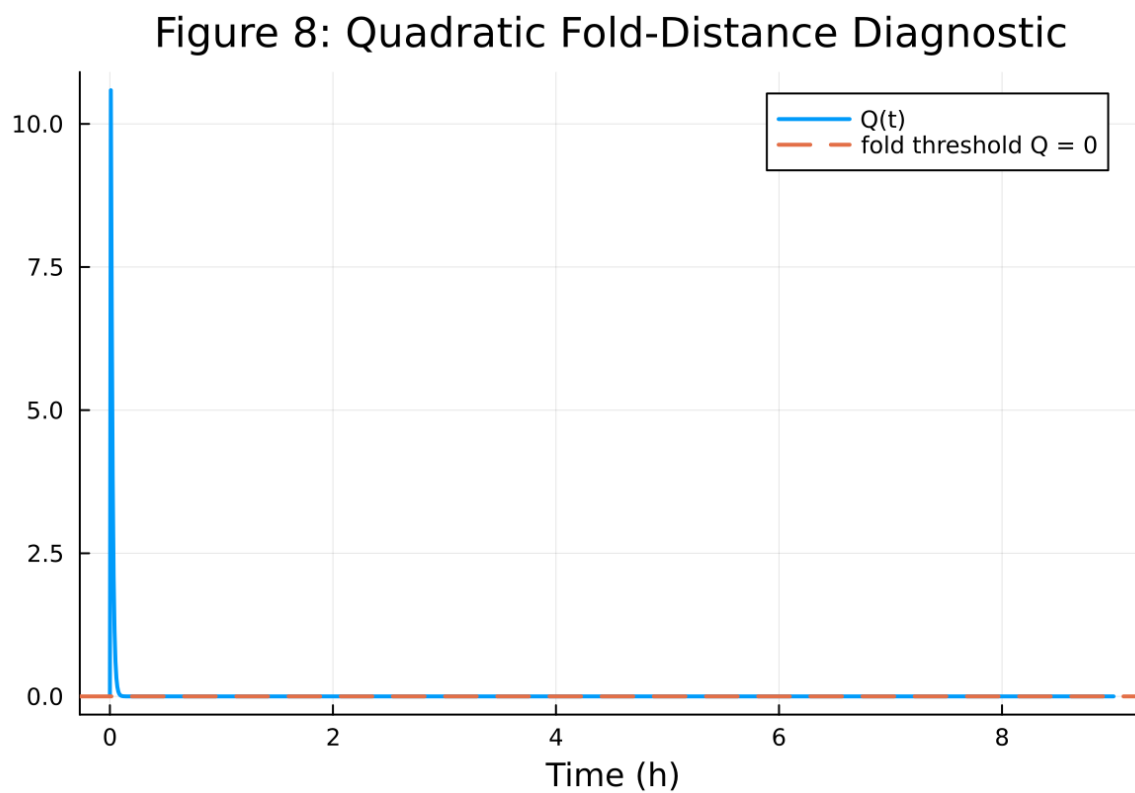


Figure 80: fig08 fold proximity.png

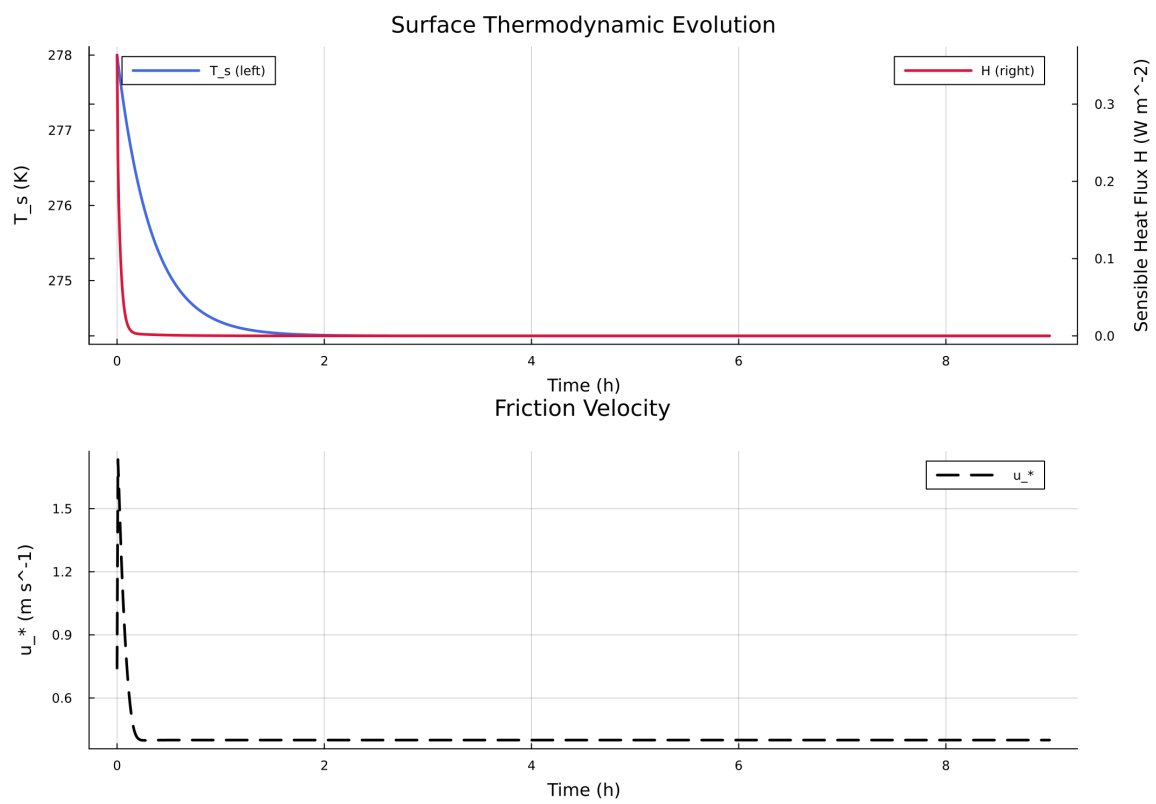


Figure 81: Time series of T_s , sensible heat flux, and friction velocity

Figure 2: Time-Height Wind Speed

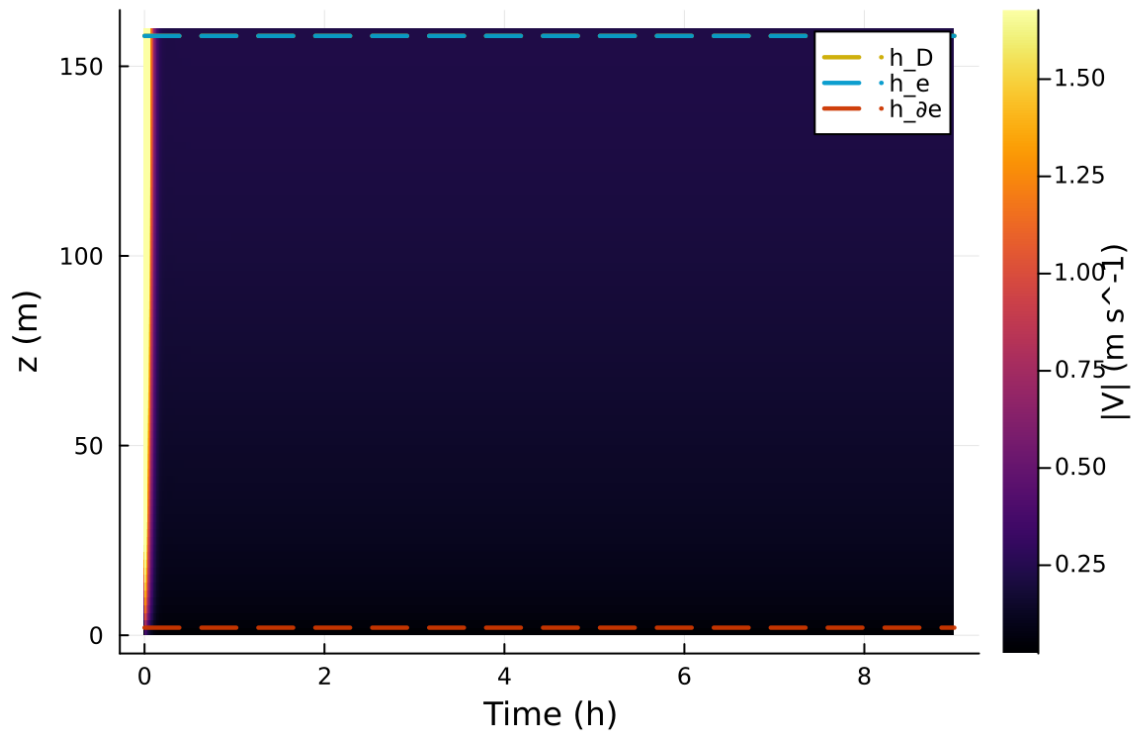


Figure 82: Time-height contour of wind speed with LLJ evolution

Figure 3: Time-Height Potential Temperature

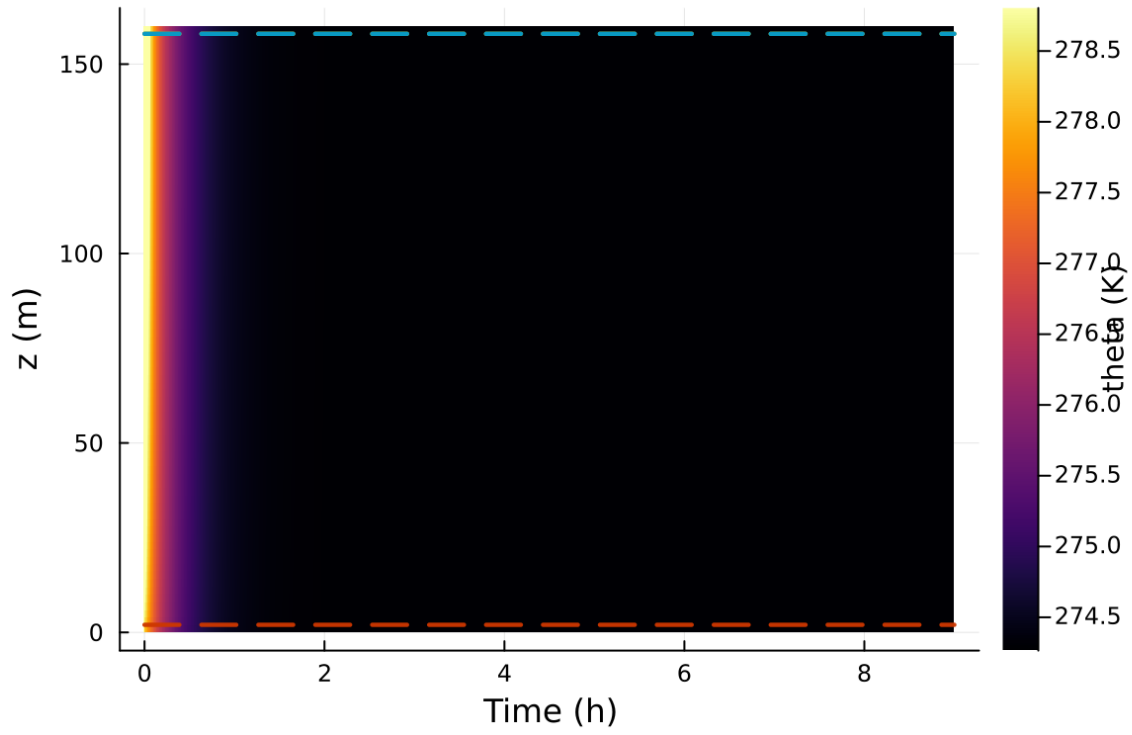


Figure 83: Time-height contour of potential temperature and inversion growth

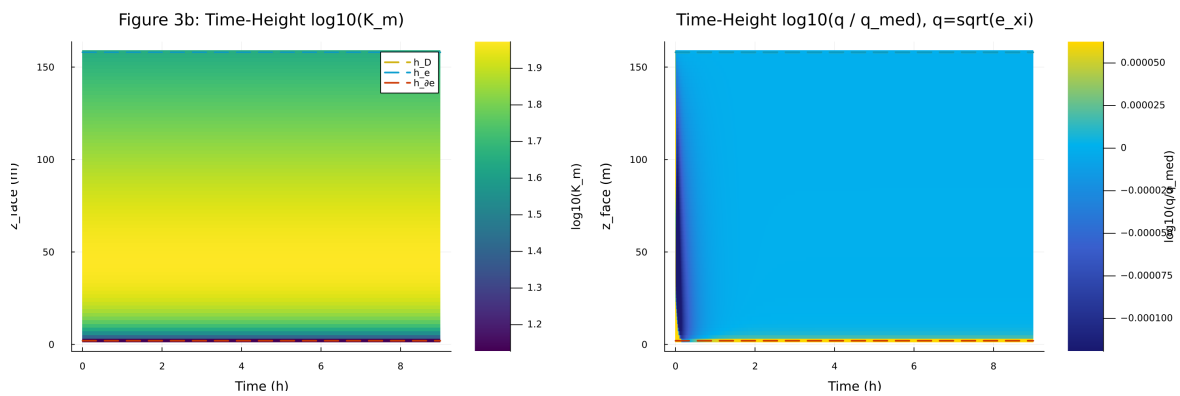


Figure 84: Vertical profiles of U , θ , and K_m at representative times

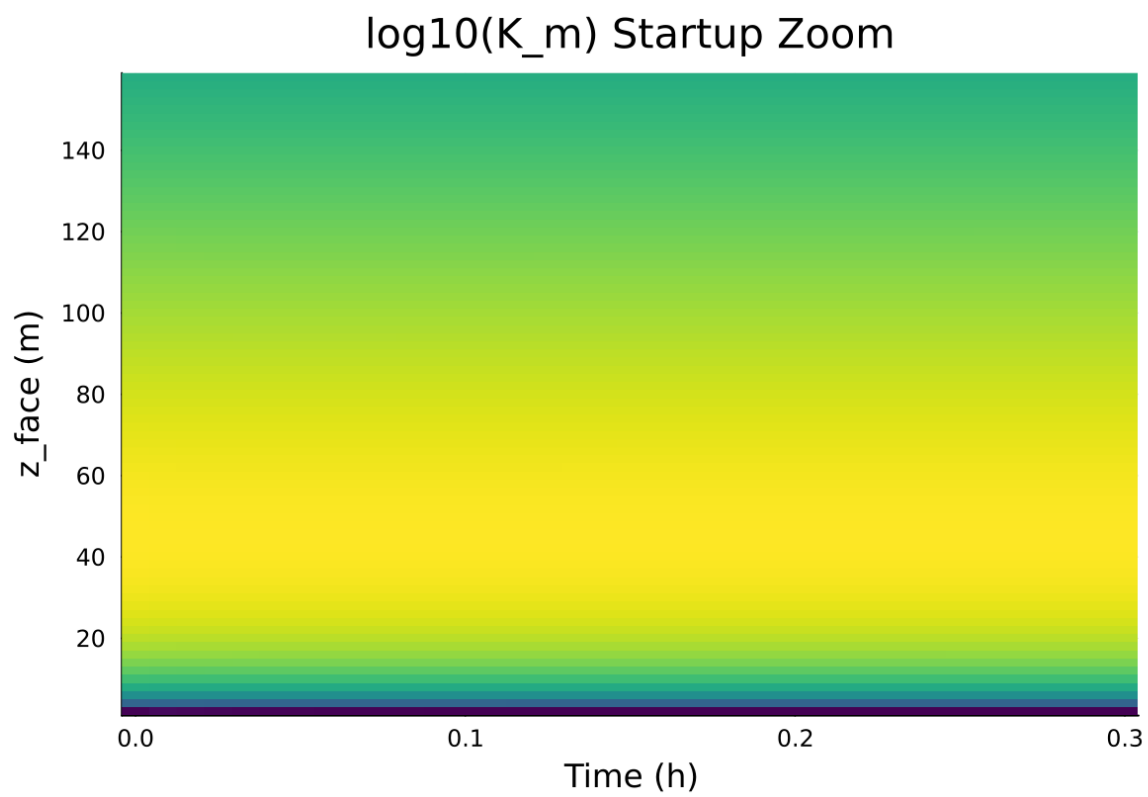


Figure 85: Surface energy budget components (R_n , H , G , storage)

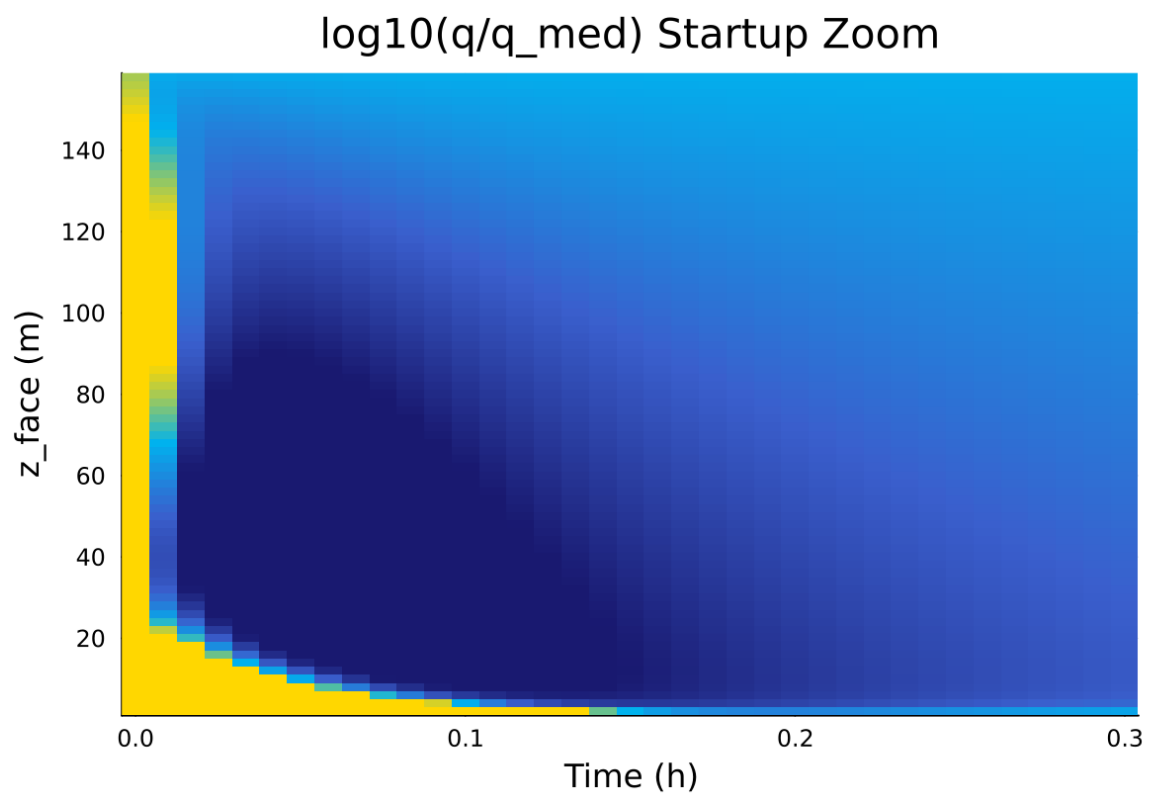


Figure 86: Phase portrait on regularized manifold (Delta vs e_{xi})

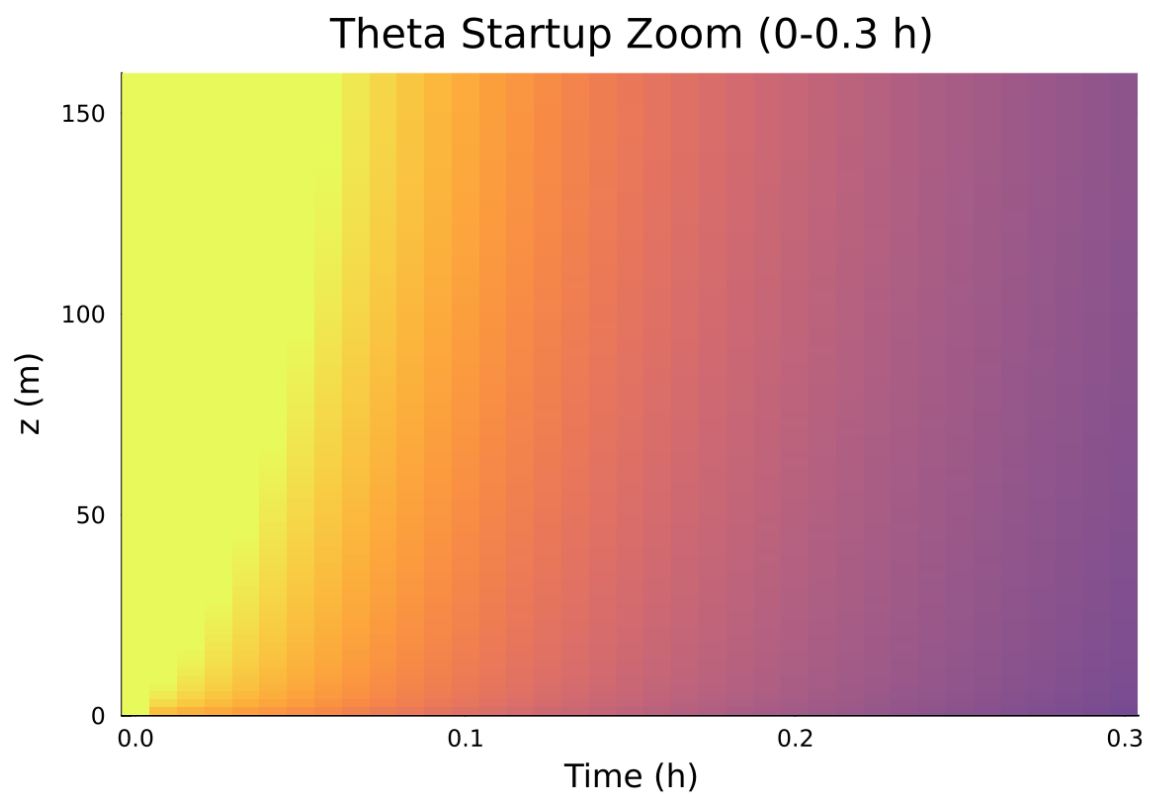


Figure 87: Eddy diffusivity response (K_m , K_h) versus stability/Richardson number

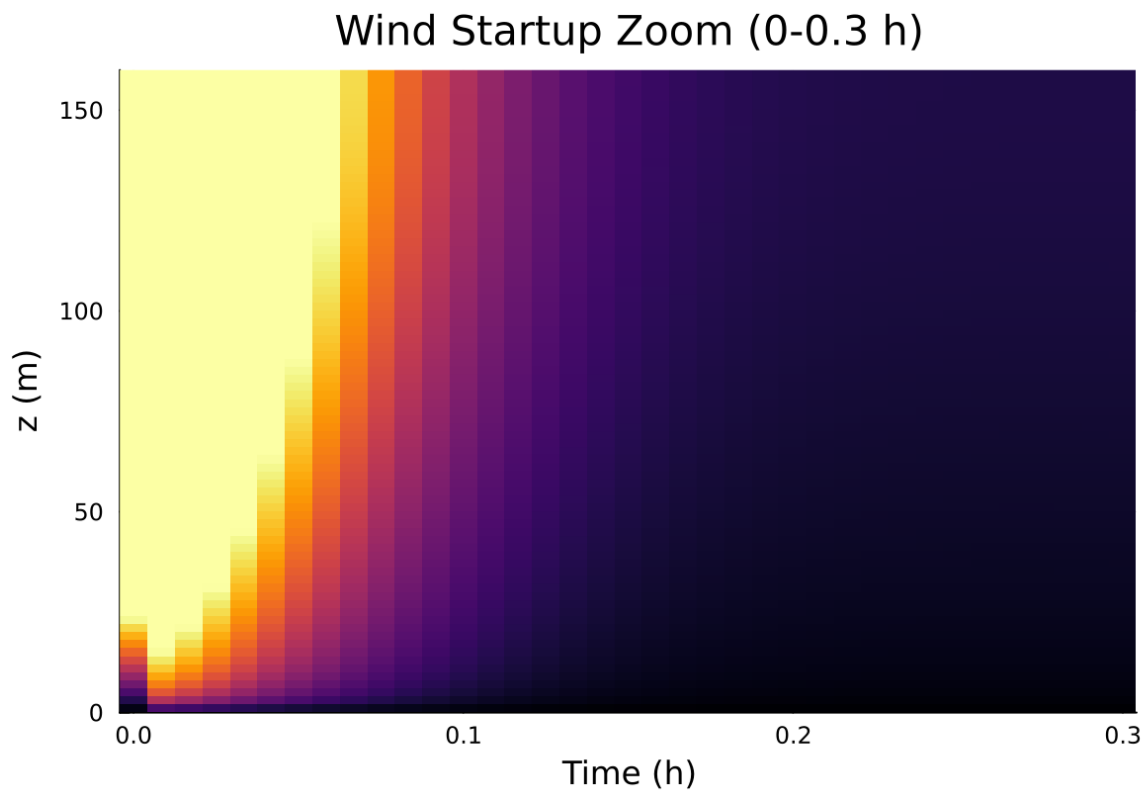


Figure 88: Fold proximity diagnostics versus time

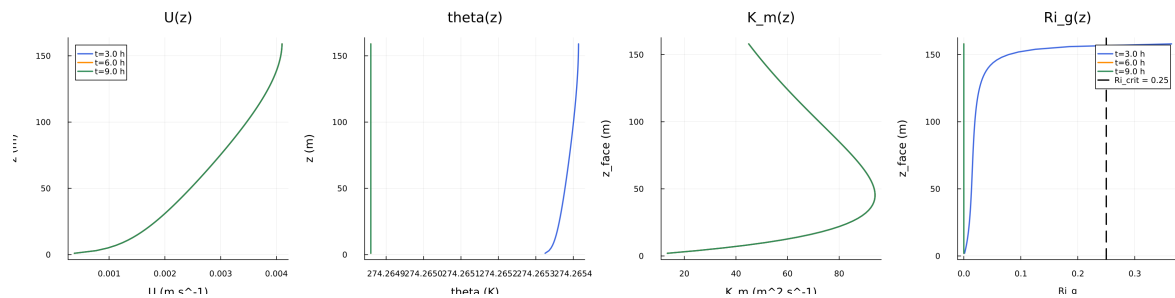


Figure 89: fig04 profiles u theta km.png

Figure 5: Surface Energy Budget

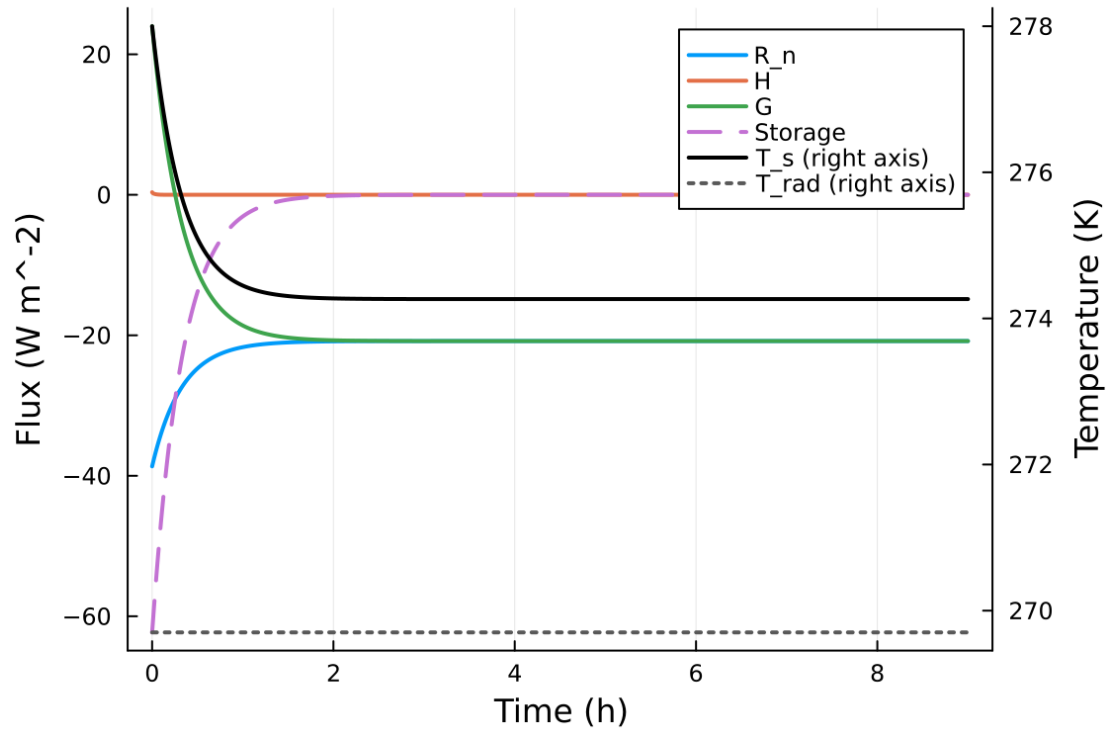


Figure 90: fig05 surface energy budget.png

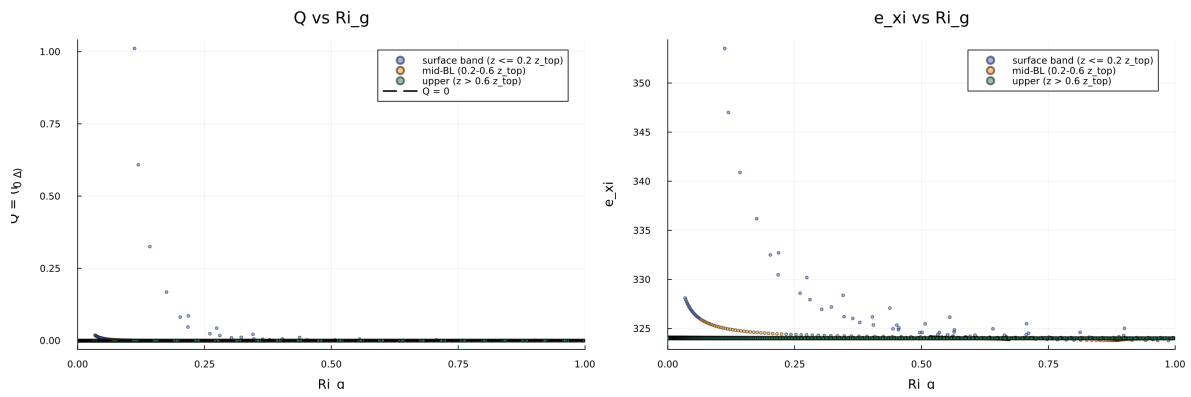


Figure 91: fig06 phase delta exi.png

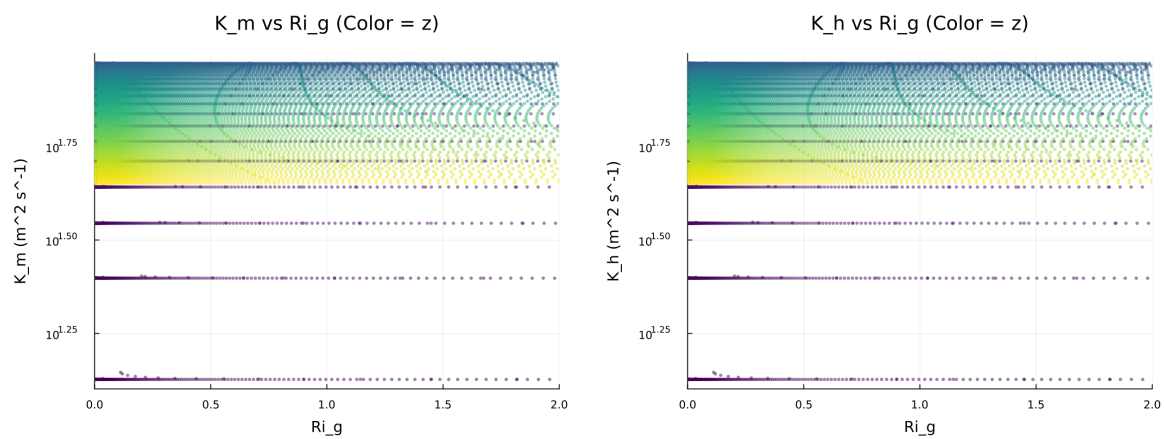


Figure 92: fig07 diffusivity vs ri.png

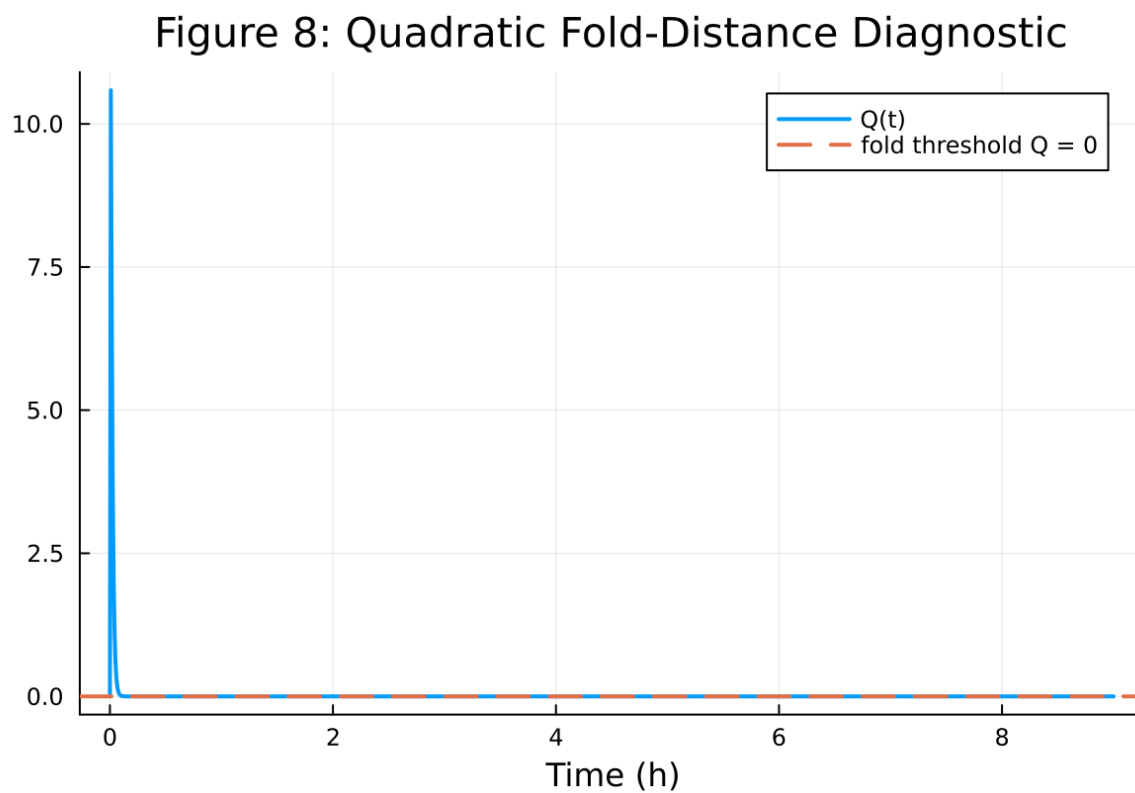


Figure 93: fig08 fold proximity.png